SUBBALAKSHMI LAKSHMIPATHY COLLEGE OF SCIENCE

(AN AUTONOMOUS INSTITUTION)

"Affiliated to the Madurai Kamaraj University and Re-Accredited with B+ Grade by NAAC"

TVR NAGAR, ARUPPUKOTTAI ROAD, MADURAI - 625 022

DEPARTMENT OF COMPUTER SCIENCE (SSS)

Cyber Forensics Lab - 21CS508P

2023 - 24

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III B.Sc Computer Science(Security System Specialization)

Semester	**V**	**Batch**	**2021-2024**
Course Code	**21CS508P**	**Course Name**	**Cyber Forensics Lab**
CIA Marks	**40**	**End Semester Marks**	**60**
Part	**III [Elective Practical]**	**No. of Hours / Week**	**6**
No. of Credits	**3**		

Course Objective:

To enable the students to understand the usage Cyber Forensics Tools and the techniques.

Course Outcomes:

After the completion of the course the students will be able to,

CO 1: Analyze the practical implementation of Cyber Forensics

CO 2: Determine the need of data acquisition using tools

CO 3: Estimate the use of windows registry and file system during investigation

CO 4: Inspect the password strength using application password cracking

CO 5: Develop Investigation on Mobile and Email crimes

Ex.No:1
DATE: **DISK PARTITION**

AIM:

To do disk partition using Minitool partition wizard.

REQUIREMENT:

Minitool partition wizard.

REQUIREMENT DESCRIPTION:

MiniTool Partition Wizard Free is free partition management software for Windows that can perform lots of different tasks on hard drives and partitions. It can copy, format, delete, wipe, extend, and resize partitions.

DISK PARTITION:

Disk partitioning or disk slicing is the creation of one or more regions on secondary storage, so that each region can be managed separately. These regions are called partitions. It is typically the first step of preparing a newly installed disk, before anyfile system is created.

TOOL DESCRIPTION:

MINI TOOL:

Mini tool partition wizard free helps users to manage disks and partitions,check file system,alignssdpartition,migrateos to ssd,clone disk

TASKS:

- Create a partition.
- Extend a partition.
- Split a partition
- Merge two partitions.

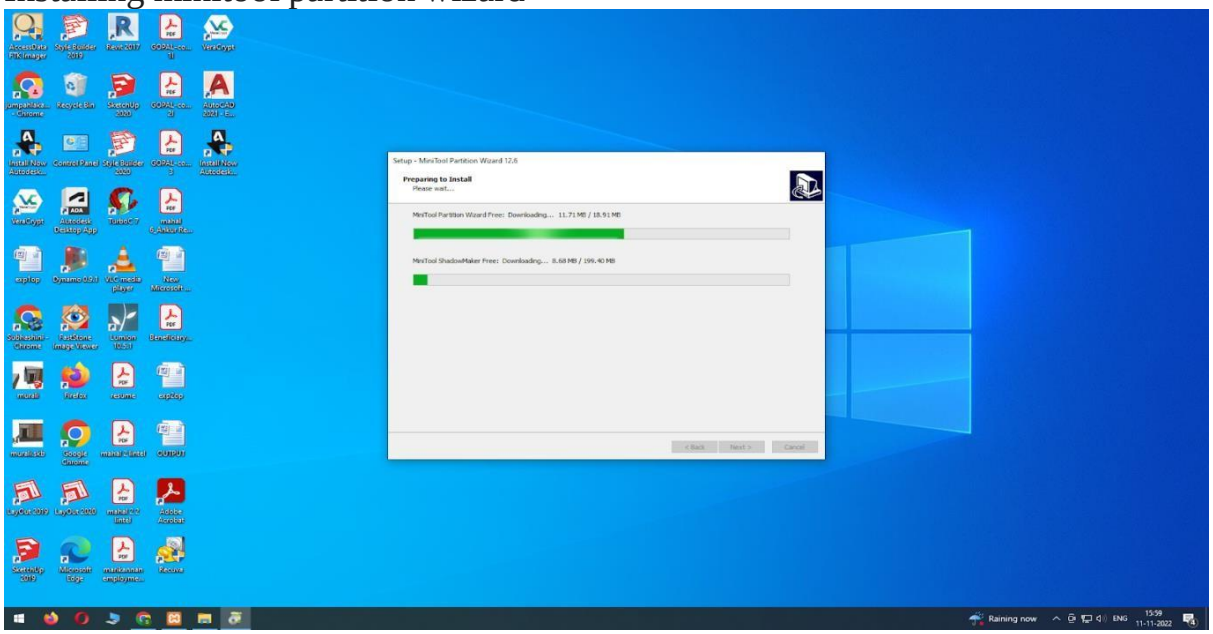
PROCEDURE:

- Open google chrome.
- Download and install minitool partition wizard.

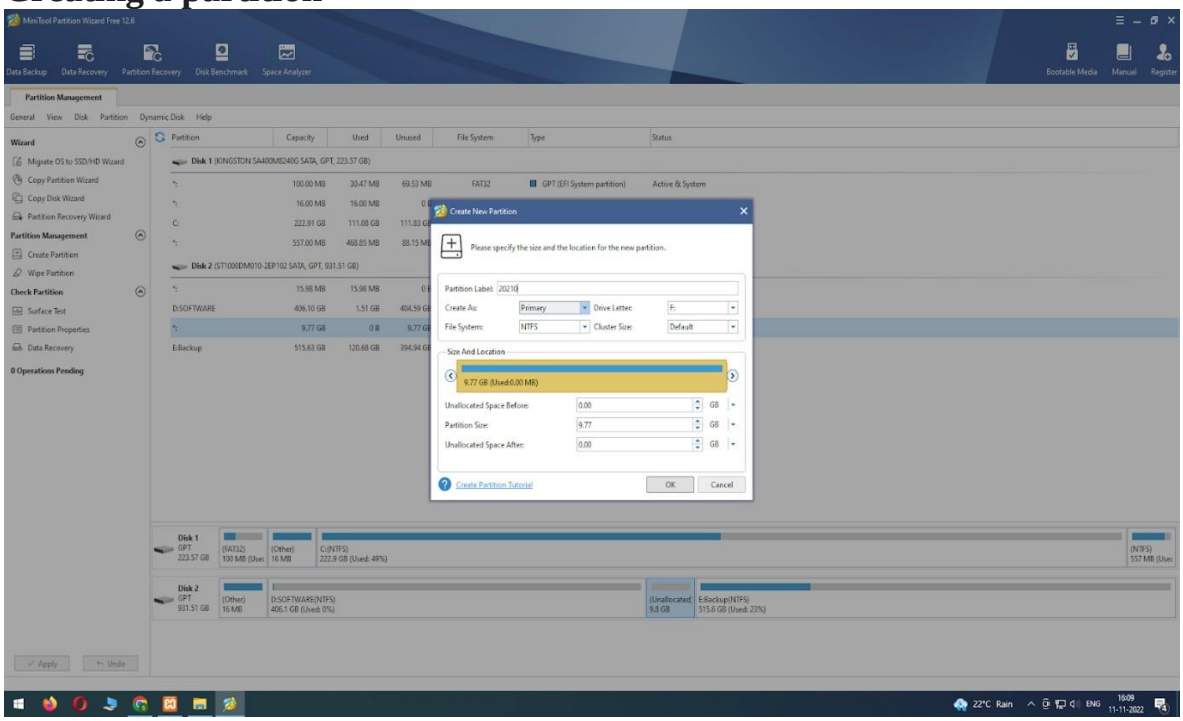
(i) To create a partition, right click the unallocated space and click create partition.
(ii) To extend a partition, right click any partition and click extend option.
(iii) To split a partition, right click the partition which you want to split, and click the split option.
(iv) To Merge a partition, right click the partition and click merge. Then a wizard will open. In that wizard type the partition which you want to merge with. And click ok. Then the partitions will be merged.
(v) To delete a partition, right click the partition and click delete, then the partition will be deleted.

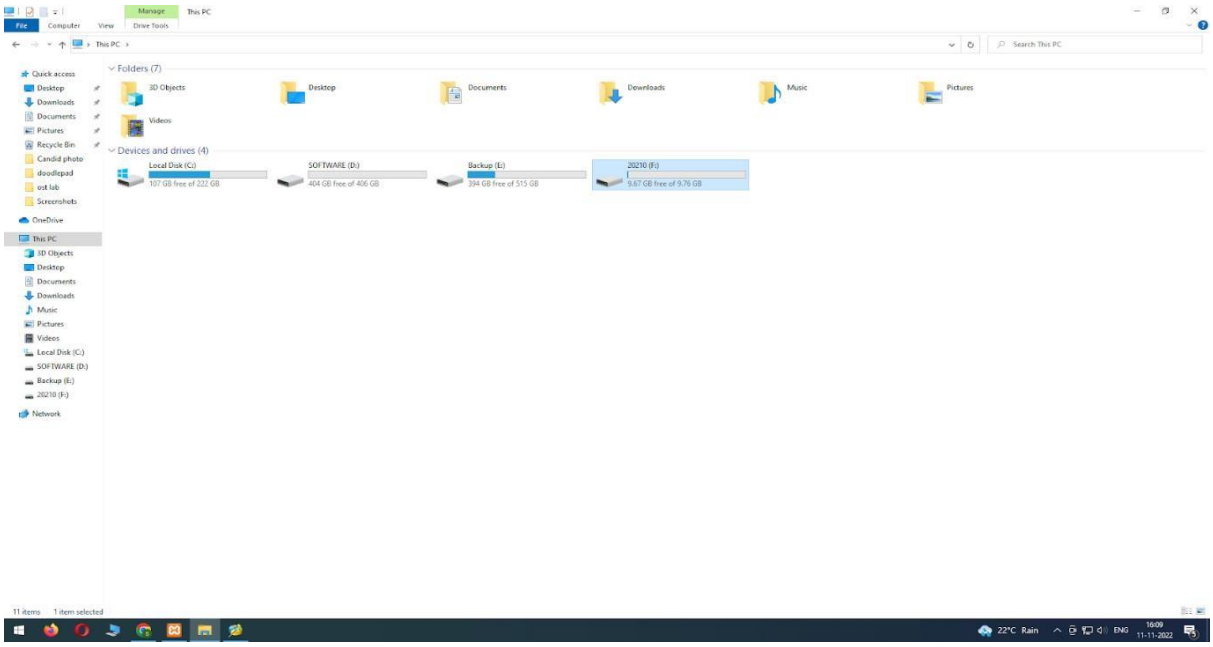
OUTPUT:

Installing minitool partition wizard

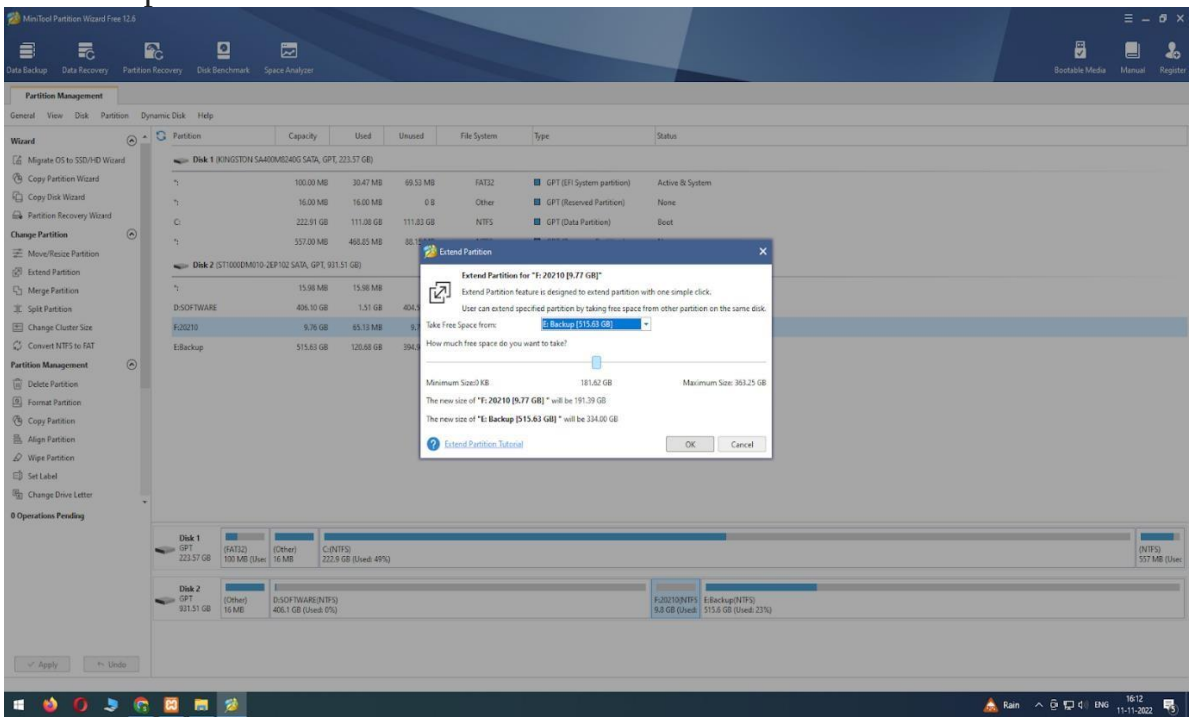


Creating a partition

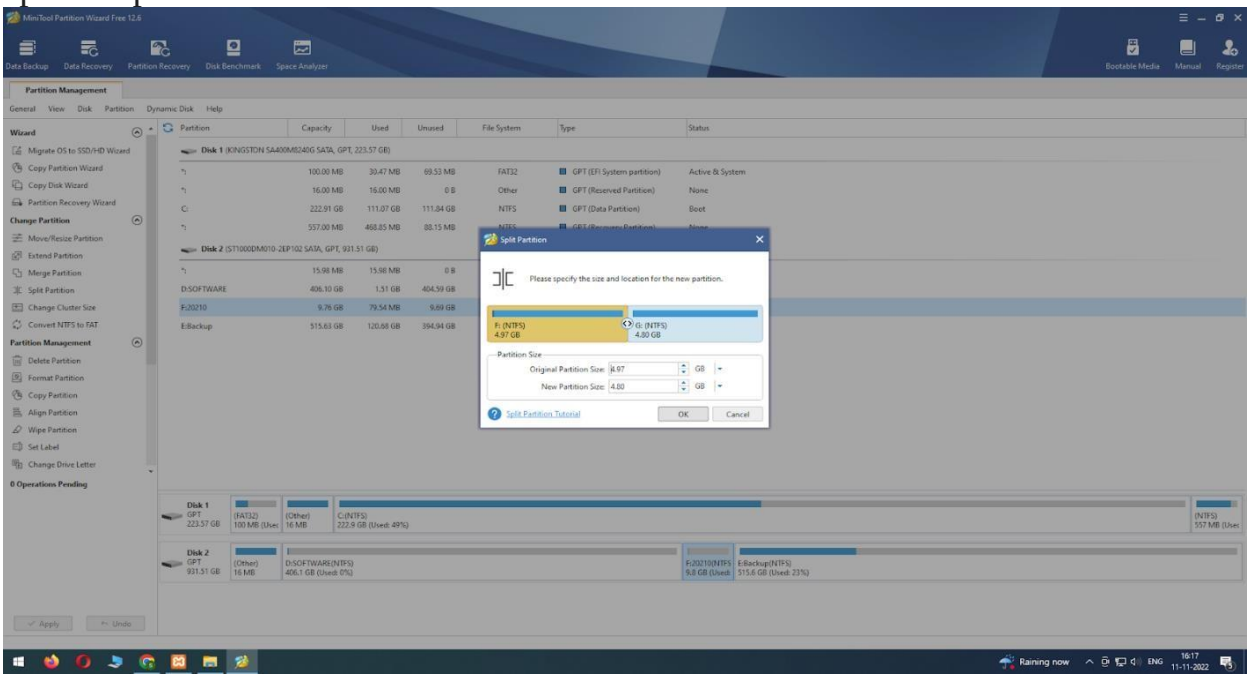


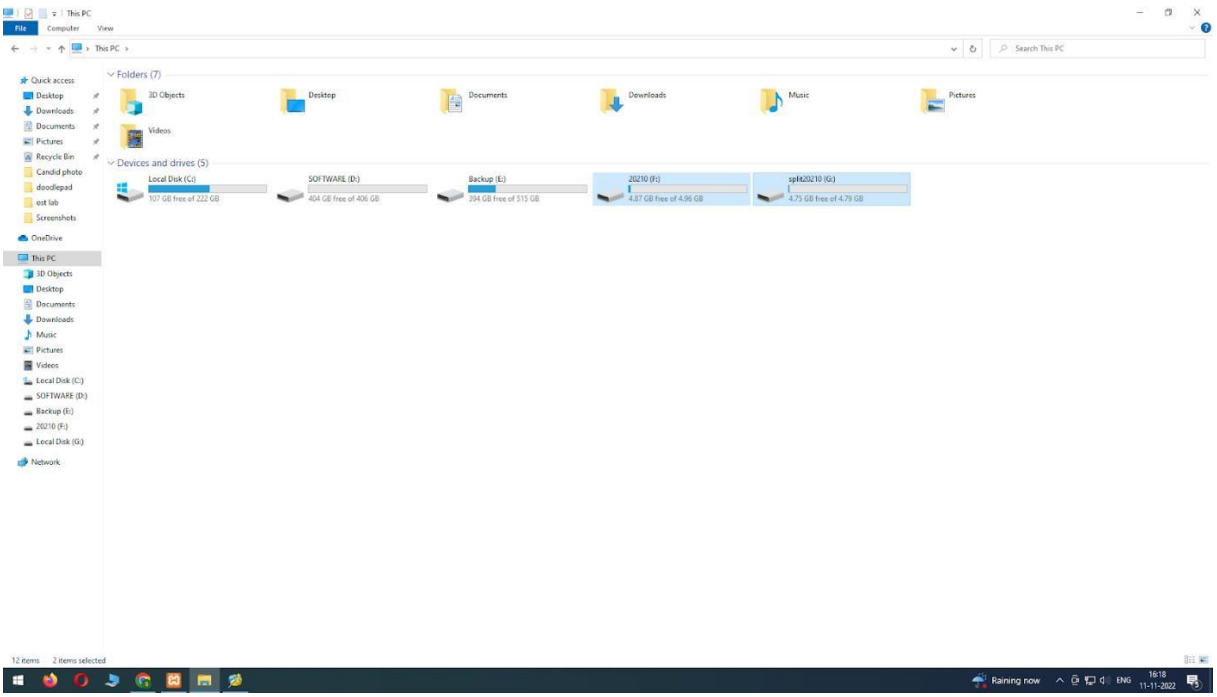


Extend a partition

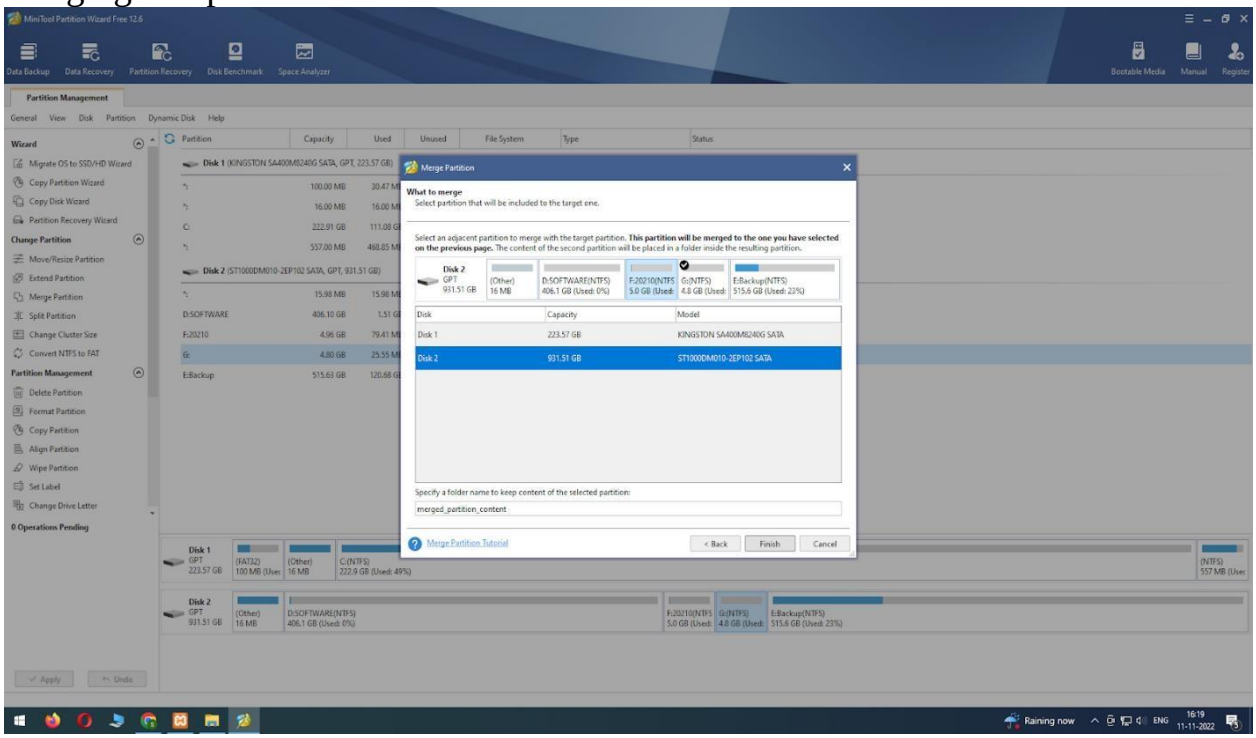


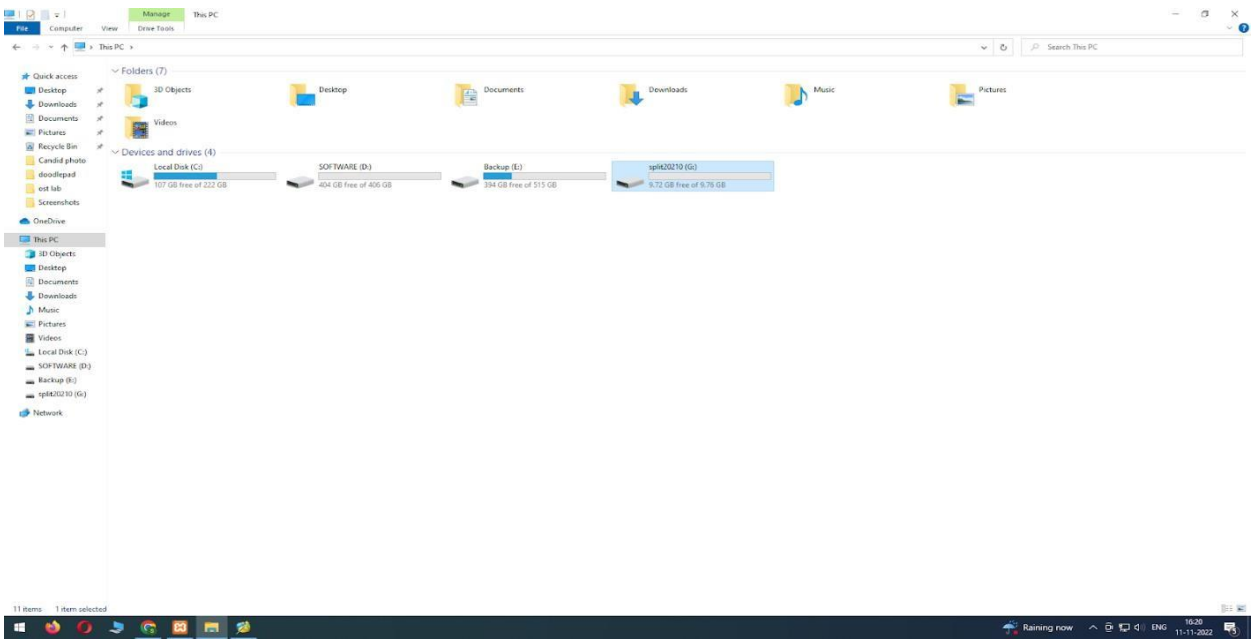
Split the partition





Merging two partitions





RESULT:

Thus, the disk partition using minitool wizard has been completed successfully.

Ex.No: 2
DATE: **DATA RECOVERY**

AIM:

To recover data using recuva tool

REQUIREMENTS:

Recuva tool

DATA RECOVERY:

Is the process of restoring data that has lost,accidentallydeleted,corrupted made inaccesible,formatted data from secondary storage,removable media or filesRecovery required due to physical damage to the storage devices or logical damage to the file system that prevents it from being mounted by the host os .

TOOL DESCRIPTION:

RECUVA:

Can recover pictures,music,documents,videos or any other file type you have lost. Recover from any writable media you have memory cards external hard disk drives.

TASK:

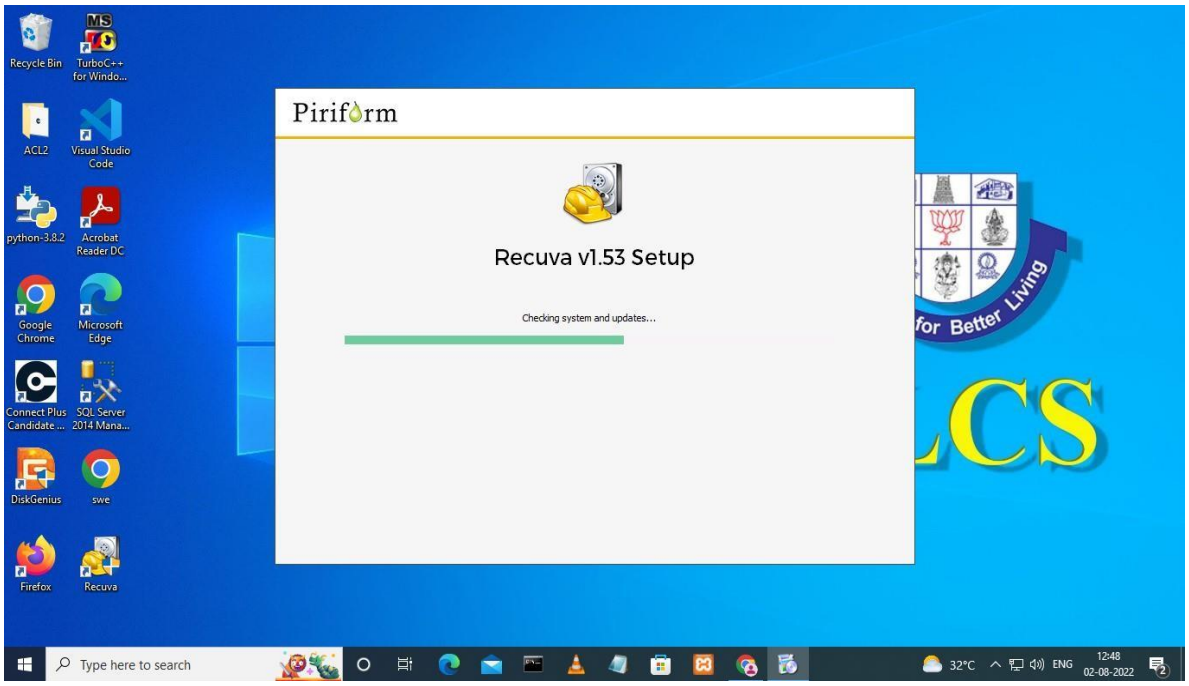
Use the recuva tool to recover deleted data in storage device search as text,imageetc

PROCEDURE:

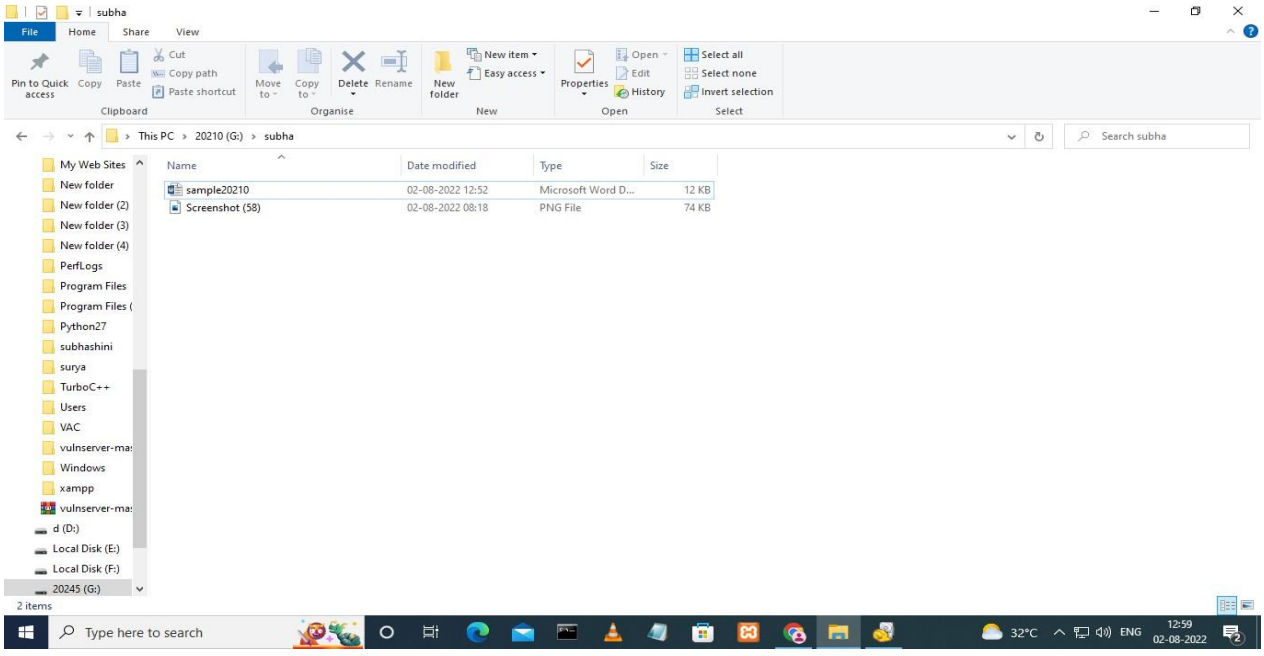
- Select the partition
- Save any text document ,image or other type of files you want
- Delete the file
- Also you need to delete it in recycle bin
- Install recuva tool and open it
- On the top select your partition ,select file type in drop down option in right corner
- And hit scan option
- It starts scanning and wait for few minutes
- After you get your lost file screen click on that and right click and choose recover option
- It asks for a path where to recover the file given path as you need.
- Now your lost file has recovered.

OUTPUT:

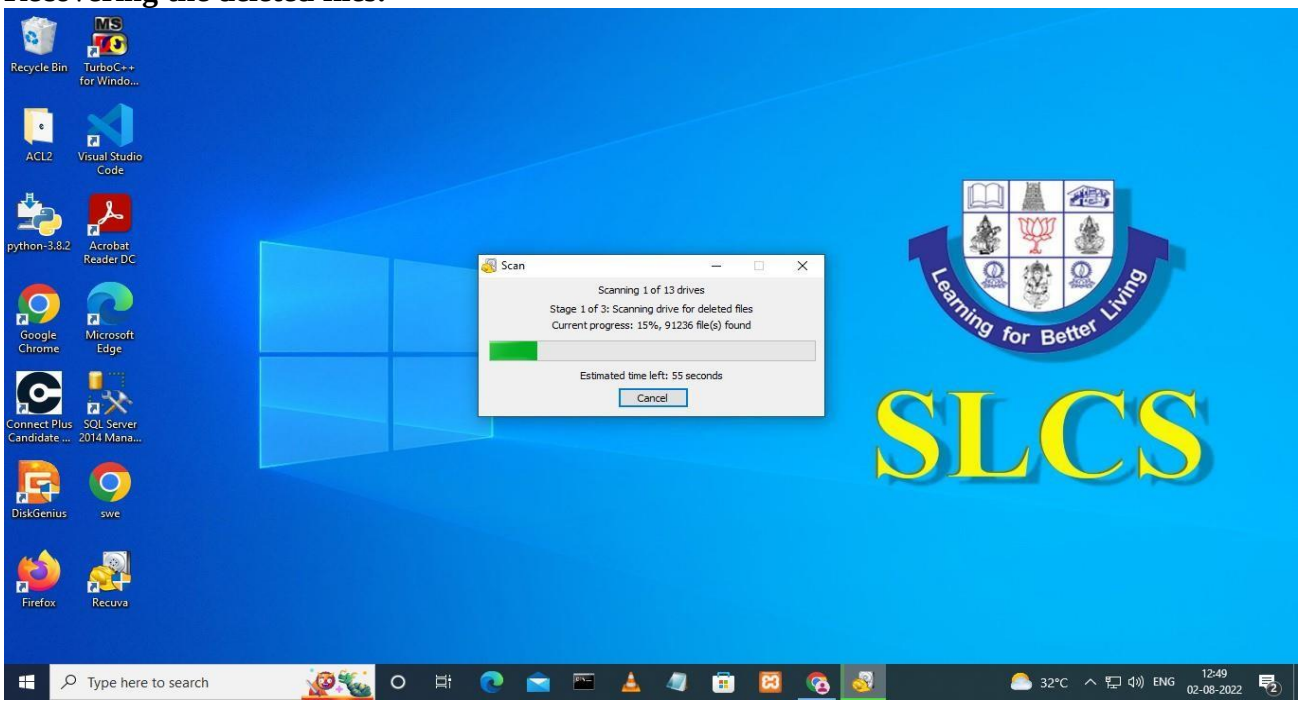
Install recuva:



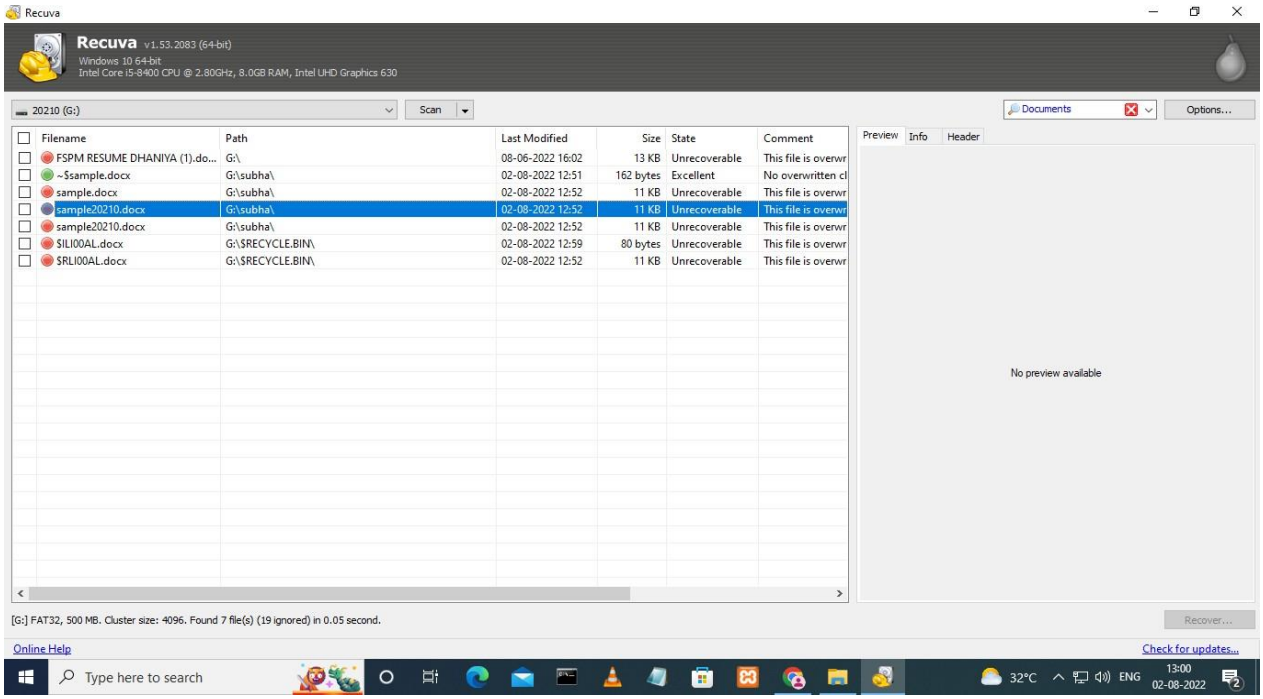
Deleted files:

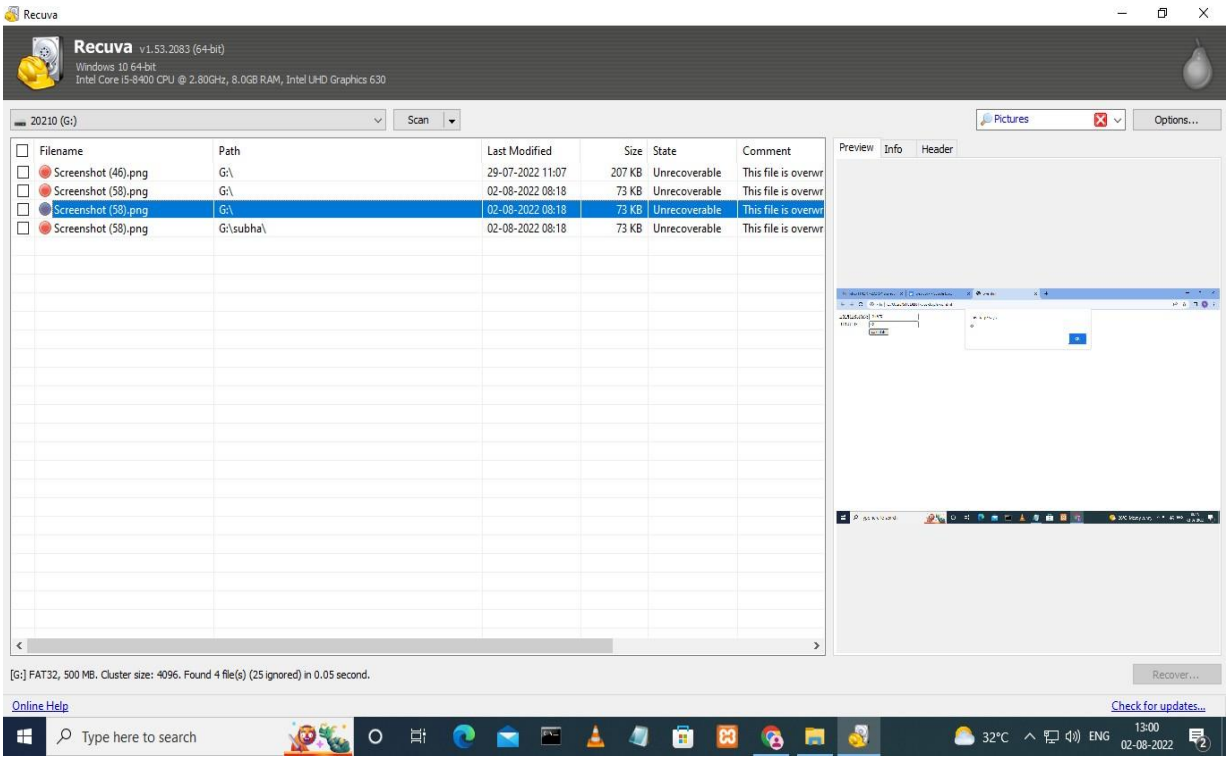


Recovering the deleted files:



Recovered files:





RESULT:

Thus, the data recovery has been done successfully using recuva tool.

Ex.No:3
DATE: WINDOWS FORENSICS – REGISTRY, EVENT VIEWER

AIM:

To perform windows forensics using registry editor and event viewer.

REQUIREMENTS:

- Windows os
- Registry editor
- Event viewer

WINDOWS FORENSICS:

Windows forensics include the process of conducting or performing forensics investigations of system which run on windows operating system

REQUIREMENT DESCRIPTION:

REGISTRY EDITOR:

Registry editor is a graphical tool in the Microsoft Windows operating system that enables authorized users to view the windows registry and make changes.

EVENT VIEWER:

The Windows 10 Event viewer is an app that shows a log detailing information about significant events on your computer. This information includes automatically downloaded updates, errors and warnings.

TASKS:

- Windows registry analysis
  1. Find the key link which maintains a list of entries executed using run command
  2. Find the key link Which maintains a list of most recently used files in windows
  3. Investigate the key link which maintains windows virtual memory
  4. Find the key link to display the list of installed program in the computer
  5. Find the key used for detecting the networking evidence from registry
- Event viewer analysis
  1. List out the last file activities that occurs in the system
  2. List out the last file events of system login and logout.
  3. Investigate the application and service log

4. Investigate the security log of Audit success and failure.

PROCEDURE:

WINDOWS REGISTRY ANALYSIS:

- Open run
- Type ‘regedit’ and run it . Registry editor will be opened.
- Open the path“**Computer\HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\E xplorer\RunMRU**” to find the key link which maintains a list of entries executed using run command

- Open the path "**Computer\HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Ex plorer\ComDlg32\OpenSavePidlMRU\docx**" to find the key link Which maintains a list of most recently used files in windows
- Open the path "**Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\ Session Manager\Memory Management**" to investigate the key link which maintains windows virtual memory
- Open the path "**Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\ Uninstall** " to find the key link to display the list of installed program in the computer
- Open the path "**Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\ Network**" to find the key used for detecting the networking evidence from registry

EVENT VIEWER ANALYSIS:

- Select start → event viewer.
- Select windows logs→ system to view the last file activities that occurs in the system.
- Select windows logs→ security to view the last event of system login and logout .
- Select application and services log to investigate the application and service log.
- Select windows logs → security to investigate the security log of audit success and failure.

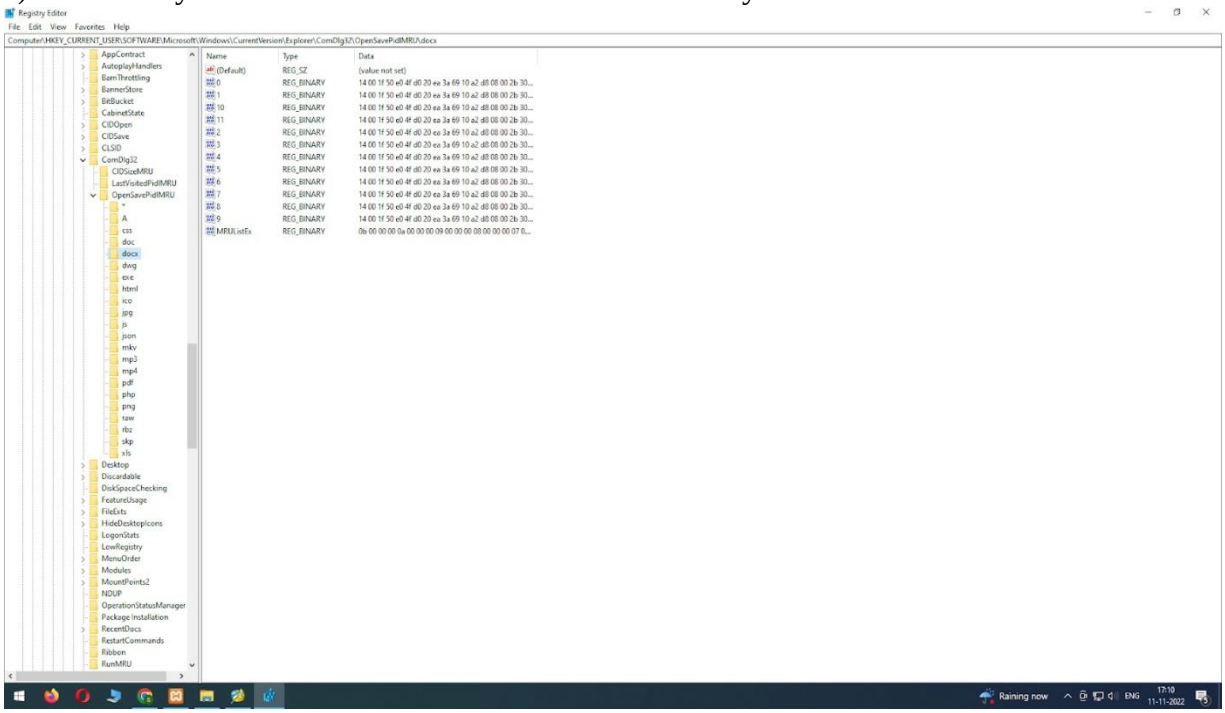
OUTPUT:

Windows registry analysis:

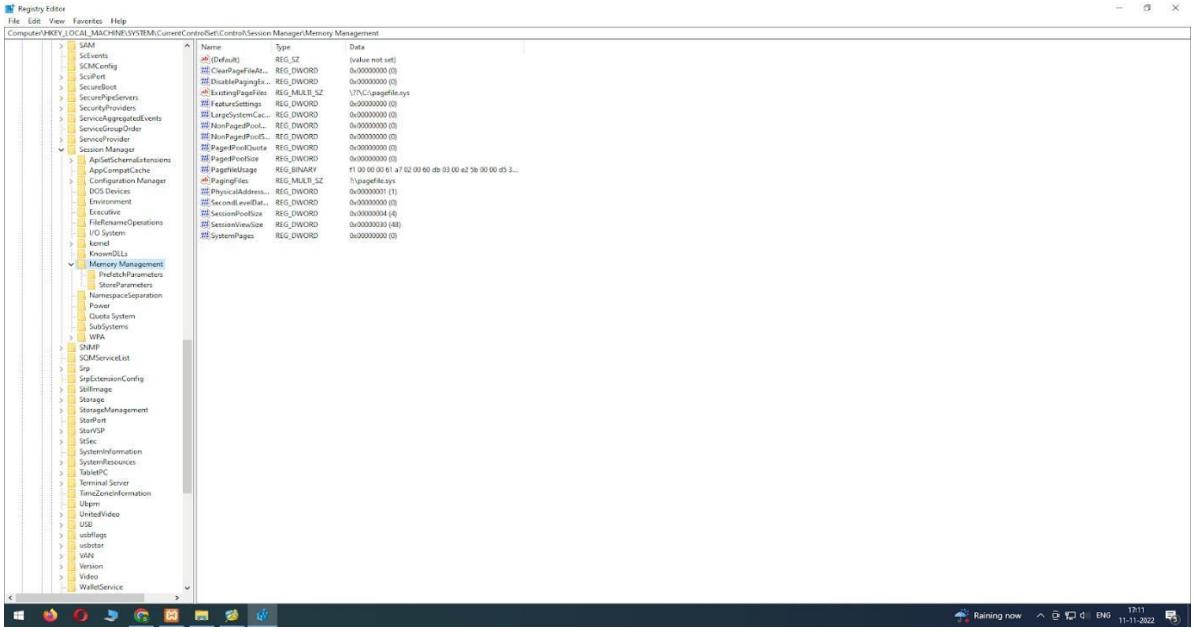
1) Find the key link which maintains a list of entries executed using run command



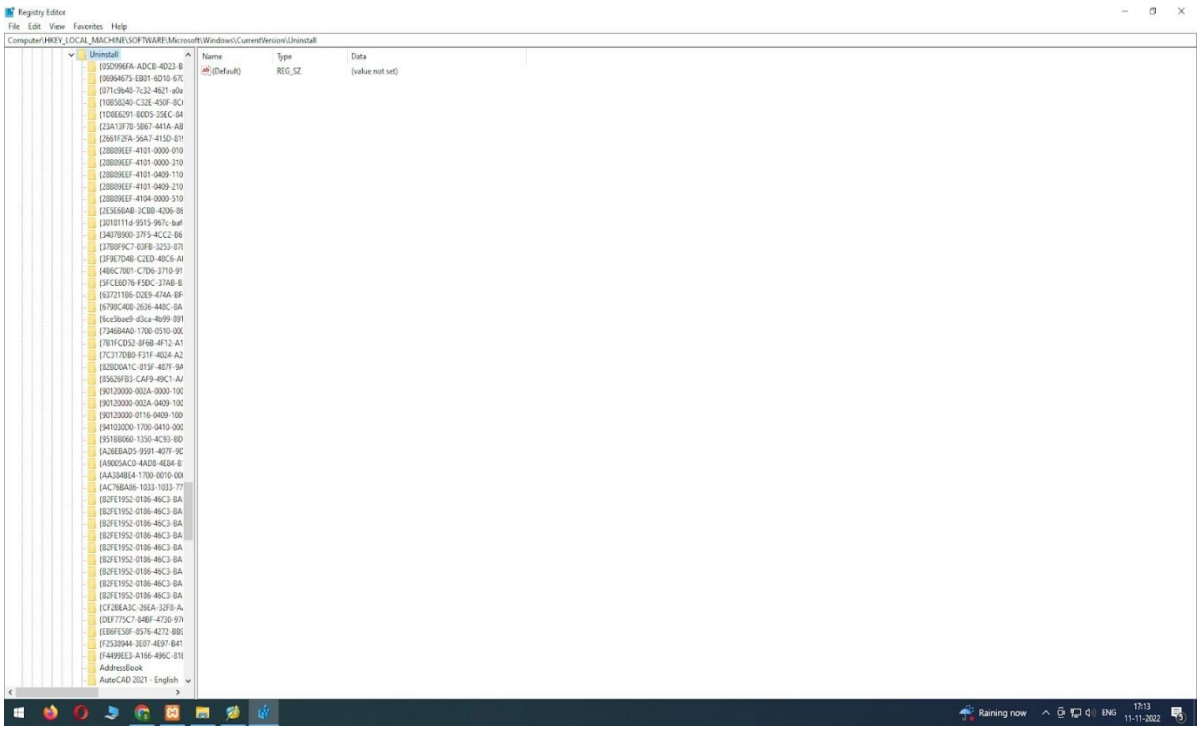
2) Find the key link Which maintains a list of most recently used files in windows



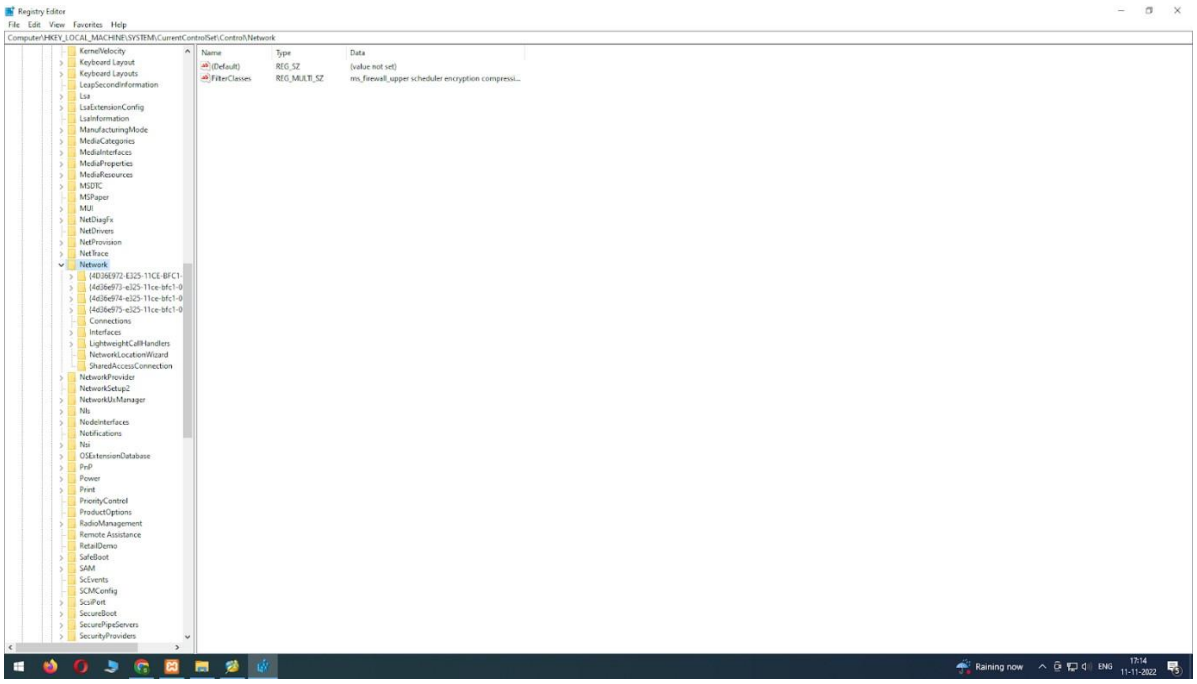
3) Investigate the key link which maintains windows virtual memory



4. Find the key link to display the list of installed program in the computer

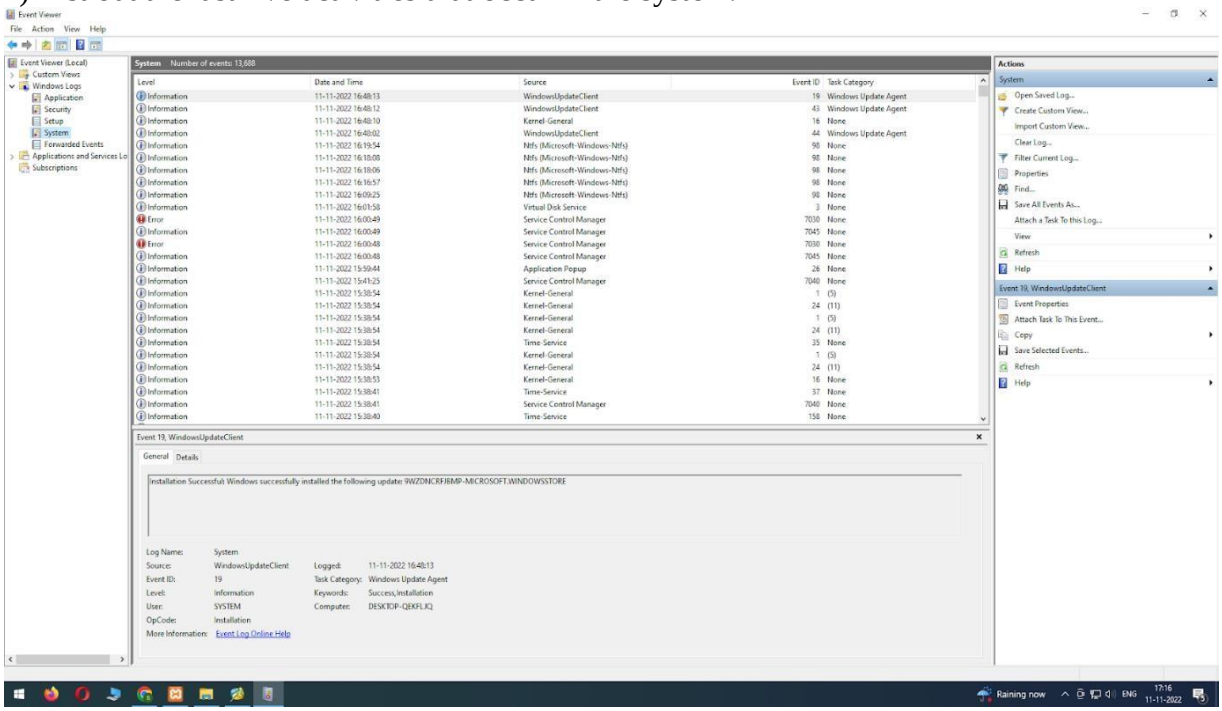


5. Find the key used for detecting the networking evidence from registry

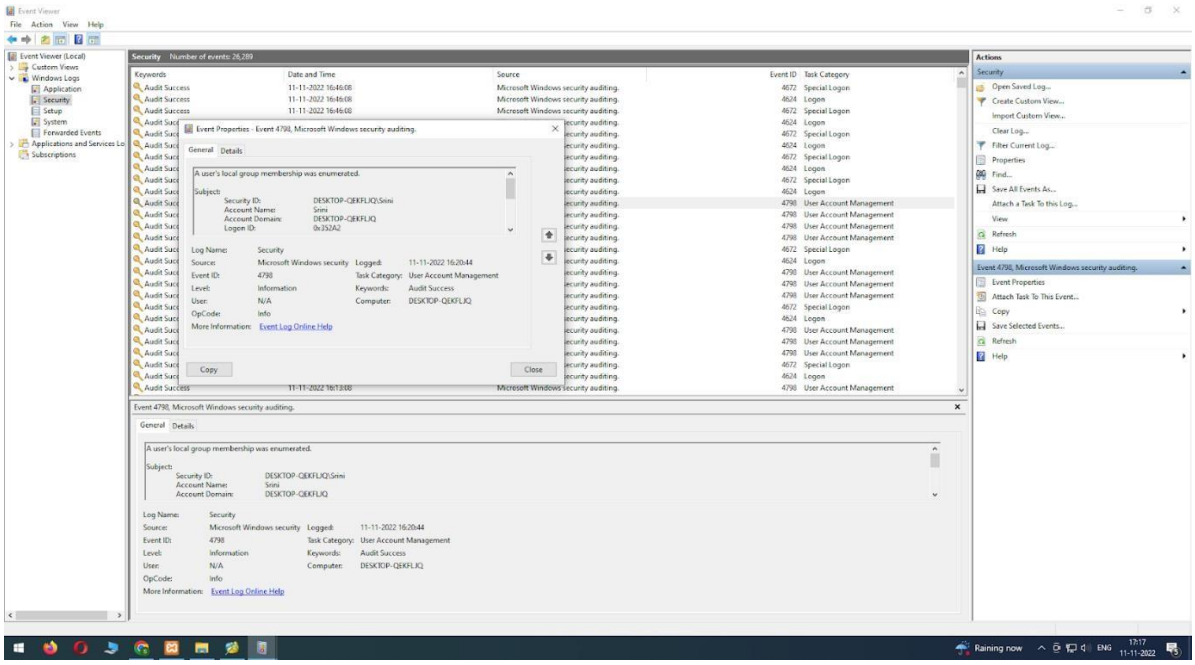


EVENT VIEWER ANALYSIS:

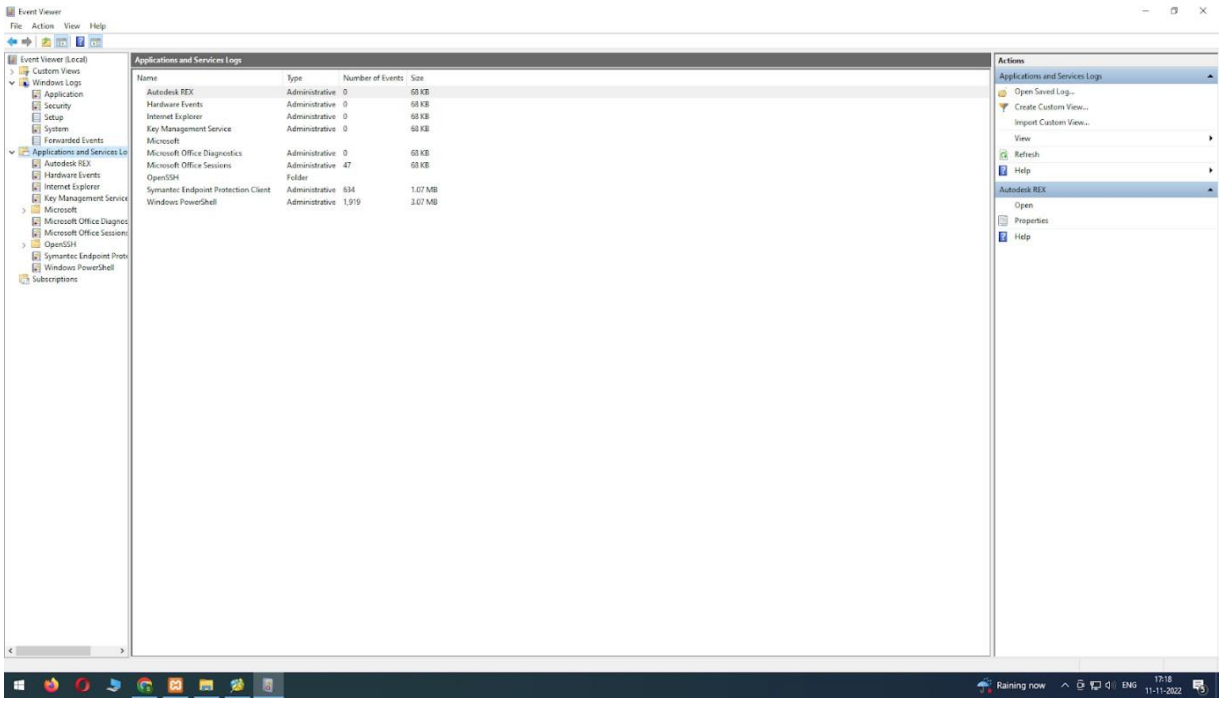
1) List out the last five activities that occur in the system.



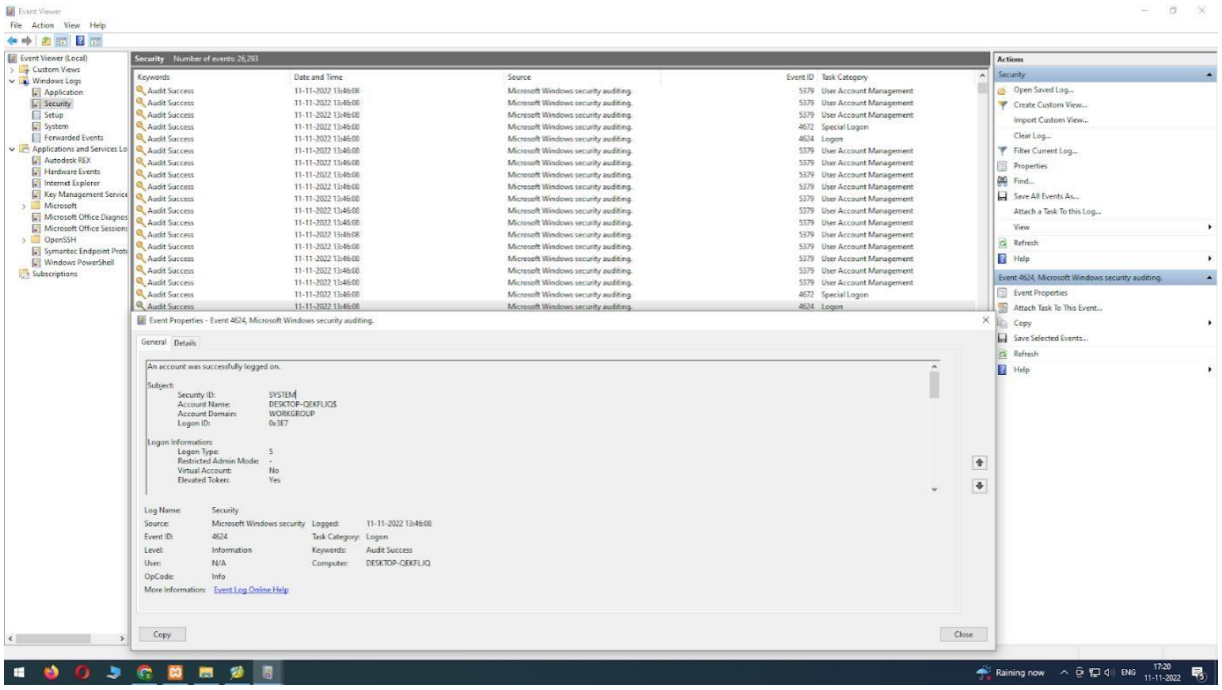
2)List out the last event of system login and logout.



3)Investigate the application and service log



4) Audit success and failures



RESULT:

Thus, the given windows forensics task is completed successfully using registry editor and event viewer.

Ex.No: 4
DATE: **DATA ACQUISITION – FTK TOOL**

AIM:

To perform data acquisition using FTK toolkit.

REQUIREMENTS:

FTK tool

TOOL DESCRIPTION:

FTK or forensic toolkit is a computer forensics software made by Accessdata. It scans a hard drive looking for various information. For example, potentially locate deleted emails and scan a disk for text strings to use them as a password dictionary to crack encryption.

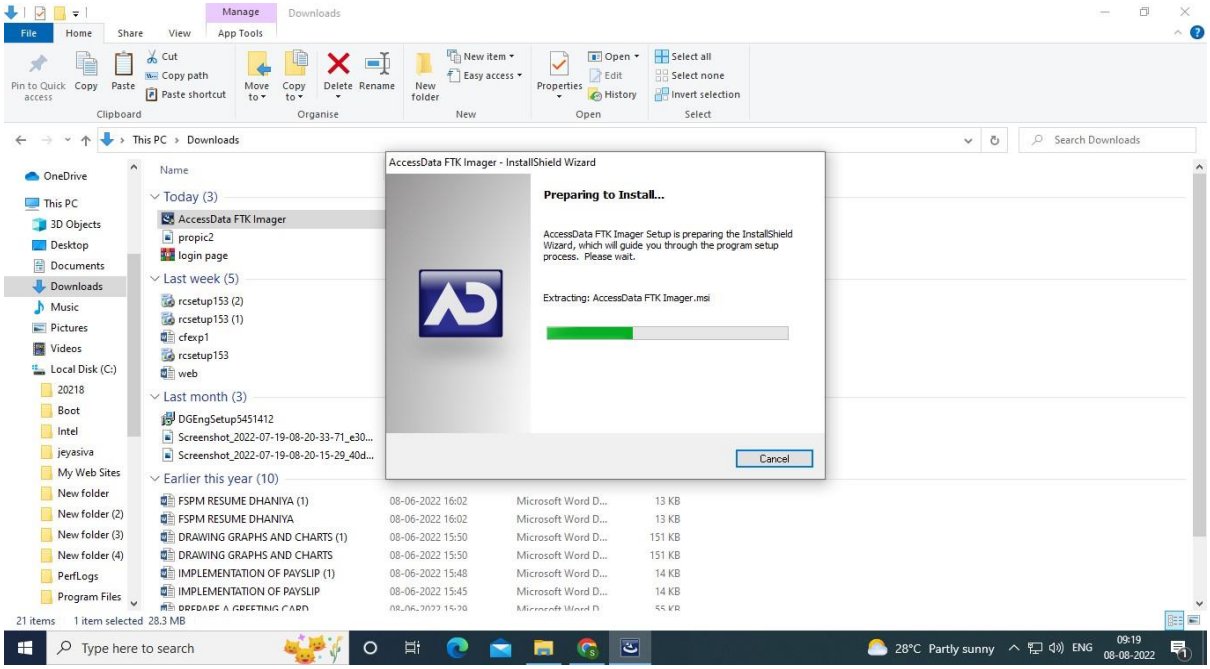
TASKS:

- Perform data acquisitions of physical drive.
- Perform data acquisitions of logical drive.
- Perform data acquisitions of image files.

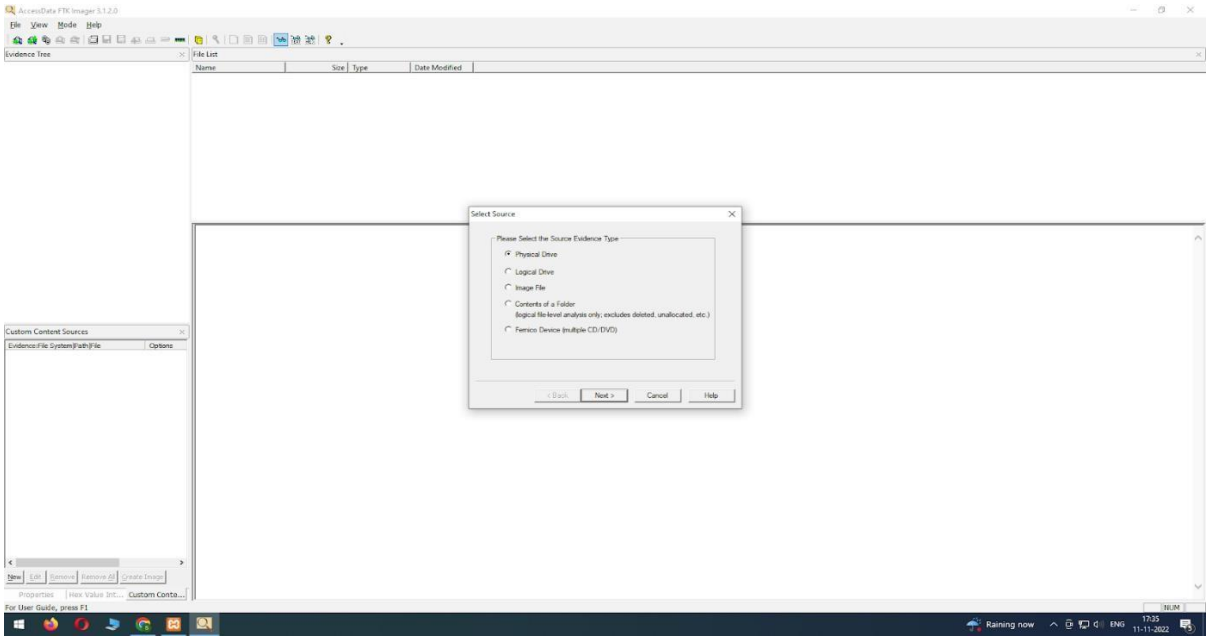
PROCEDURE:

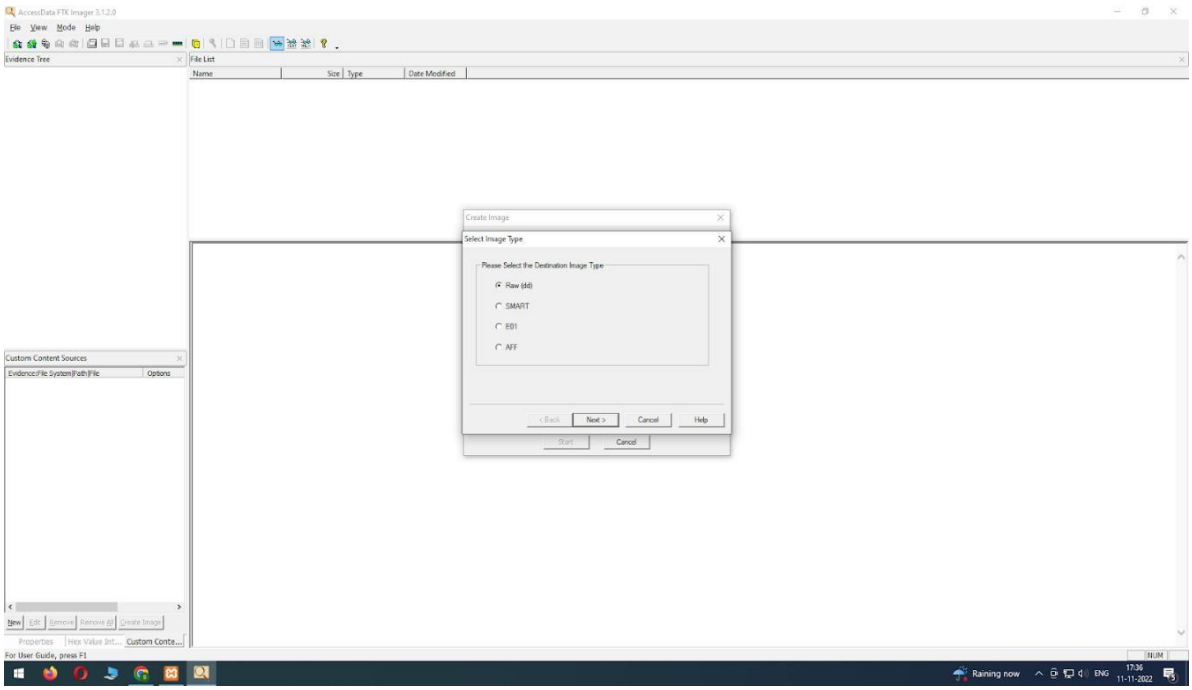
- Download and install FTK imager tool.
- Open file.
- Click create disk image.
- Now select the source that we need to acquire.
- Select the physical drive option.
- Select the drive you need to acquire and click finish.
- Now add a destination folder to store the data acquired.
- Now enter the case details.
- Add an image destination.
- Click start to start acquiring.
- Now select logical drive option and repeat the same process to acquire the data in the logical drive.
- Next select the image file inserted and repeat the same process to acquire it.

OUTPUT:

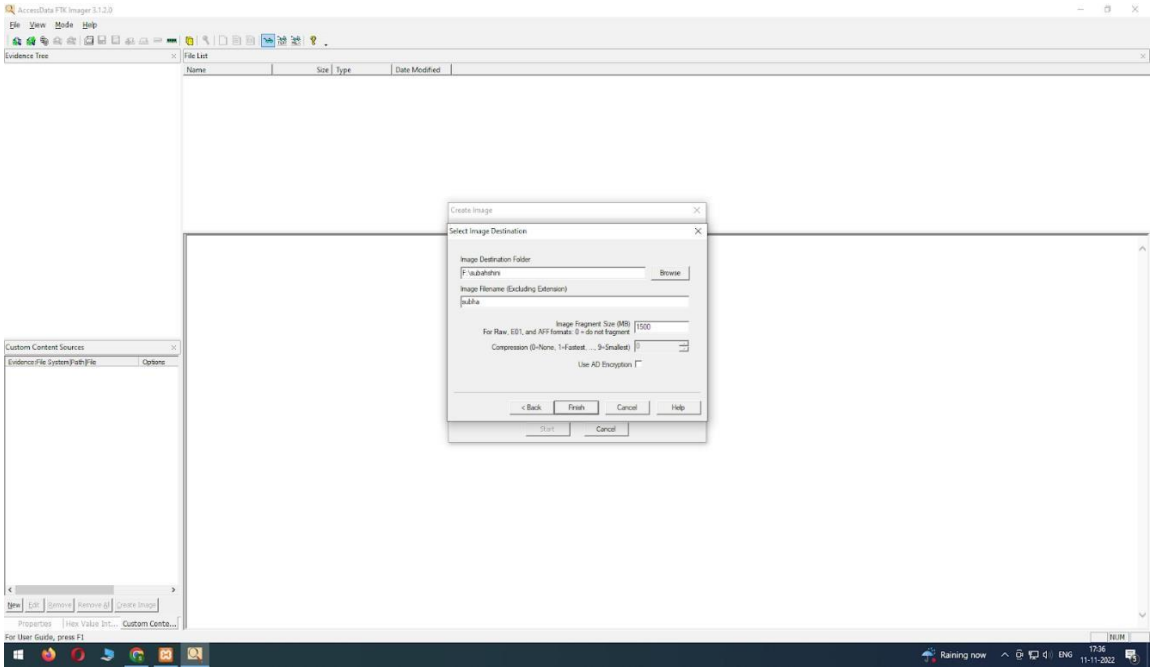


1)Perform Data acquisition of Physical drive where the suspected evidence reside.
Selecting physical drive.

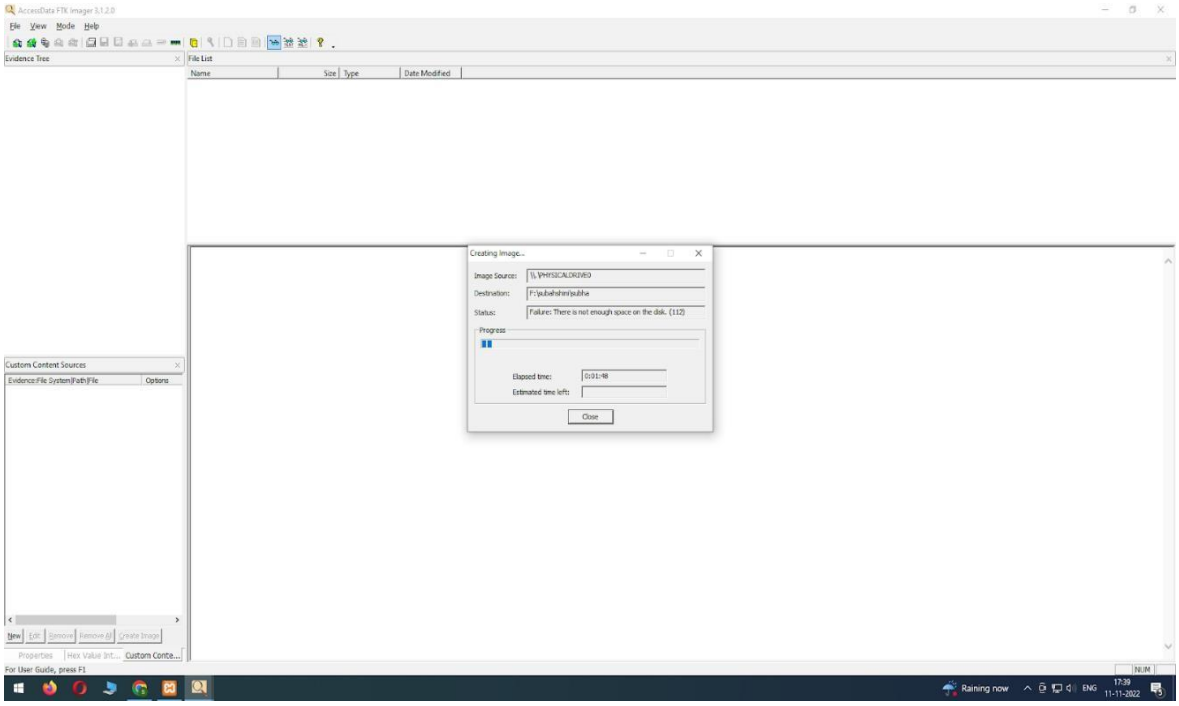




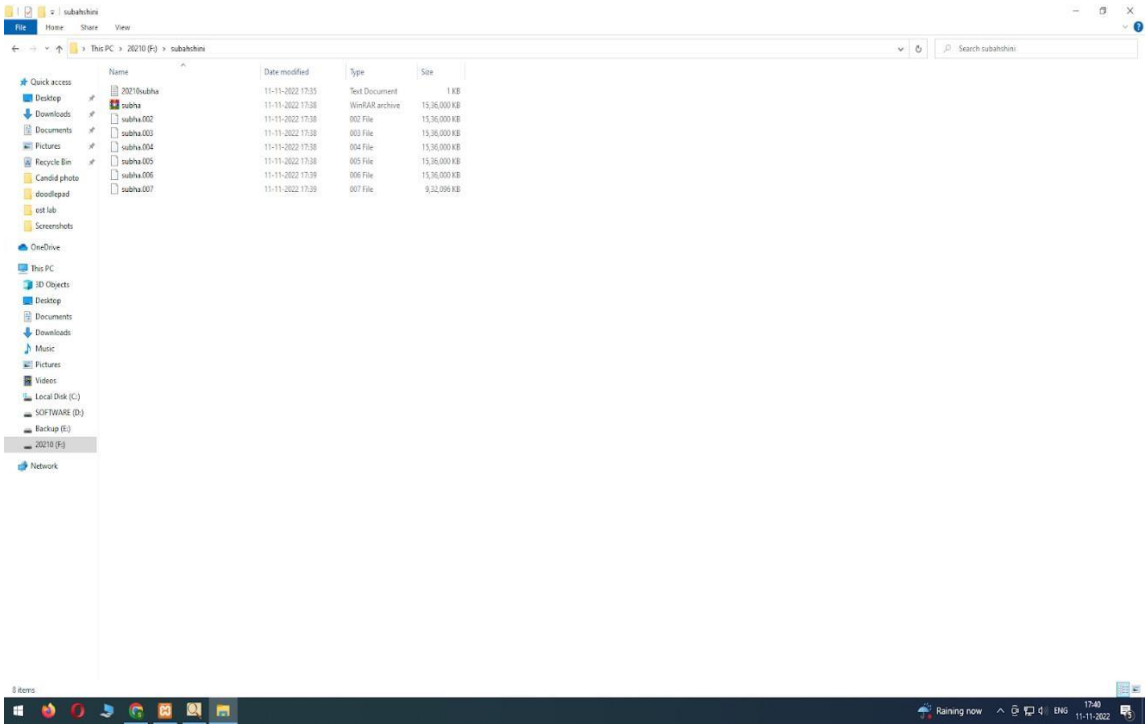
Selecting image destination folder

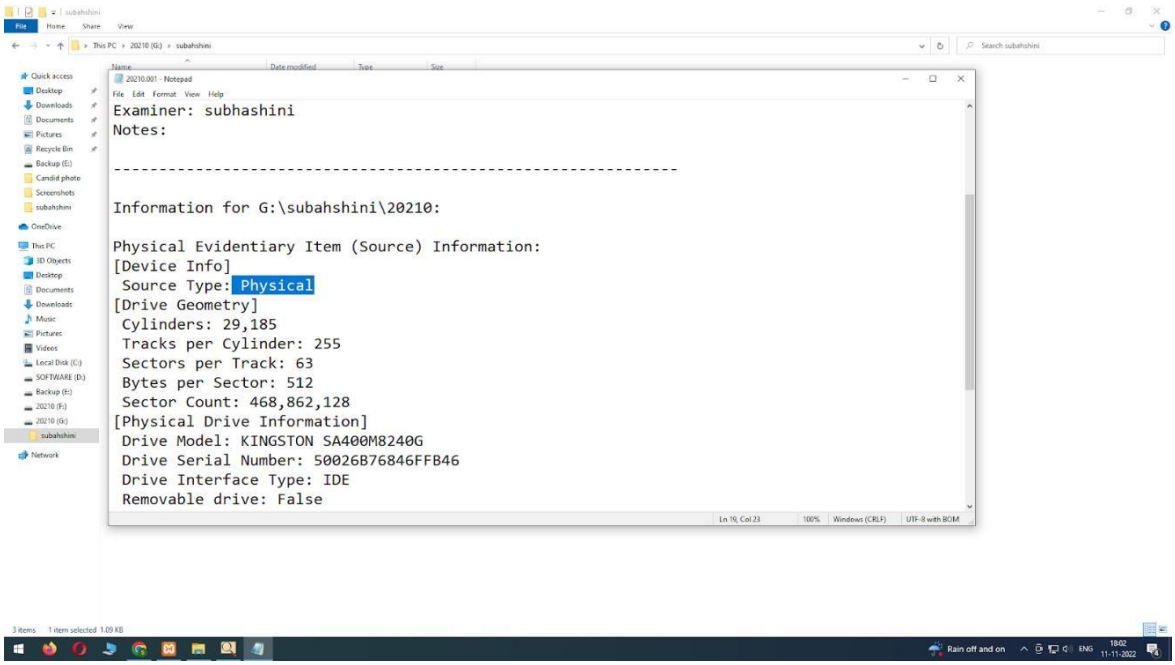


Creating images



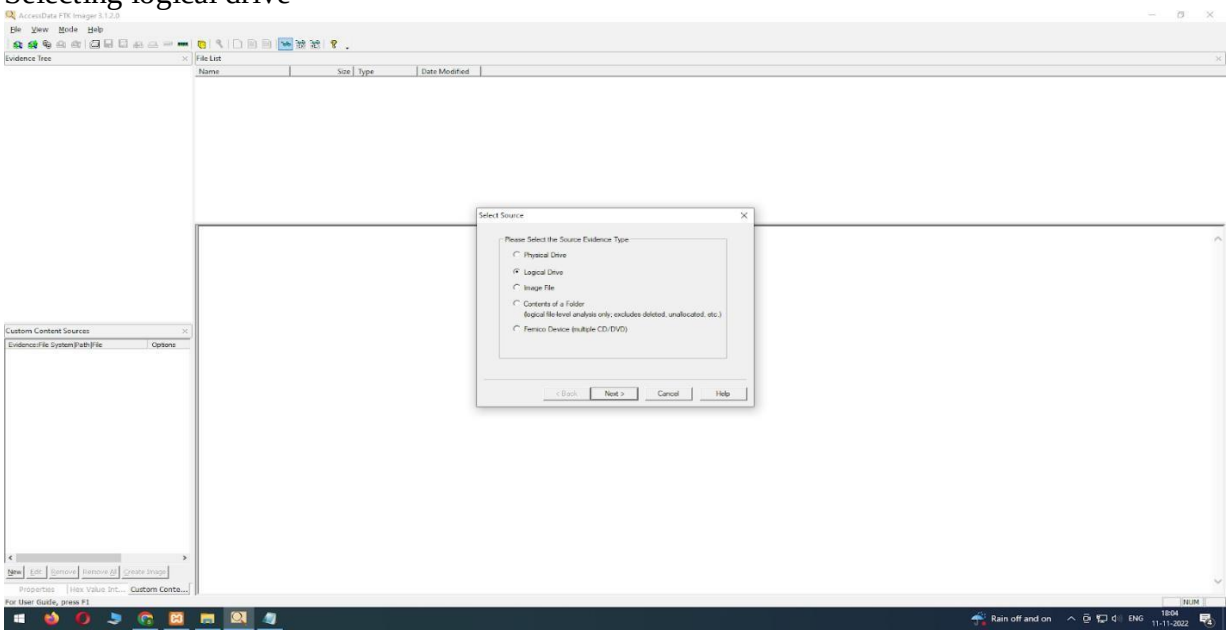
Acquired images:

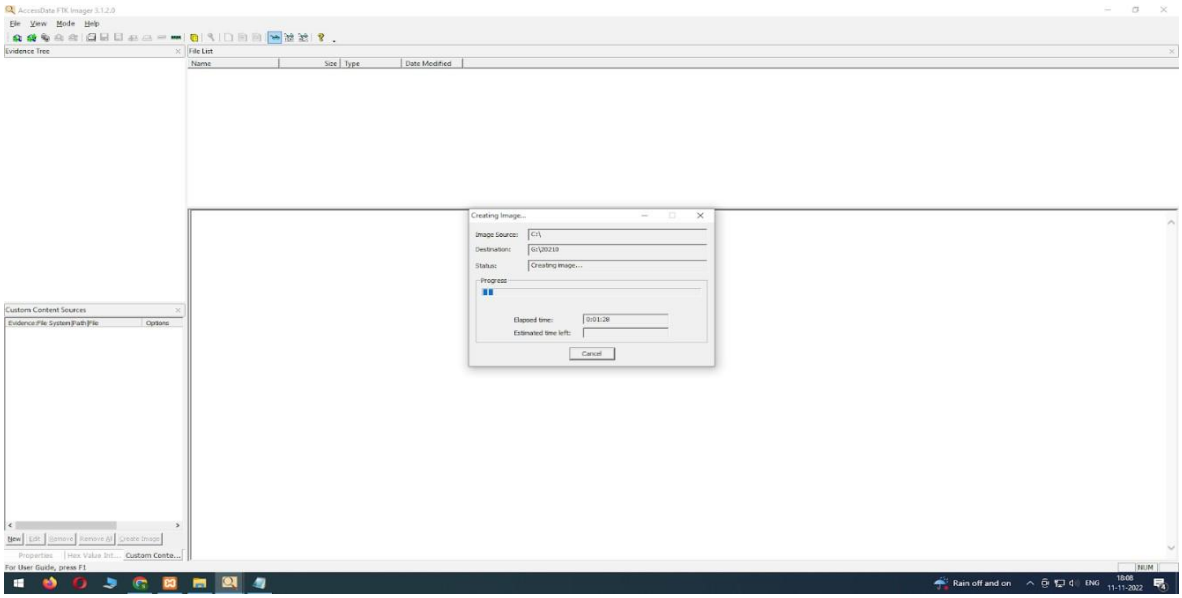


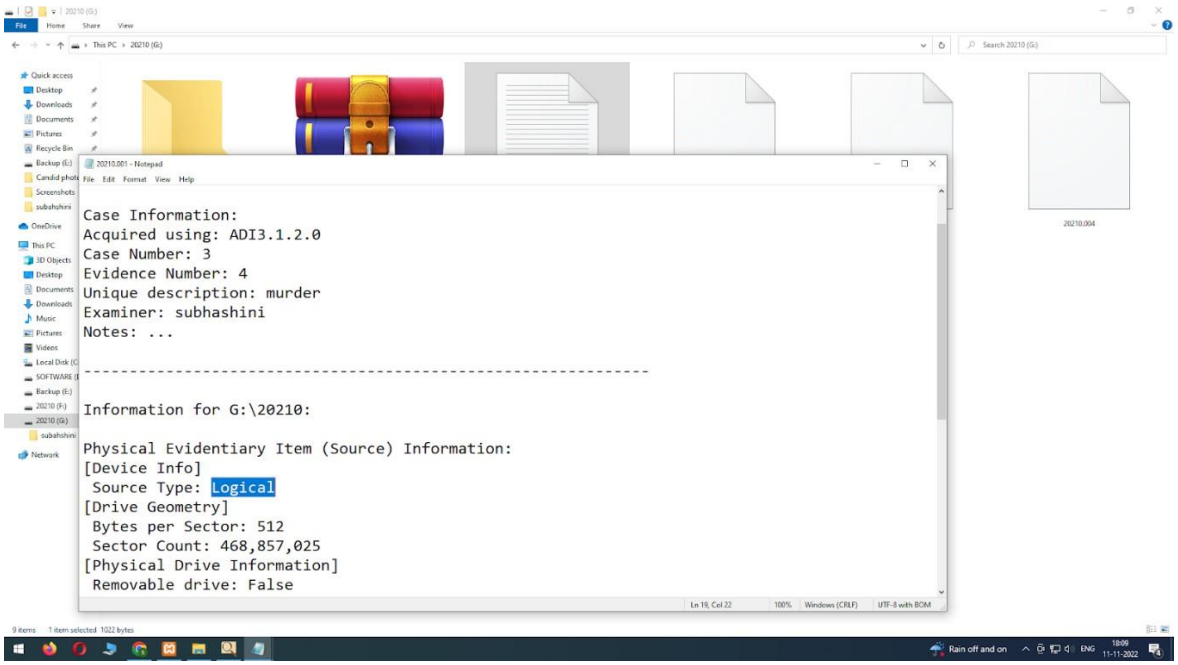


2.Perform data acquisition of logical drive and also check the integrity of the file using hashes.

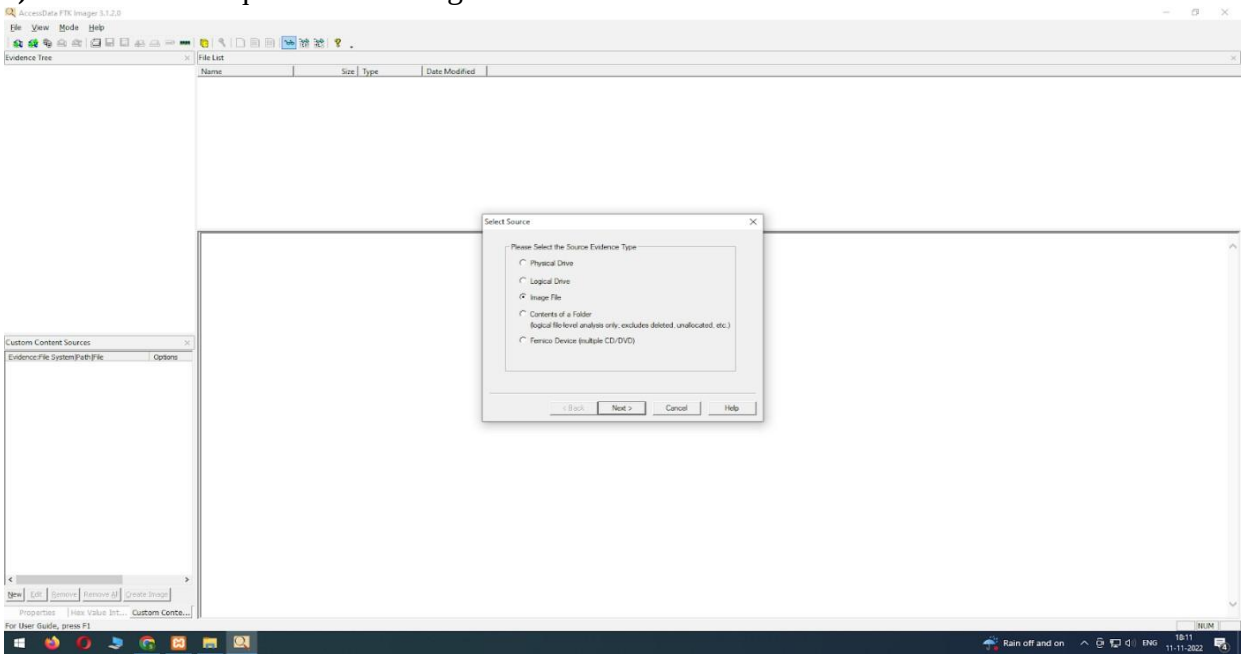
Selecting logical drive

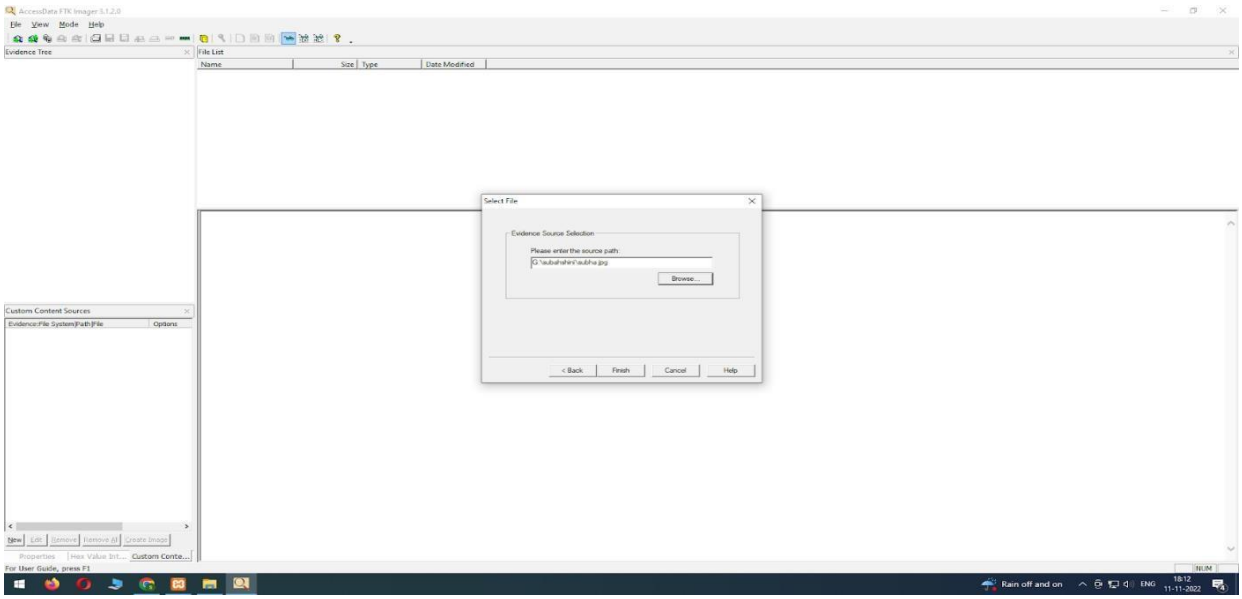





20210.001 - Notepad
File Edit Format View Help
Case Information:
Acquired using: ADI3.1.2.0
Case Number: 3
Evidence Number: 4
Unique description: murder
Examiner: subhashini
Notes: ...
Information for G:\20210:
Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Logical
[Drive Geometry]
Bytes per Sector: 512
Sector Count: 468,857,025
[Physical Drive Information]
Removable drive: False

3) Perform data acquisitions of image files.





RESULT:

Thus, Acquisition of the data has been done successfully by using the FTK tool.

Ex.No: 5
DATE: **RAM CAPTURING – FTK TOOLKIT**

AIM:

To perform RAM capturing using Access Data FTK imager.

REQUIREMENTS:

Access Data FTK imager

TOOL DESCRIPTION:

Forensic Toolkit, or FTK, is a computer forensics software made by AccessData. It scans a hard drive looking for various information.[1] It can, for example, potentially locate deleted emails[2] and scan a disk for text strings to use them as a password dictionary to crack encryption.[3]FTK is also associated with a standalone disk imaging program called FTK Imager. This tool saves an image of a hard disk in one file or in segments that may be later on reconstructed. It calculates MD5 and SHA1hash values and can verify the integrity of the data imaged is consistent with the created forensic image. The forensic image can be saved in several formats, including DD/raw, E01, and AD1

RAM CAPTURING:

Capturing volatile data in a computer's memory dump enables investigators and examiners to do a full memory analysis and access data including: browsing history. encryption keys.

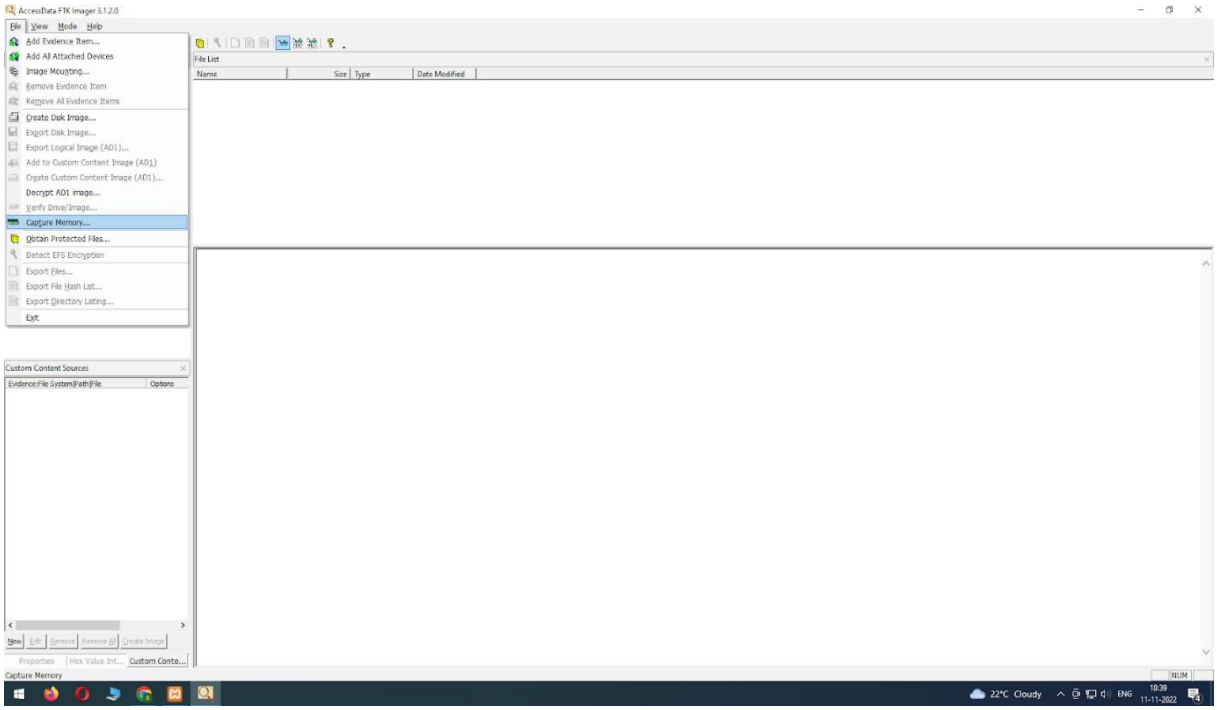
TASK:

1) Perform RAM capturing using Access Data FTK imager

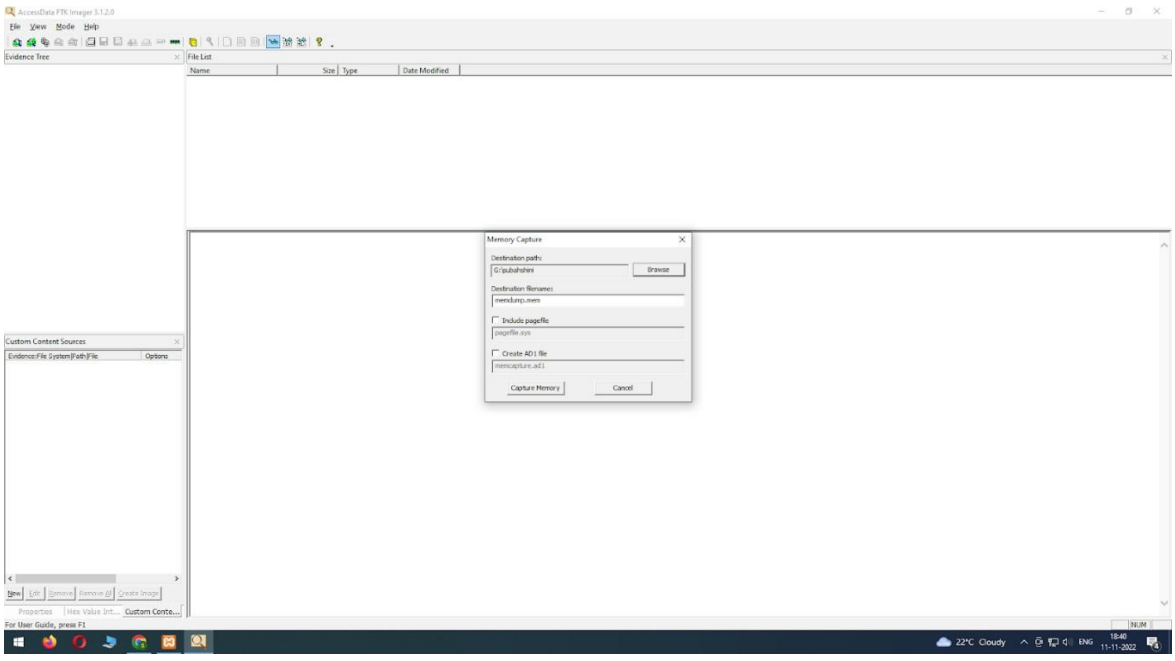
PROCEDURE:

- Download and install Access data FTR imager.
- Open the tool
- Go to file click capture ram option from the pop-up menu.
- Then a wizard will be opened. Then enter the destination place where the captured data should be stored.
- Then click ok.
- RAM will be captured in the destination place.

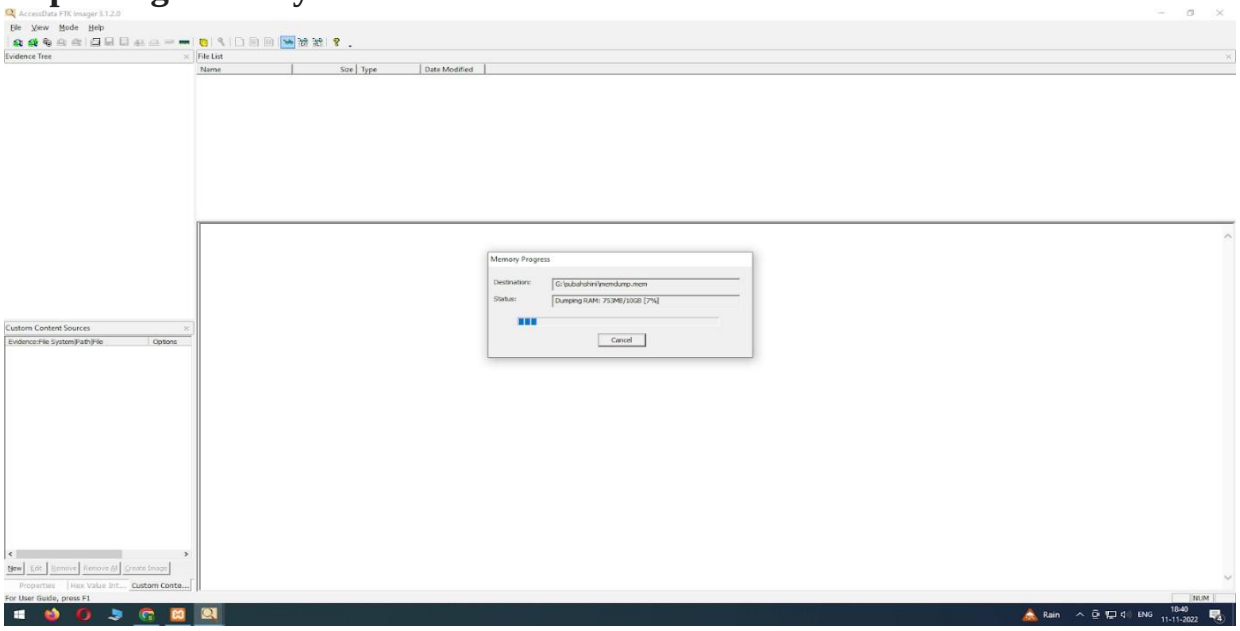
OUTPUT:



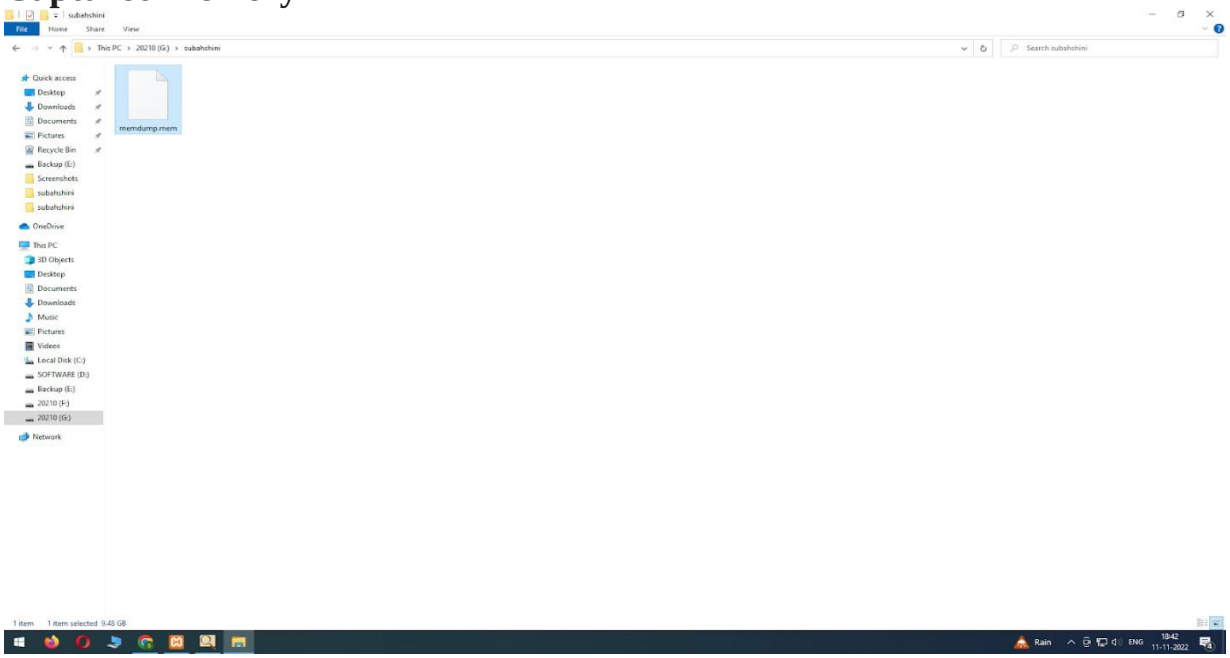
Destination folder



Capturing memory



Captured memory



RESULT:

Thus the RAM capturing using Access Data FTK imager

Ex.No: 6

DATE: **CREATING ENCRYPTED PARTITION AND DECRYPT – VERACRYPT TOOL**

AIM:

To create an encrypted partition, detecting encrypted partition in the suspected system.

REQUIREMENTS:

- Veracrypt tool
- Magnet encryption tool.

ENCRYPTING PARTITITONS:

- Encrypting files,. Partitions and entire disks prevents unauthorized access to your data and protects your confidential files and documents.

TOOL DESCRIPTION:

- Veracrypt is a free and open source utility or on-the-fly-encryption. The software can create a virtual encrypted disk that works just like a regular disk but within a file. It can also encrypt a partition or the entire storage device with pre-boot authentication.

TASKS:

- Creating encrypting partition (veracrypt tool)
- Decrypt the detected files/folders using magnet encryption disk detector.

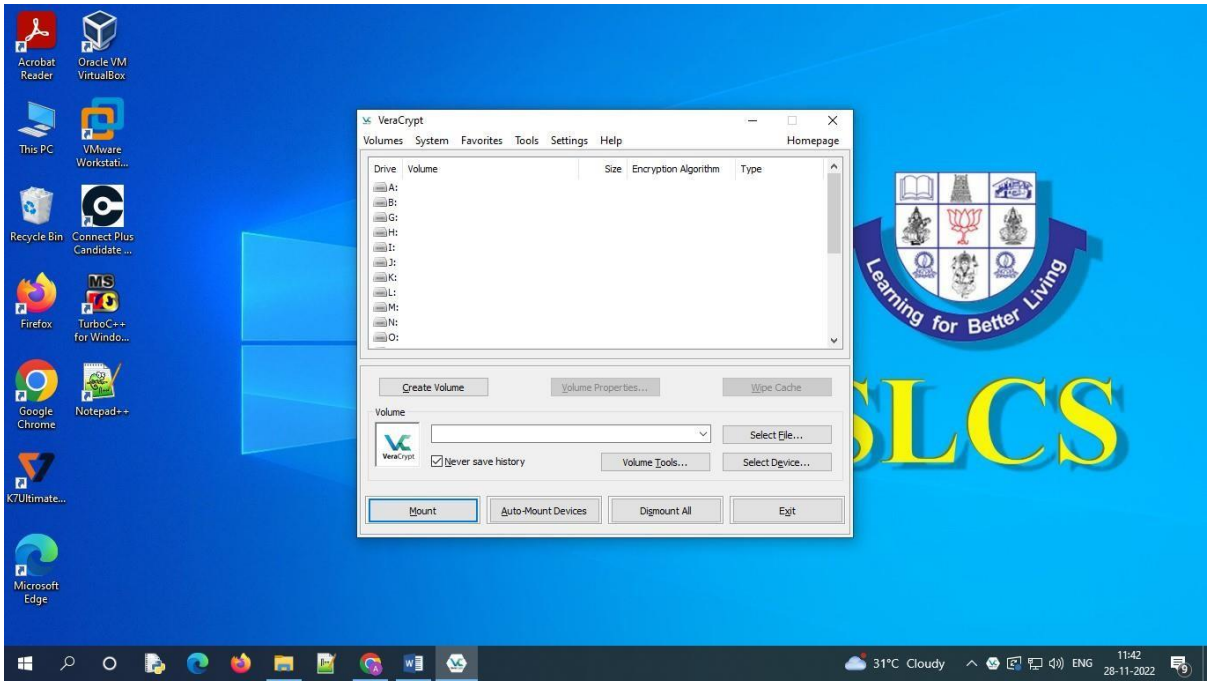
PROCEDURE:

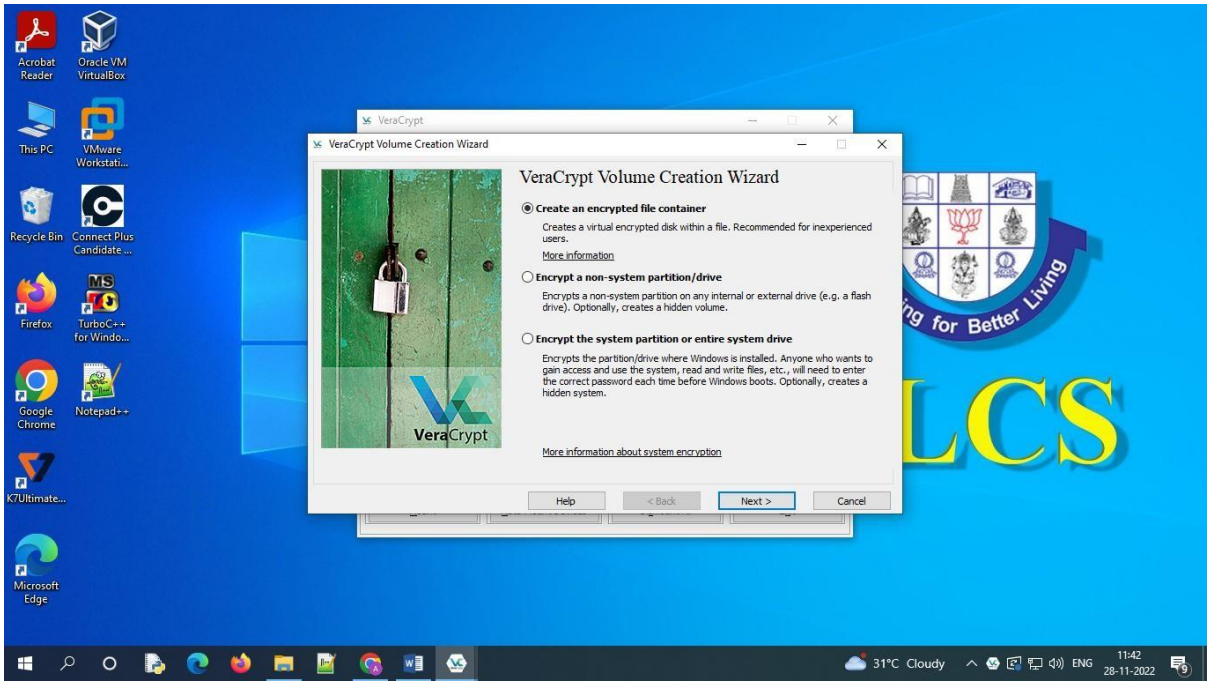
1) Creating encrypted partition
    - Download and install veracrypt tool.
    - Click create volume.
    - Create an encrypted file container.
    - Click the standard veracrypt volume.
    - Create an estimation folder and save by any name and click next.
    - Select encryption operation.
    - Provide the volume size as your wish.
    - Then create a password and click next.
    - Move the mouse point here and there to increase the strength of the key till it gets completed.
    - Click formats.
    - Check the destination folder and select file.
    - And select the disk and disk mount.
    - Copy one/two file to be encrypted.
    - And select dismount after copying to encrypt the volume.

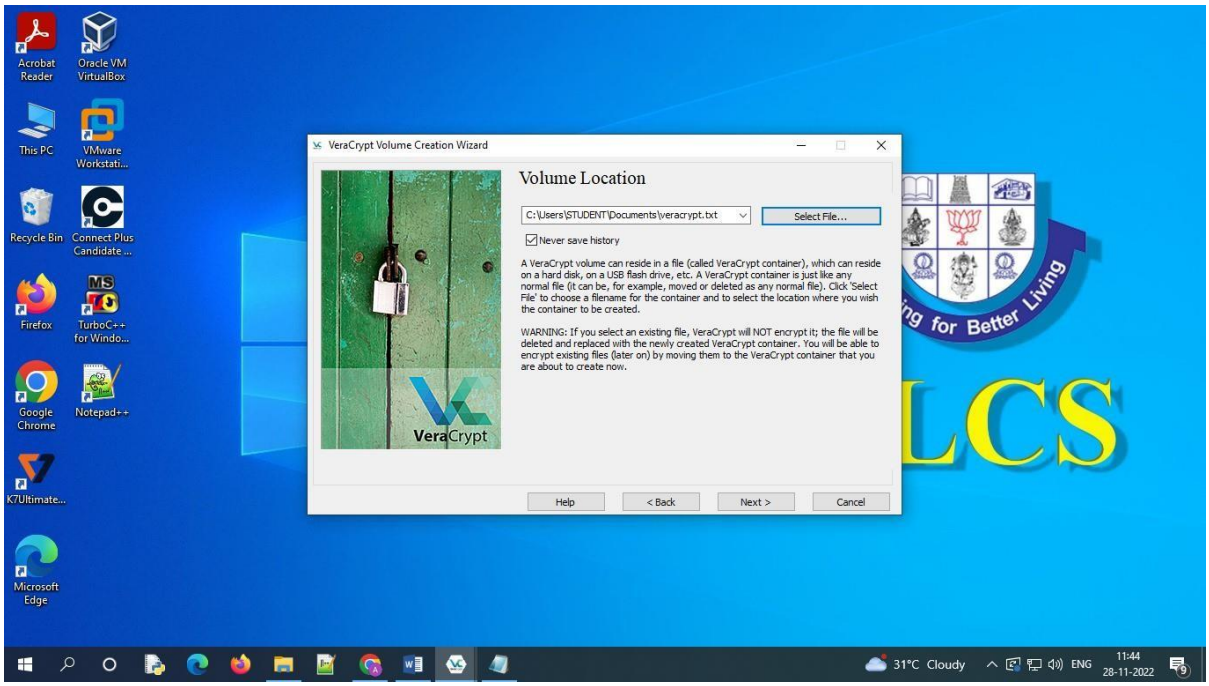
2) Select magnet encryption disk detector.
    - It detects the encrypted partitions were created and detected successfully.

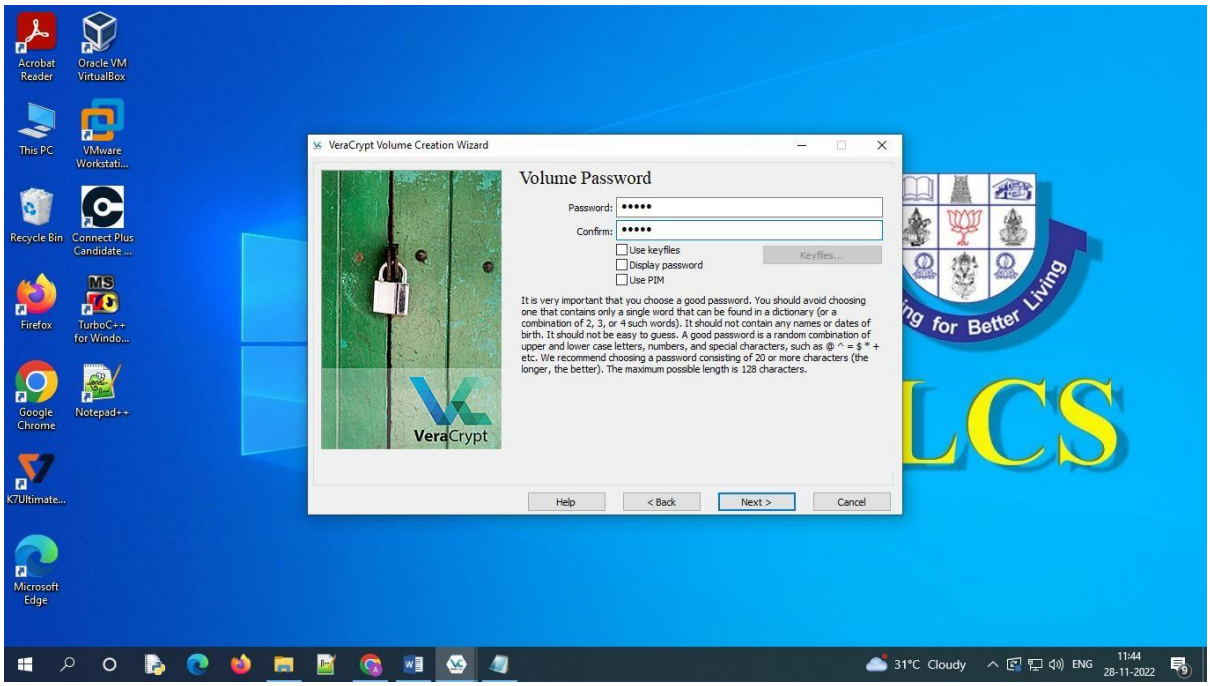
OUTPUT:

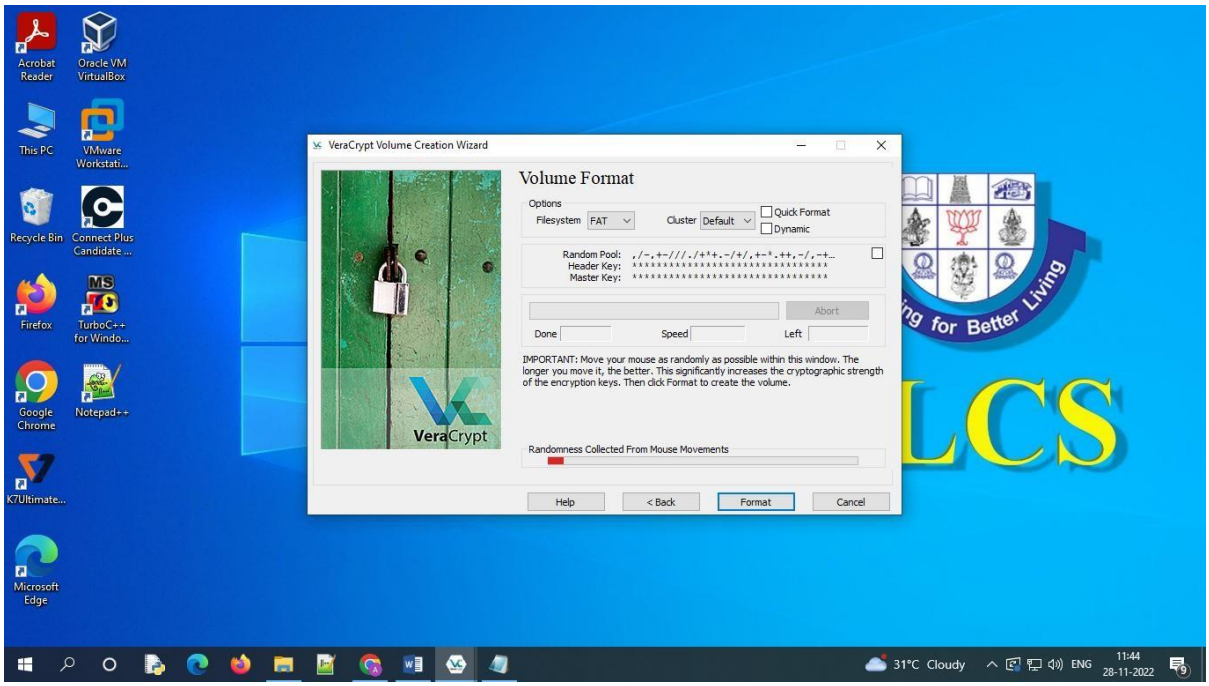
i) Creating encrypted partition



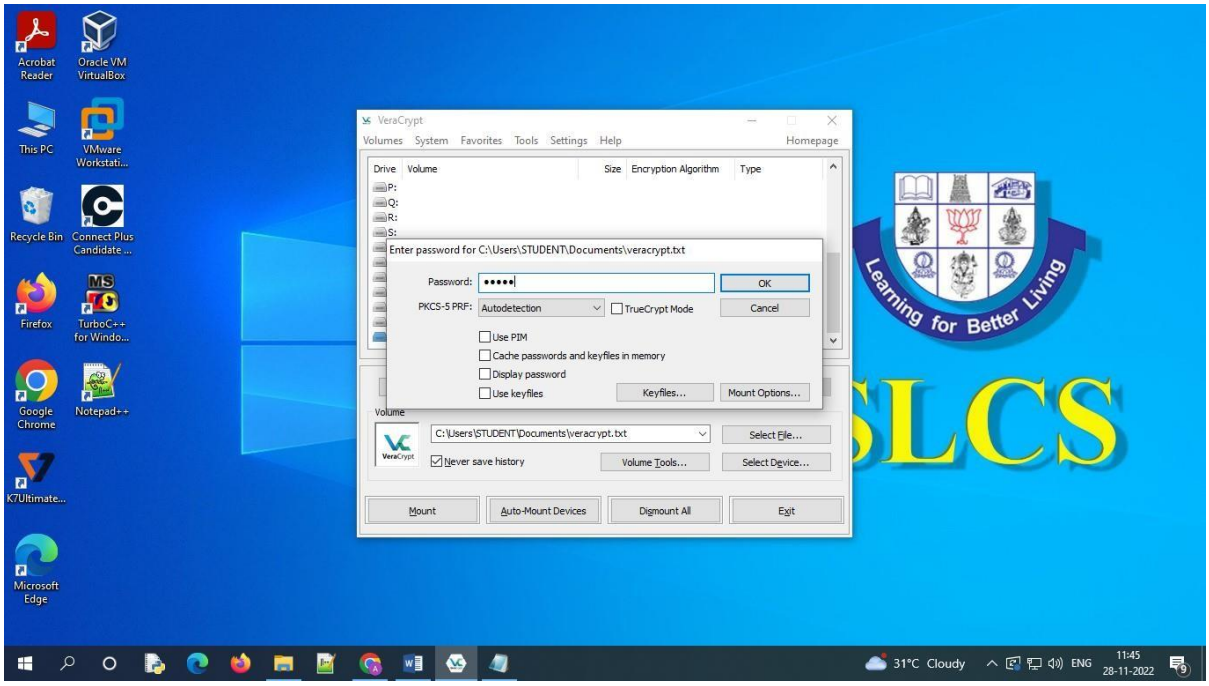


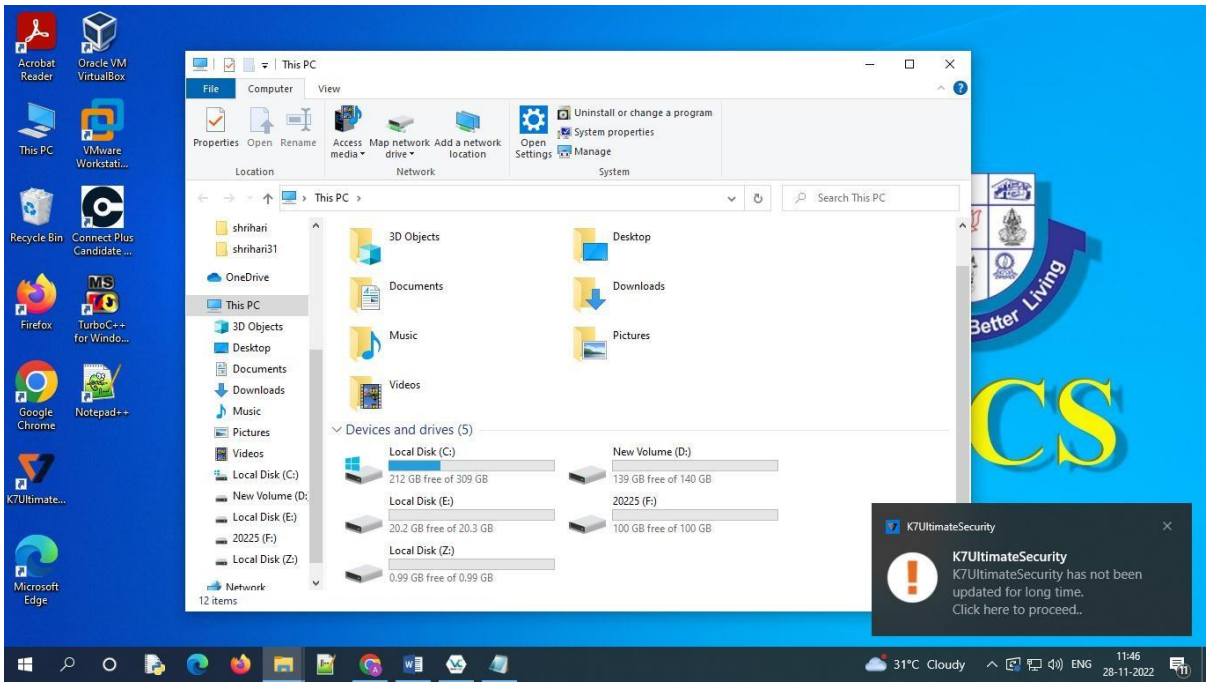

VeraCrypt Volume Creation Wizard
Volume Location
C:\Users\STUDENT\Documents\veracrypt.txt
Select File...
Never save history
A VeraCrypt volume can reside in a file (called VeraCrypt container), which can reside on a hard disk, on a USB flash drive, etc. A VeraCrypt container is just like any normal file (it can be, for example, moved or deleted as any normal file). Click 'Select File' to choose a filename for the container and to select the location where you wish the container to be created.
WARNING: If you select an existing file, VeraCrypt will NOT encrypt it; the file will be deleted and replaced with the newly created VeraCrypt container. You will be able to encrypt existing files (later on) by moving them to the VeraCrypt container that you are about to create now.
VeraCrypt
Help
< Back
Next >
Cancel
Acrobat Reader
Oracle VM VirtualBox
This PC
VMware Workstati...
Recycle Bin
Connect Plus Candidate ...
Firefox
TurboC++ for Windo...
Google Chrome
Notepad++
K7Ultimate...
Microsoft Edge
31°C Cloudy
ENG
11:44
28-11-2022

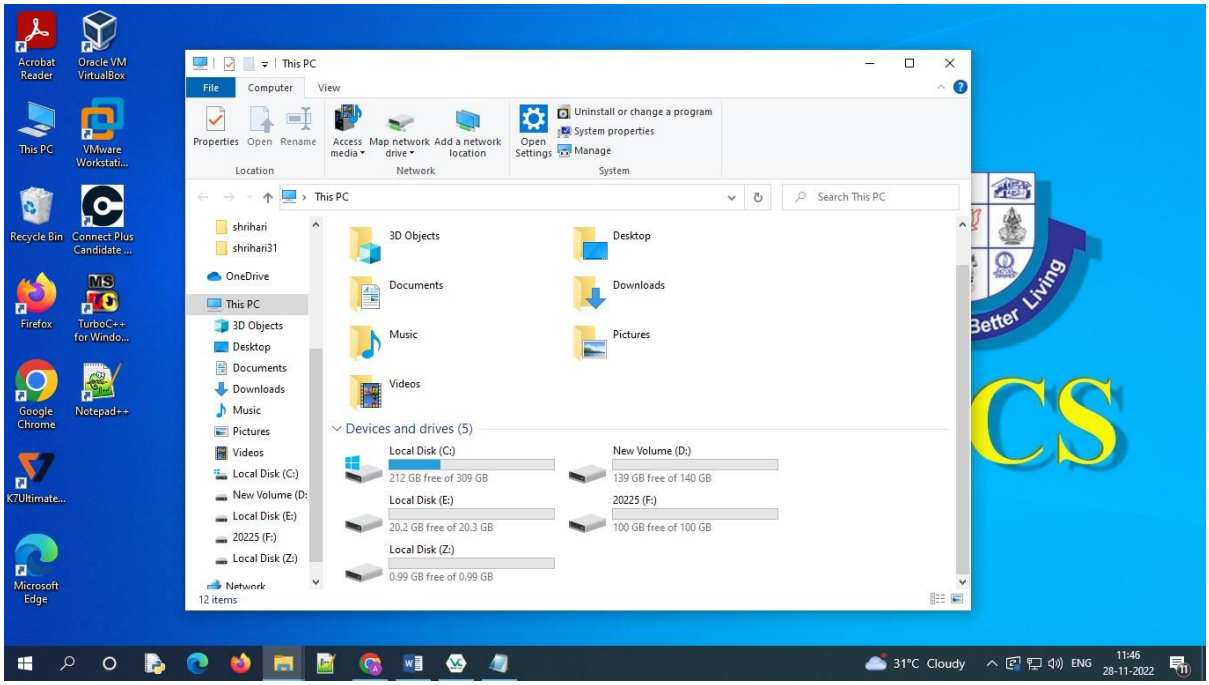

VeraCrypt Volume Creation Wizard
Volume Password
Password:
Confirm:
Use keyfiles
Keyfiles...
Display password
Use PIM
It is very important that you choose a good password. You should avoid choosing one that contains only a single word that can be found in a dictionary (or a combination of 2, 3, or 4 such words). It should not contain any names or dates of birth. It should not be easy to guess. A good password is a random combination of upper and lower case letters, numbers, and special characters, such as @ ^ = $ * + etc. We recommend choosing a password consisting of 20 or more characters (the longer, the better). The maximum possible length is 128 characters.
VeraCrypt
Help
< Back
Next >
Cancel
Acrobat Reader
Oracle VM VirtualBox
This PC
VMware Workstati...
Recycle Bin
Connect Plus Candidate ...
Firefox
TurboC++ for Windo...
Google Chrome
Notepad++
K7Ultimate...
Microsoft Edge
31°C Cloudy
ENG
11:44
28-11-2022



ii) Mounting the encrypted disk







RESULT:

Thus, the encrypted partitions were created and detected successfully.

Ex.No:7

DATE: **WEBSITE PASSWORD CRACKING**

WIRESHARK TOOL

AIM:

To capture the password of a website using wireshark tool.

REQUIREMENTS:

- Wireshark tool

TOOL DESCRIPTION:

- Wireshark is a network protocol analyser, or an app that captures packets from a network connection, such as from your computer to your home office or the internet. It is the most used packet sniffer in the world.

TASKS:

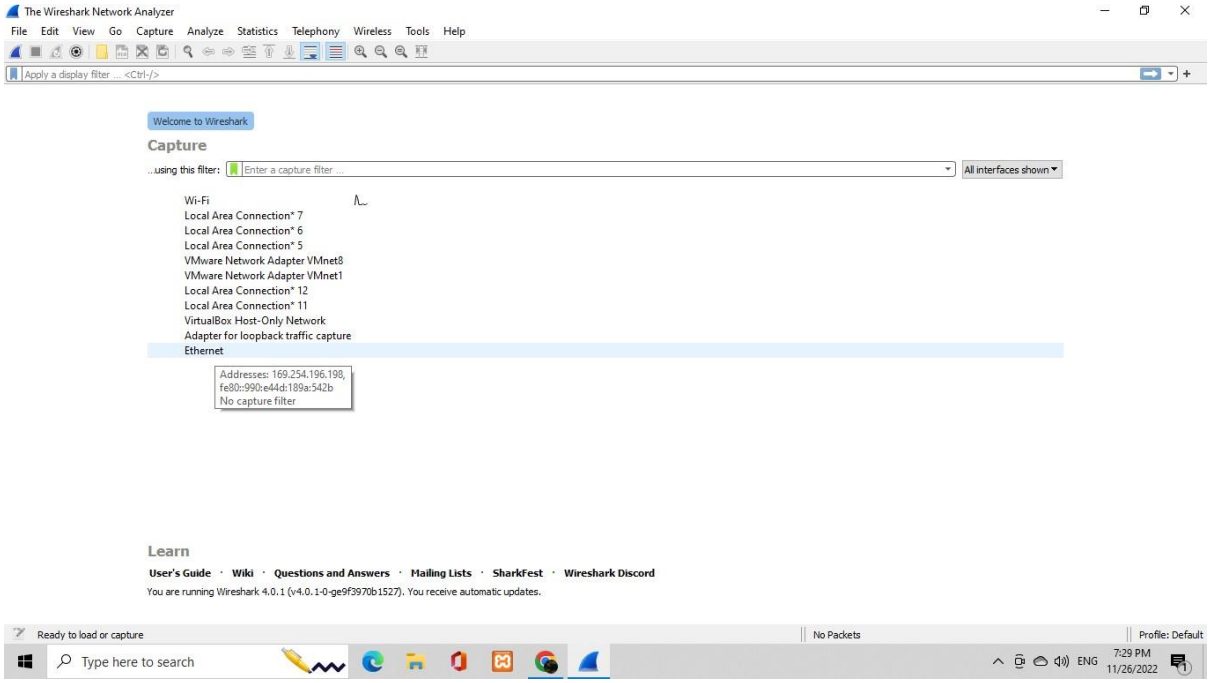
- Use wireshark tool to capture the password from a website.

PROCEDURE:

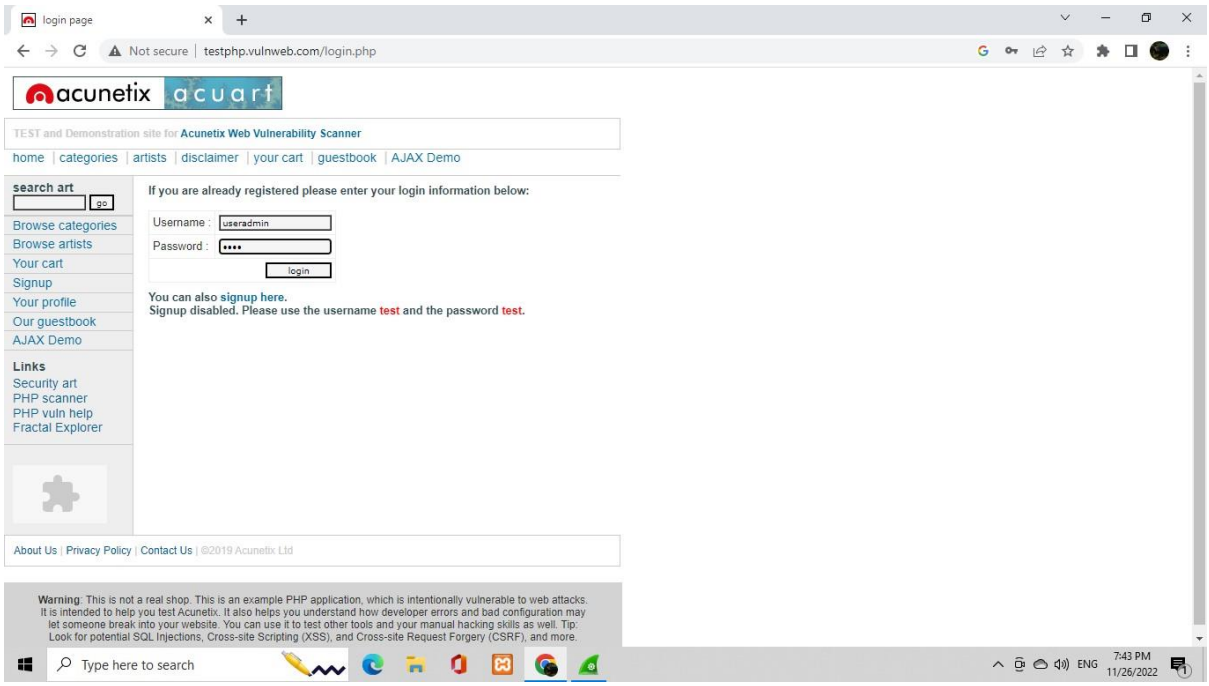
- Download and install wireshark tool for windows.
- Open google chrome and find a HTTP website and enter the login credentials after starting the tool.
- Once the credentials were entered, open the tool and choose HTTP and open in the filter space.
- Now the HTTP packets which are captured will be listed, among those find out the POST packet for capture the credential what you have entered.
- Right click that POST packet.
- Click follow.
- And then click TCP stream.
- Now, we can able to see the credentials what we have entered in that particular website.

OUTPUT:

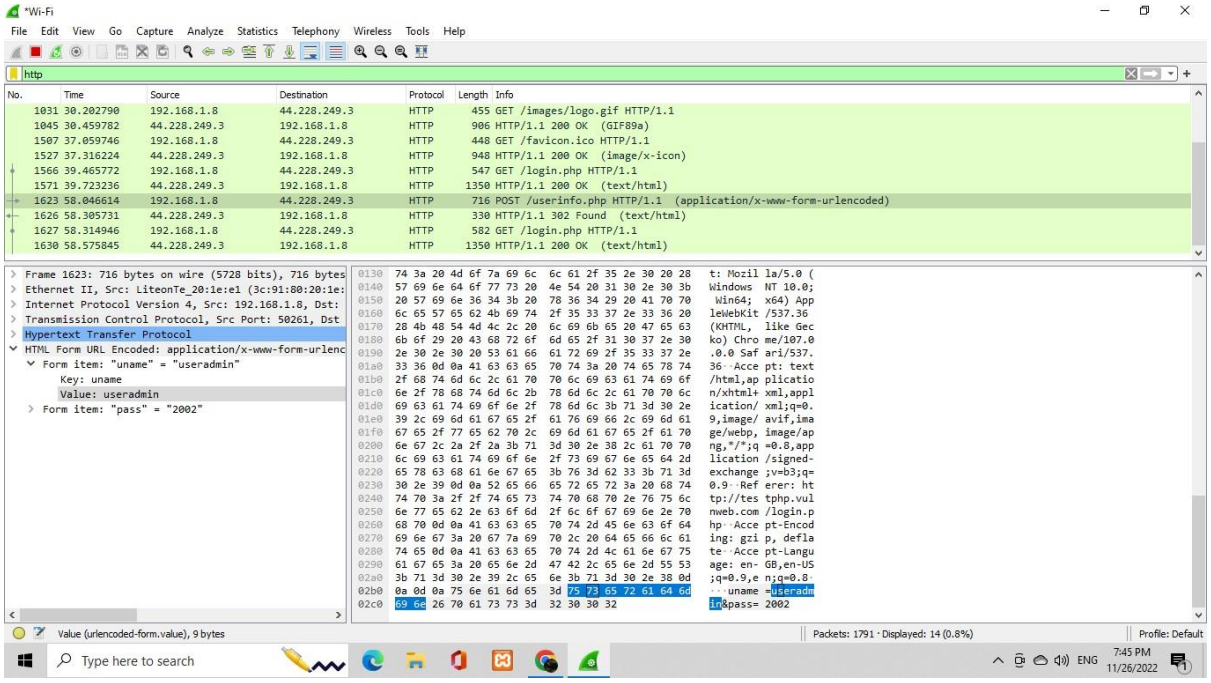
i)selecting ethernet option



ii) login into http website



iii) Viewing the cracked password



RESULT:

Thus, password cracking in a website using wireshark tool was done successfully.

Ex.No: 8

DATE: **STEGANOGRAPHY**

AIM:

To perform steganography using S-tools.

REQUIREMENTS:

- S-Tools.

TOOL DESCRIPTION:

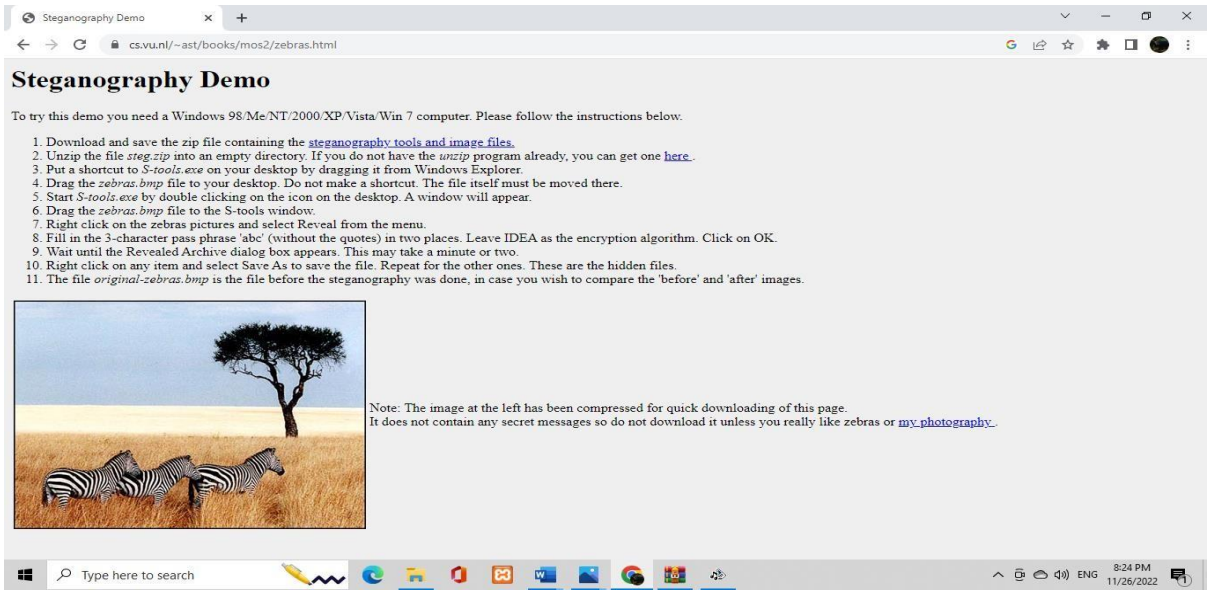
- S-Tools is the process of encrypting text file in to image and process of decrypting image file into text file.

PROCEDURE:

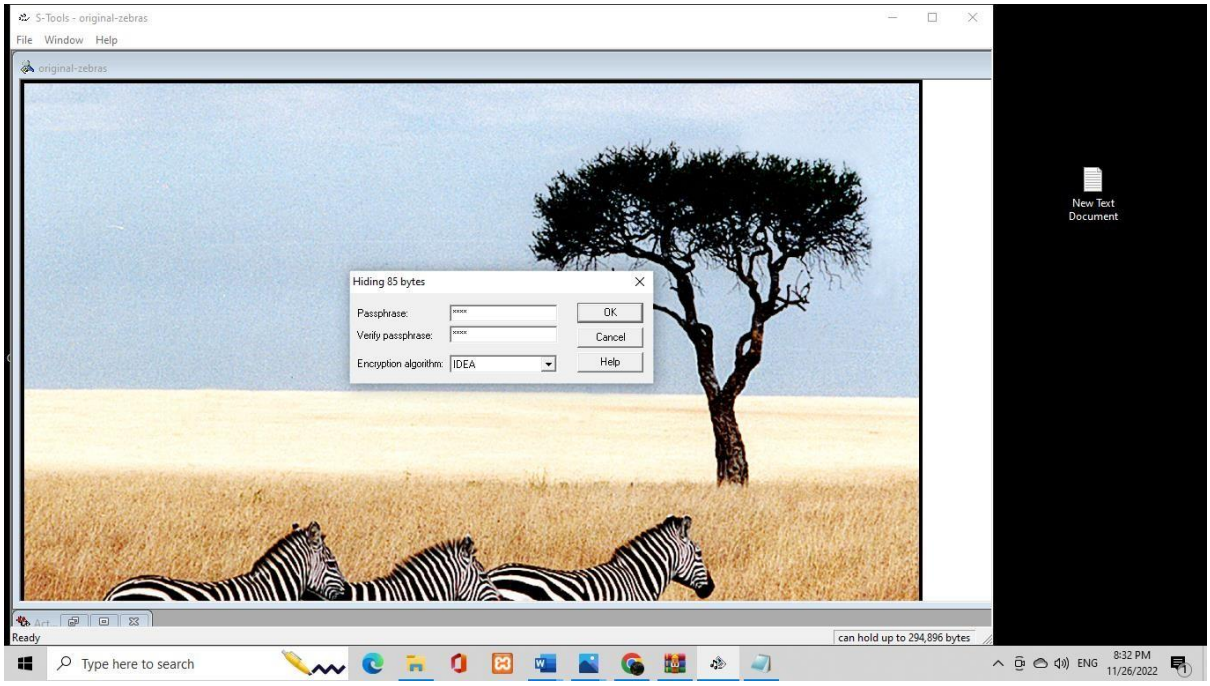
- Open chrome and install S-Tools.
- Contract the zipfile into one folder
- Open S-Tool.
- Drop the image into the S-Tools.
- Create one text document and source in the folder.
- Drop the text file on the image.
- Now, enter passphrase and verify the passphrase.
- Now, it show hidden image data.
- Right click the hidden image and click reveal and save it as text document in that particular folder.
- Click text document you can now able to view the text.

OUTPUT:

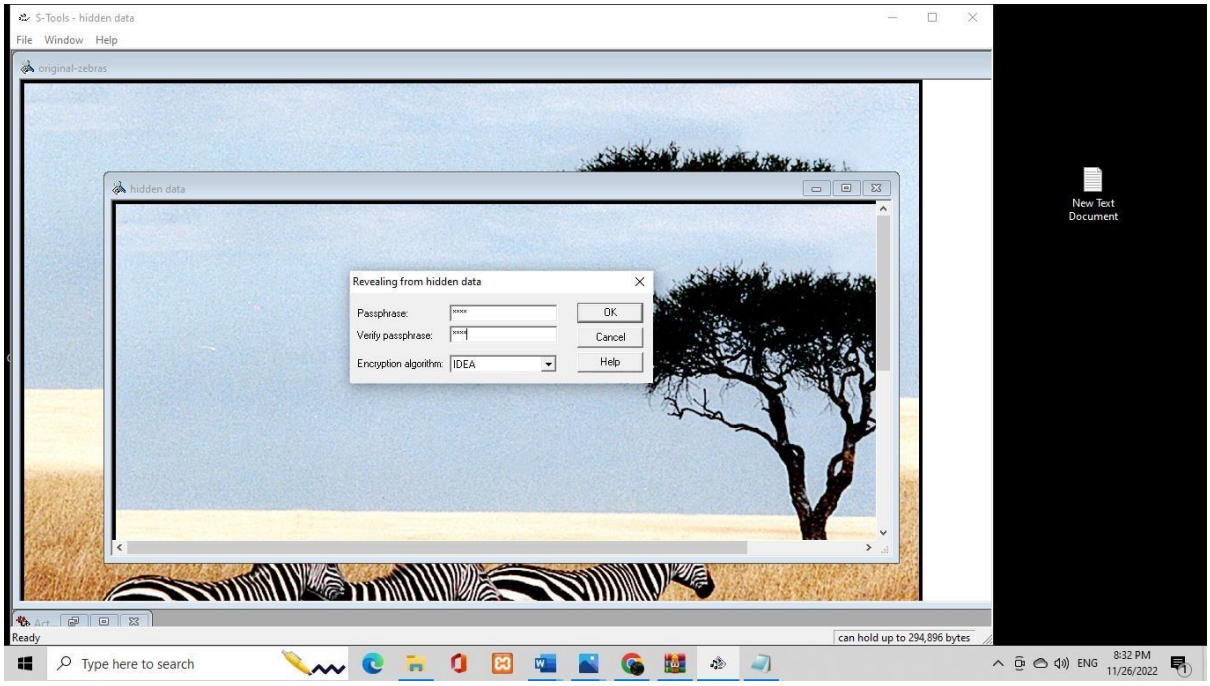
i) Installing S-tool

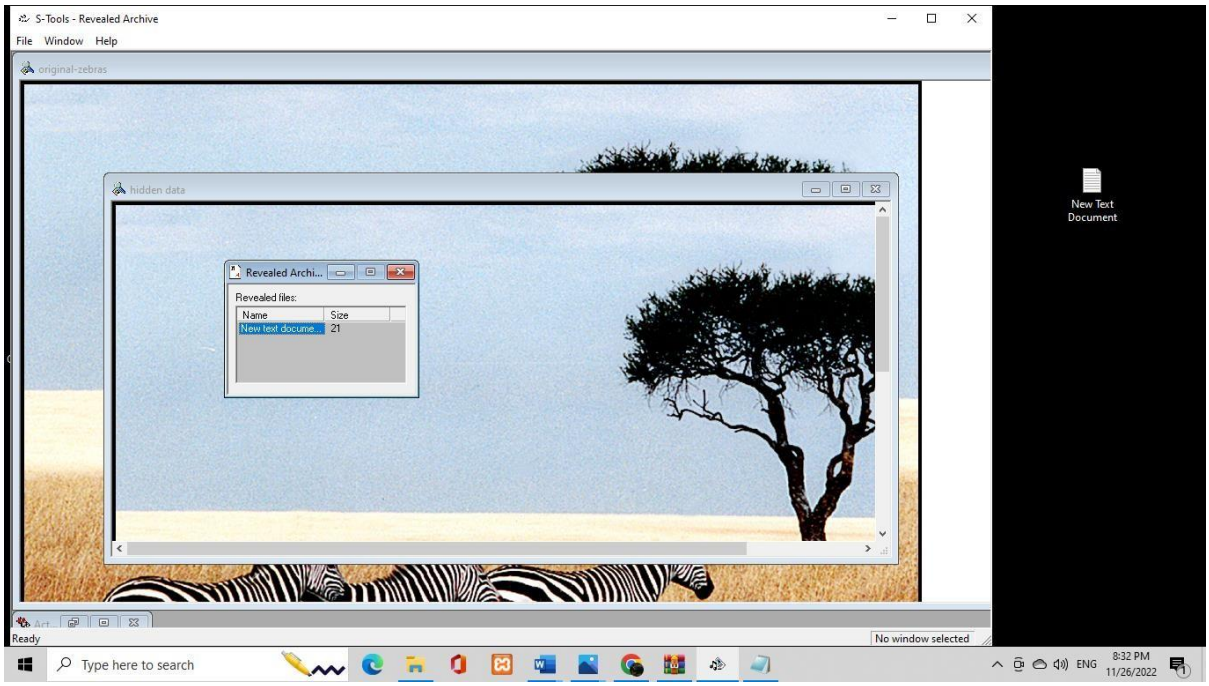


ii) Hiding the secret message



iii) Revealing the secret message





RESULT:

Thus, the steganography using S-Tools has been performed successfully

Ex.No: 9

DATE: FILE ANALYZER USING HEX EDITOR.

AIM:

To perform file analysis using hex editor.

REQUIREMENTS:

- Hex editor tools**.**

FILE ANALYSER:

- File analyser will analyse your file's content, tell you exactly what type of file it is, and suggests software that can open that specific file file

HEX EDITOR:

- A hex editor is a computer program that allows for manipulation of the fundamental binary data that constitutes a computer file.

TASKS:

- To put different types of files in hex editor and analyze the files with the hexa decimal values.
- To put a file into the hex editor for which, the type is unknown. Identify the file type with comparing to file signatures.
- To perform password cracking using wireshark and hex editor.

PROCEDURE:

1) To analyse different types of files.

- Open google chrome install hex editor tool.
- Open the tool.
- Click open file and open the different types of files like excel, word, ppt, text, images.
- Once,the file is opened, the hexa value will be displayed.
- Compare the hexa value of first file and second file.

2) To identify the file type with file signatures.

- Find the type of file.
- Open browser and search for file signatures.
- Compare the fetched signature with the existing file.
- If that matched, then nothing modified.
- If that doesn't matches, then the content of the file should be changed.

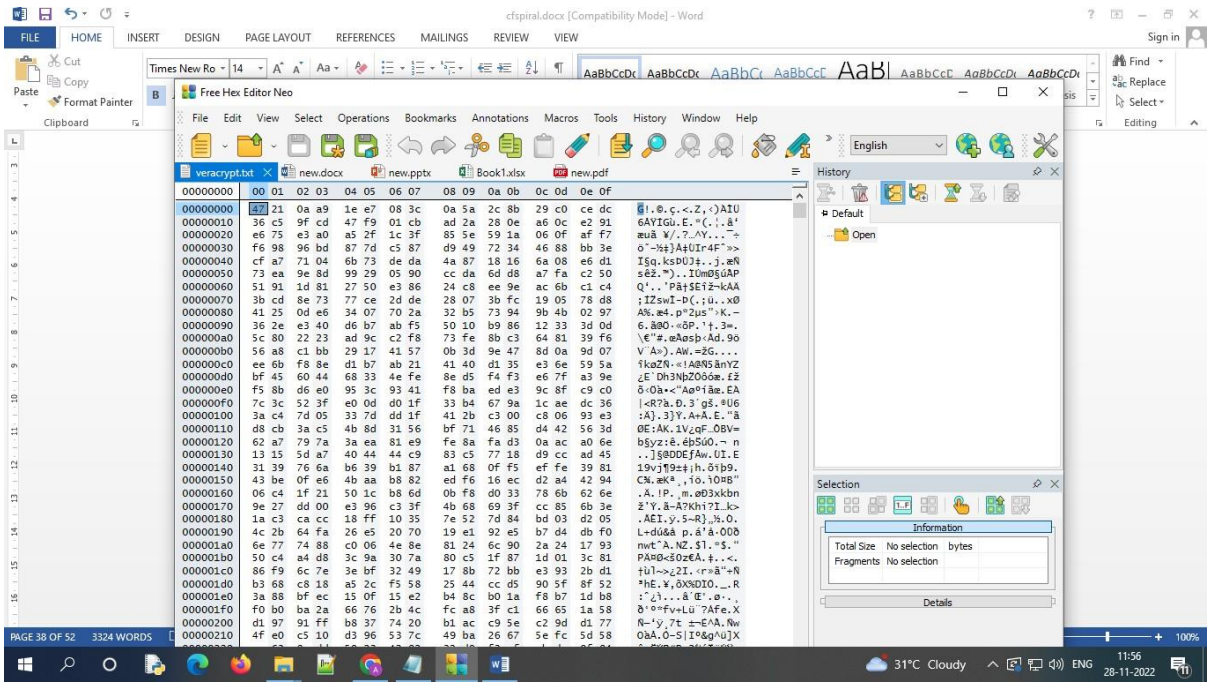
3) To perform password cracking using hex editor

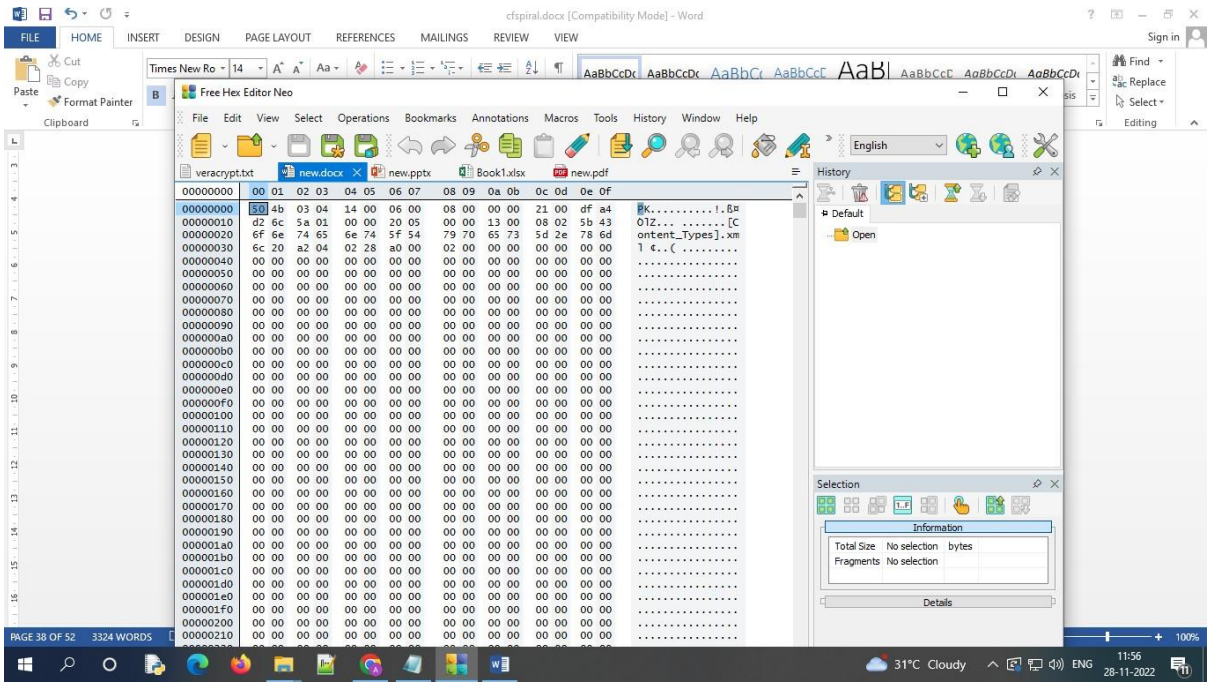
- Open browser.
- Search for the website and login to it.
- Open task manager. Right click browser .
- Take dump and it will be stored.

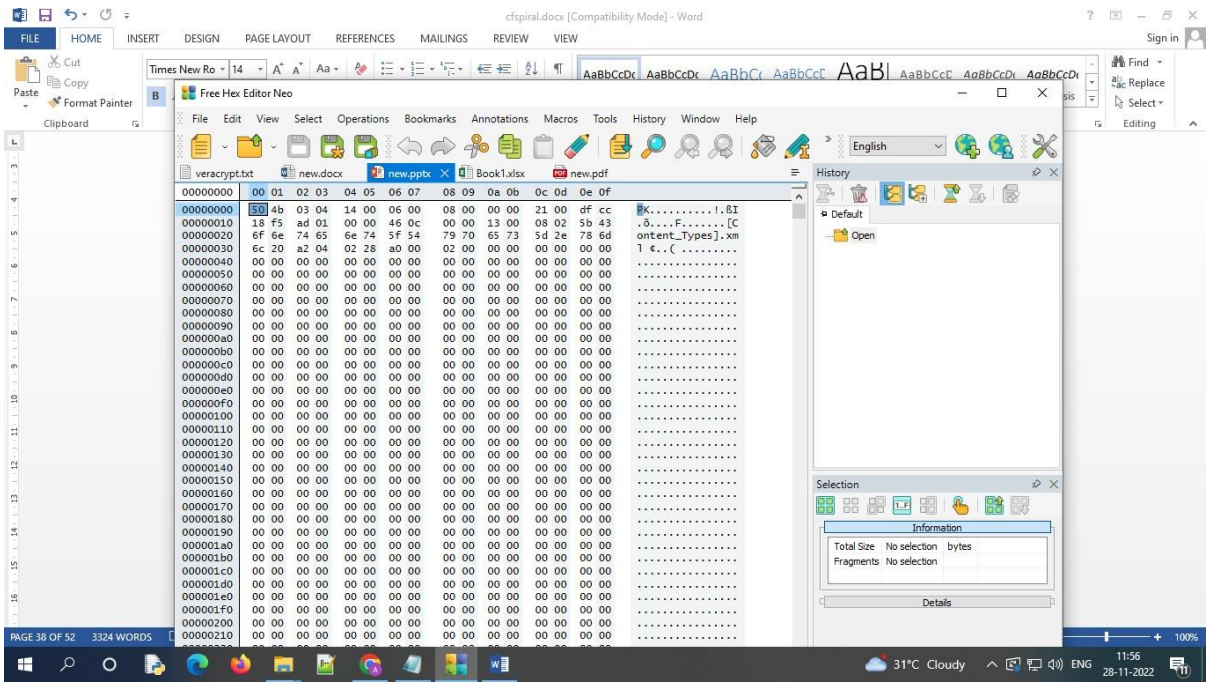
- Open it using Hex Editor.
- Press ctrl+F find prompt will be opened . search for password.
- Password will be found.

OUTPUT:

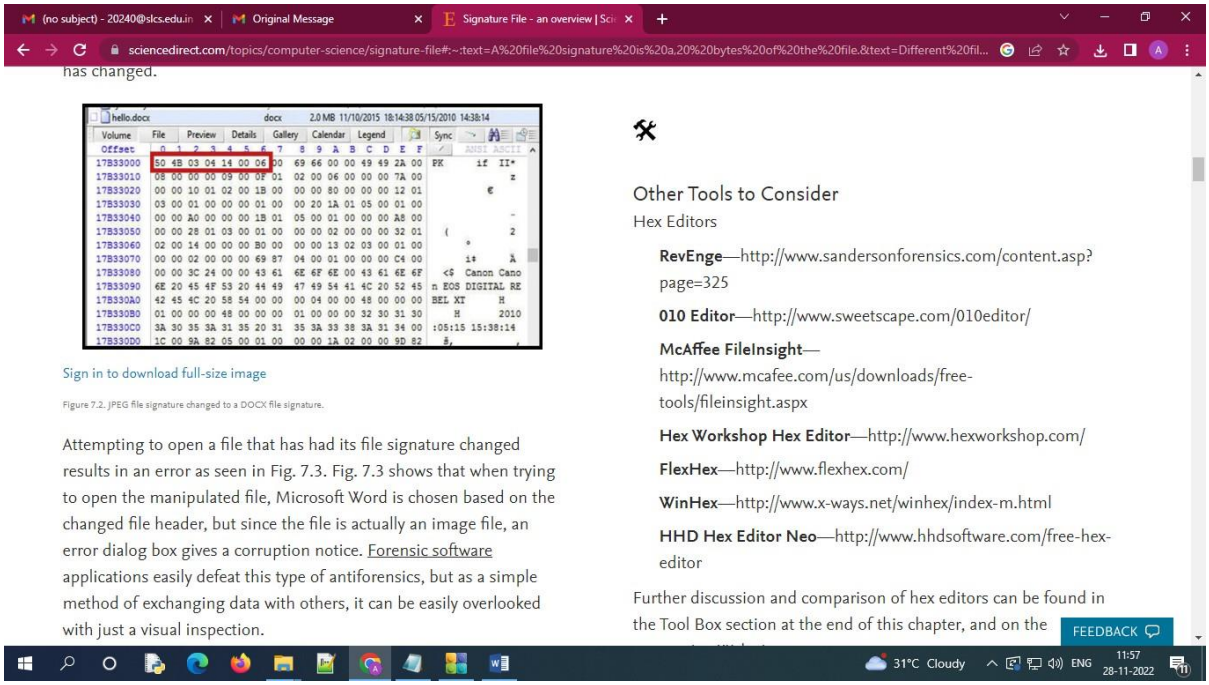
i) To analyse different types of files.

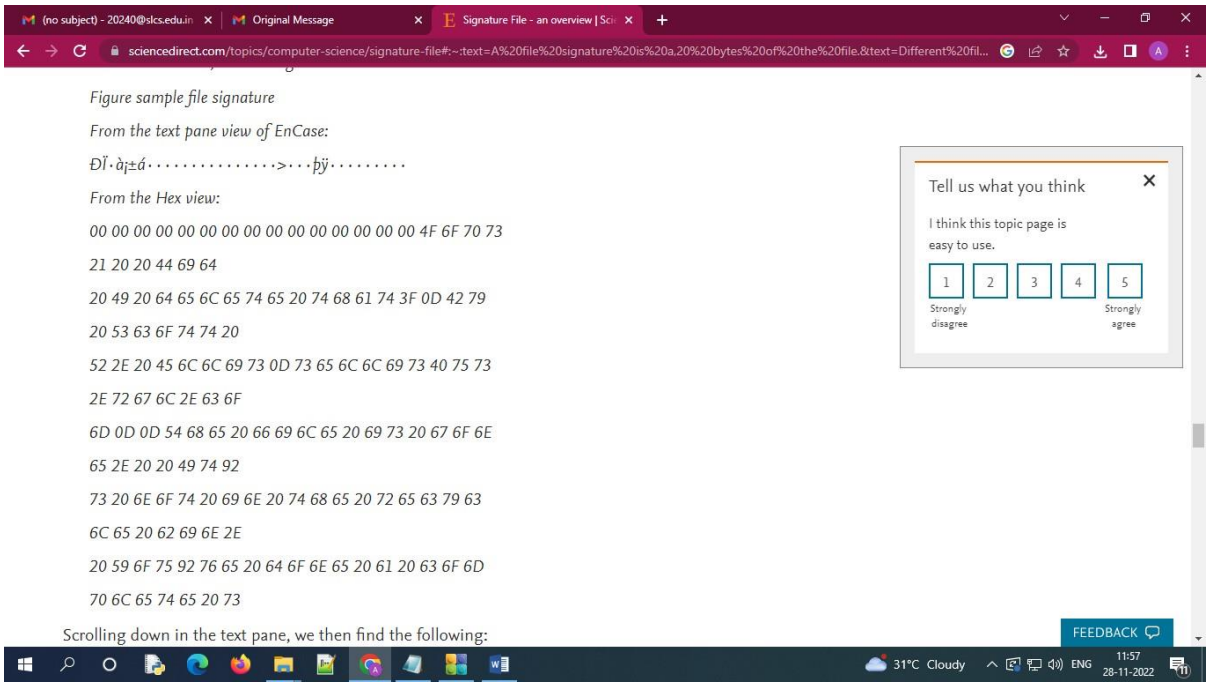




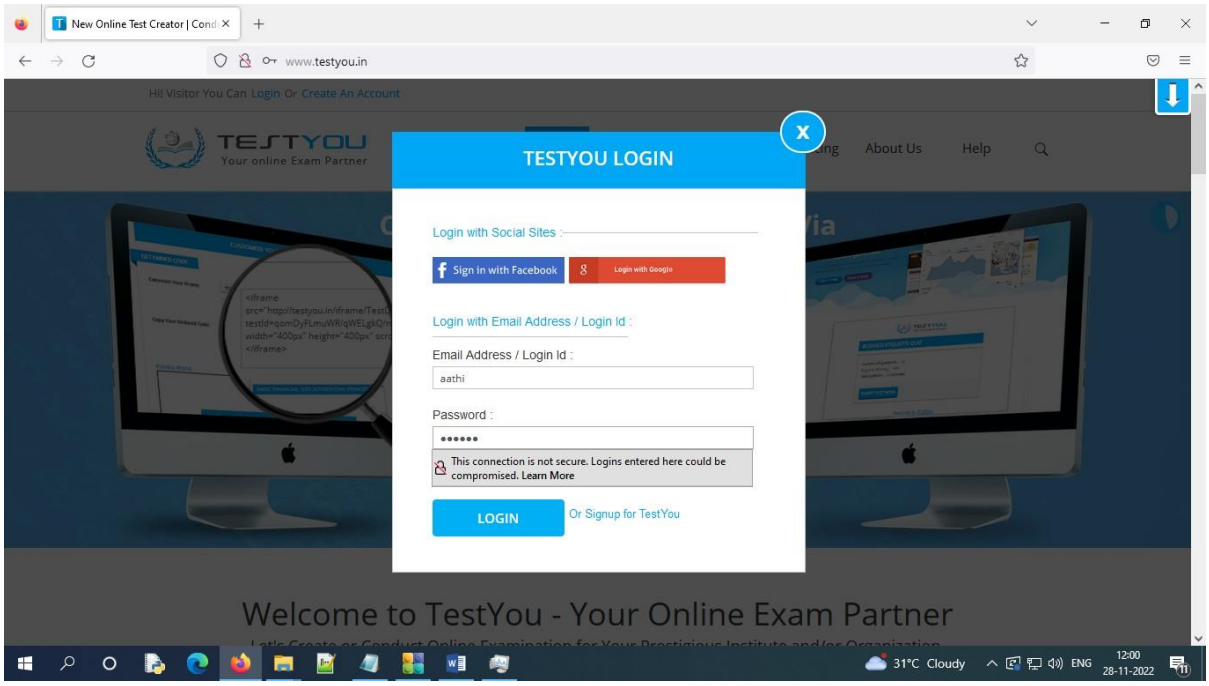


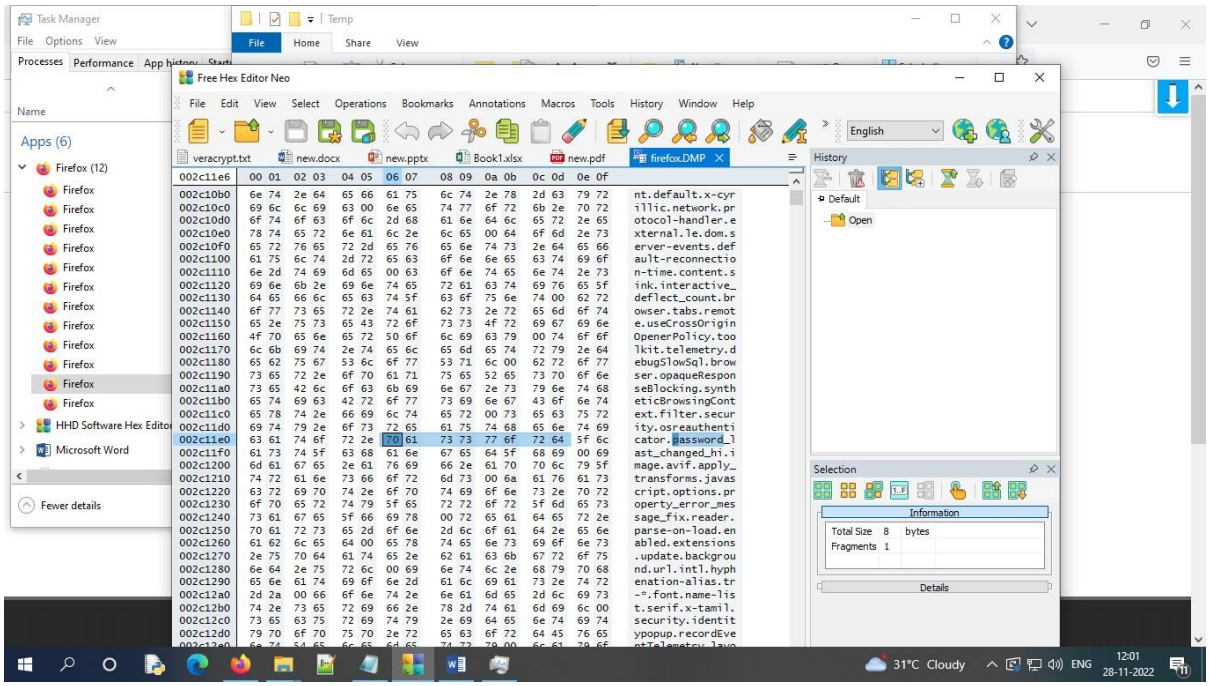
ii) To identify the file type with file signatures.





iii) Password cracking using Hex editor





RESULT:

Thus, file analysing using hex editor was done successfully.

Ex.No: 10

DATE: WIRESHARK – NETWORK AND PACKET ANALYSIS

AIM:

To perform network and packet analysis using wireshark tool.

REQUIREMETNS:

- Wireshark tool.

TOOL DESCRIPTION:

- Wireshark is a network protocol analyser, or an app that captures from a network connection, such as from your computer to you home office or the internet. It is the most used packet sniffer in the world.

TASKS:

- To perform network analysis.
- To perform packet analysis.
- To perform traffic analysis.

PROCEDURE:

➢ To perform network analysis

(ii) tcp.port==80 || udp.port==80

Using this command, we can see the network traffic passed through TCP protocol.

(iii) icmpV6

Using this command, we can analyse packets passed through icmpV6 protocol with type as multicast listener report

➢ To perform packet analysis

Open wireshark and select ethernet and start the tool.

(i) ip.addr==’any ip address’

Using this command, we can fetch all the packets received or sent from the given IP address.

(ii) ip.src == ‘source ip address’

Using this command, we can analyse what are all the packets sent from the given source address.

(iii) ip.dest == ‘destination ip address’

Using this command, we can analyse what are all the packets received by the destination address.

(iv) http

- Open browser.
- Search for any HTTP site.
- Using this HTTP command in wireshark, we can analyse the HTTP traffic.

(v) http.request

Using this command , HTTP.request , it filters the request packets.

(vi) http.response

Using this command, we can analyze the responses received through HTTP protocol.

(vii) http.response.code

Using this command, we can analyse the responses received through HTTP protocol.

- **To perform traffic analysis**

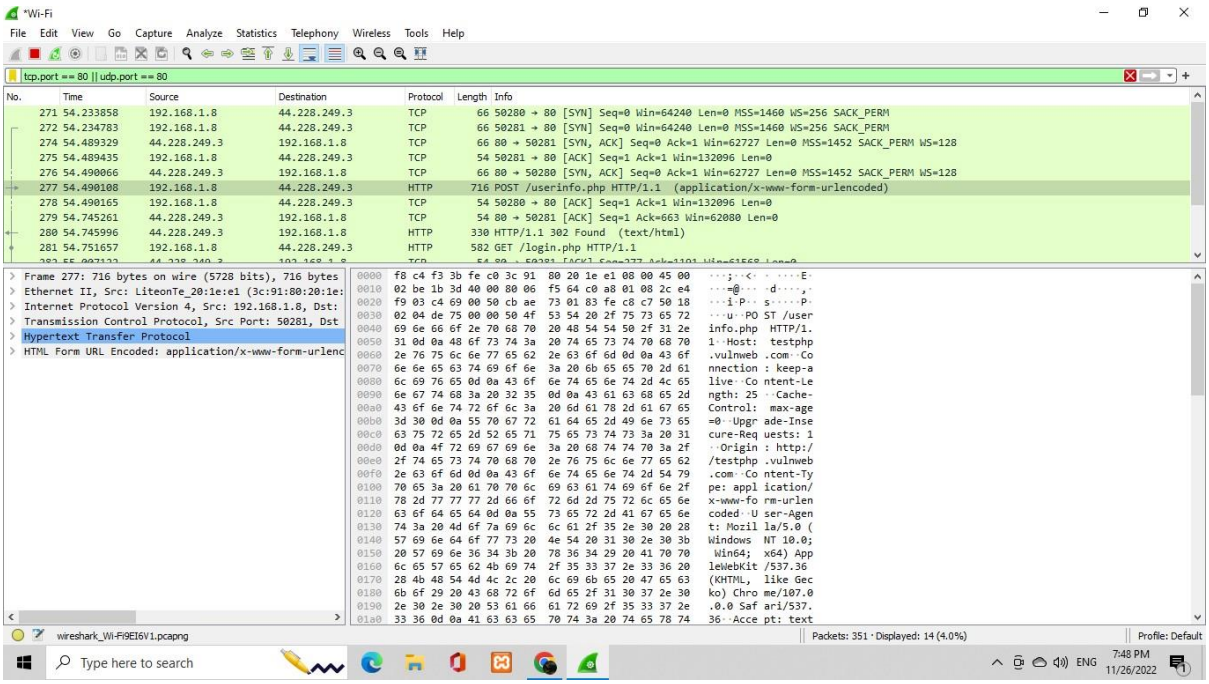
(i) tcp.flags.ack ==1

Using this command, we can analyse TCP packets that has sent ACK packet to the host.

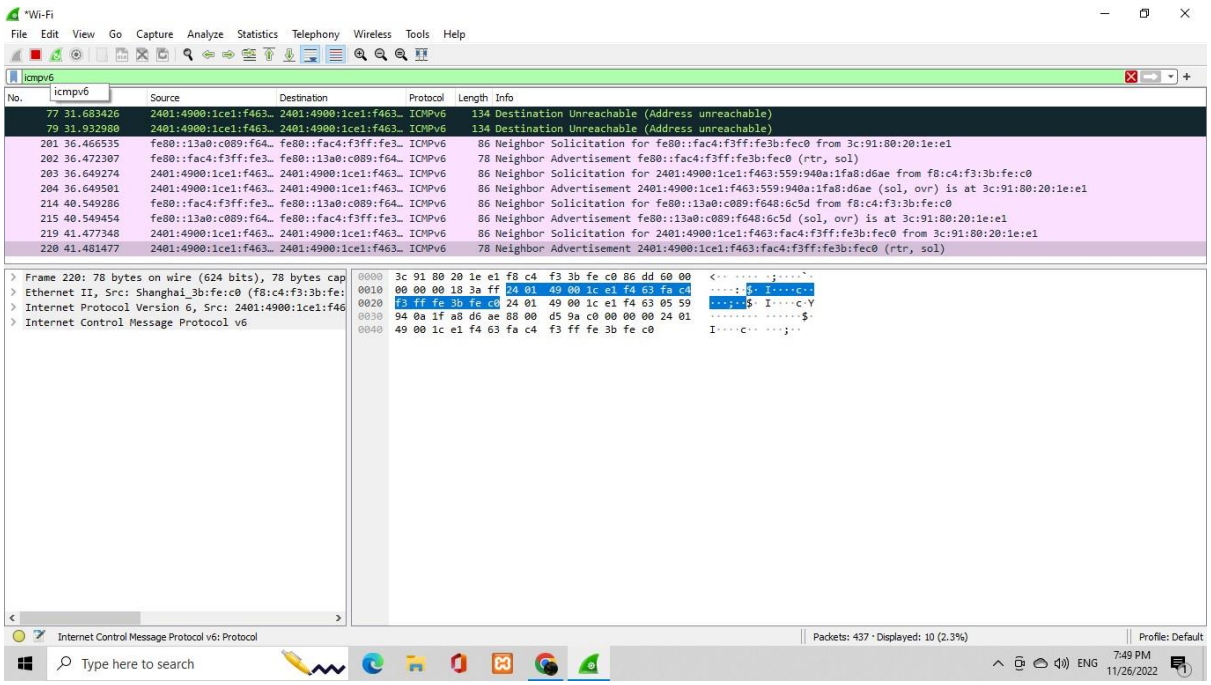
OUTPUT:

NETWORK ANALYSIS

i) tcp.port==80||udp.port==80

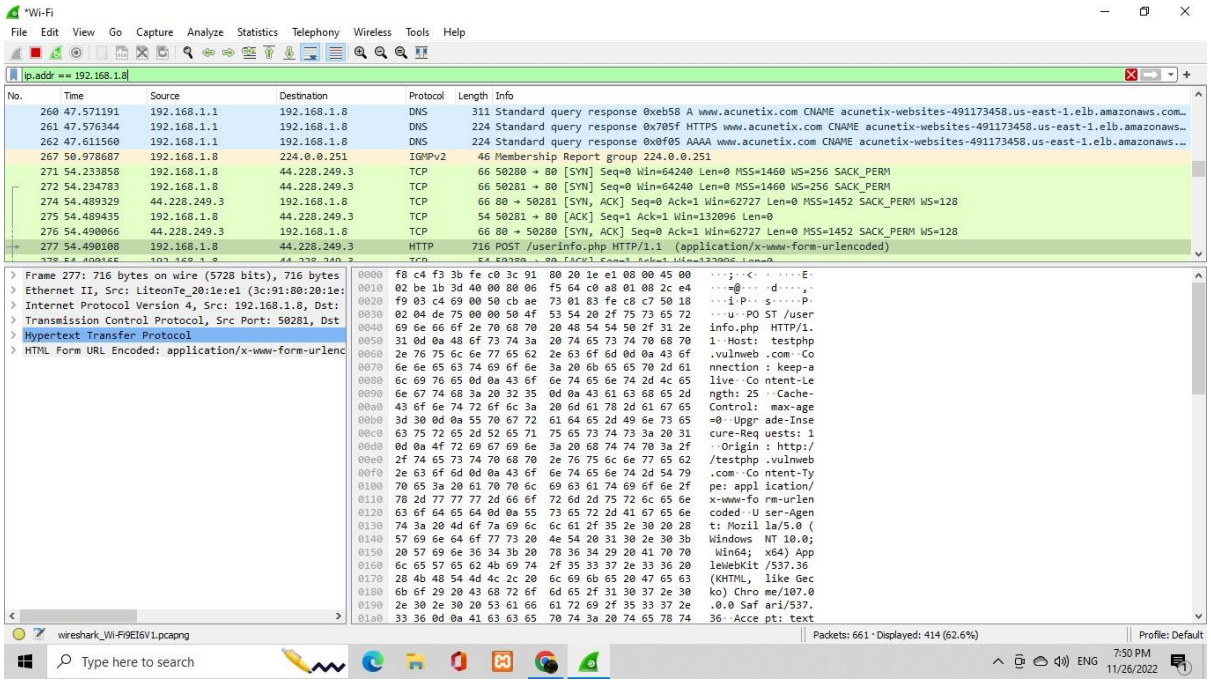


ii) icmpv6

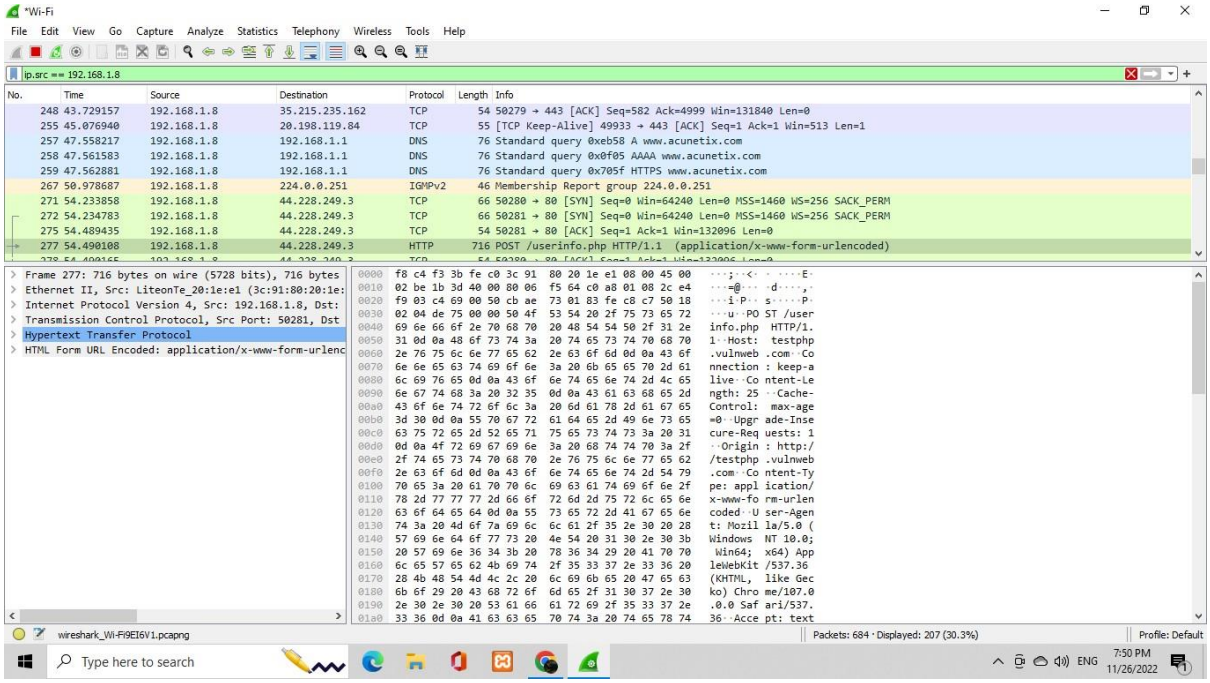


PACKET ANALYSIS:

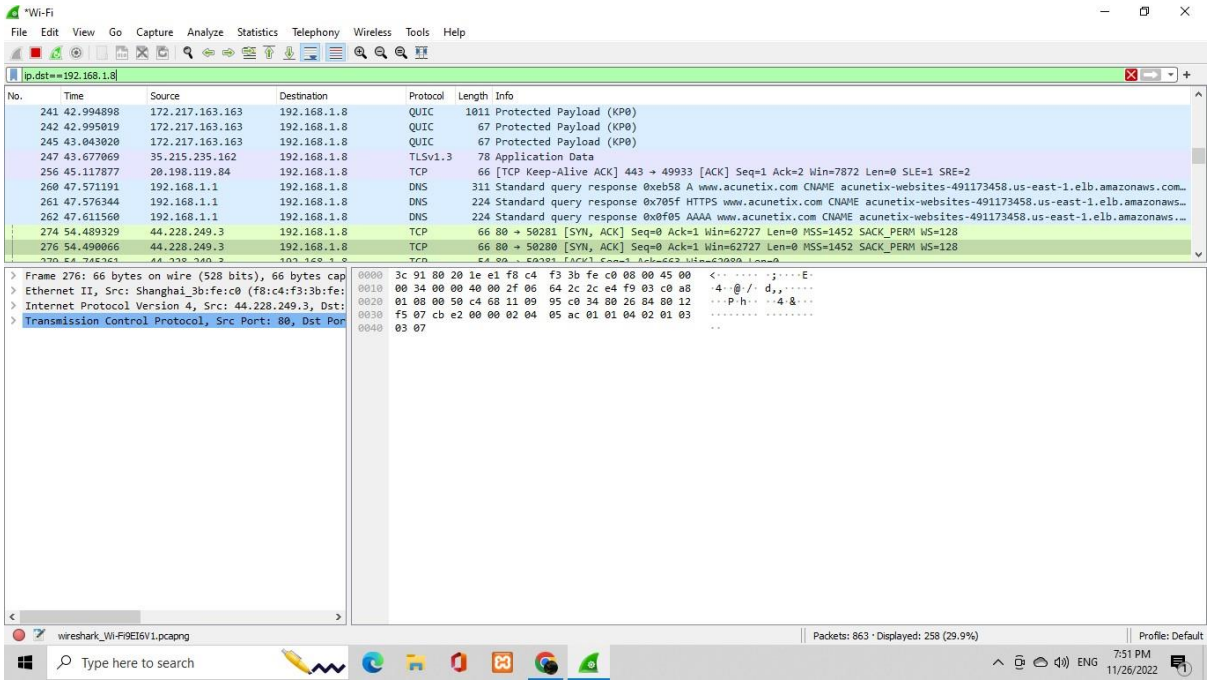
i)ip.addr=’ip address’



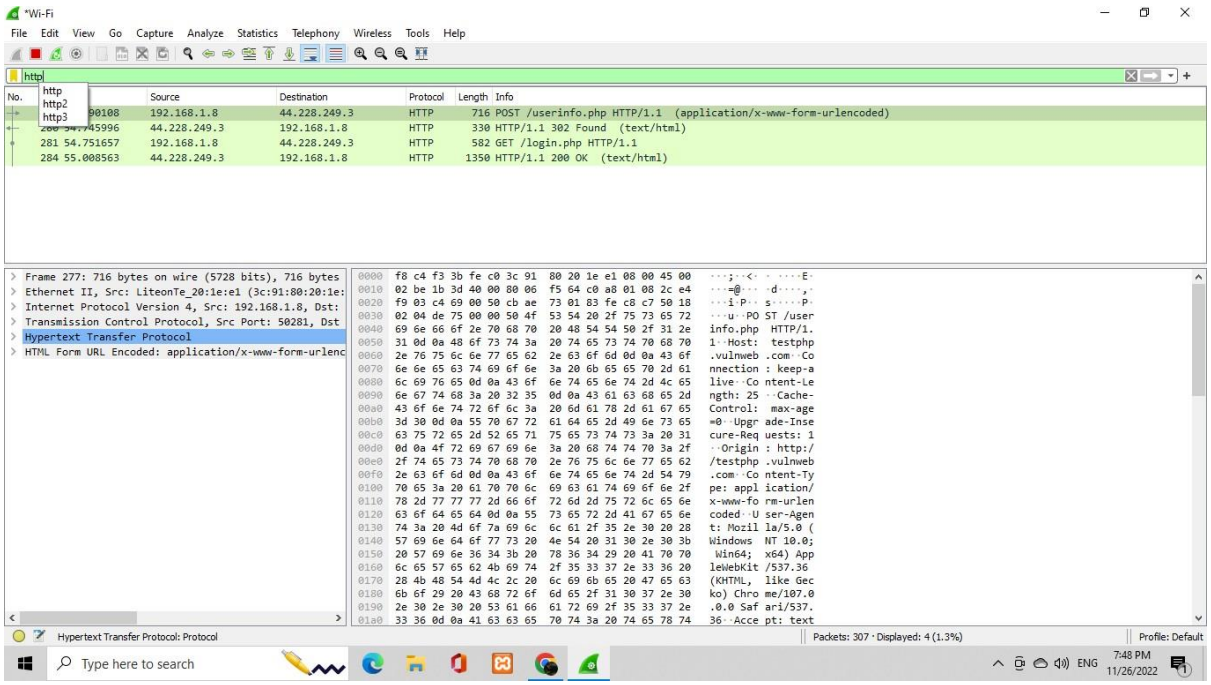
ii) ip.src=’ip address’



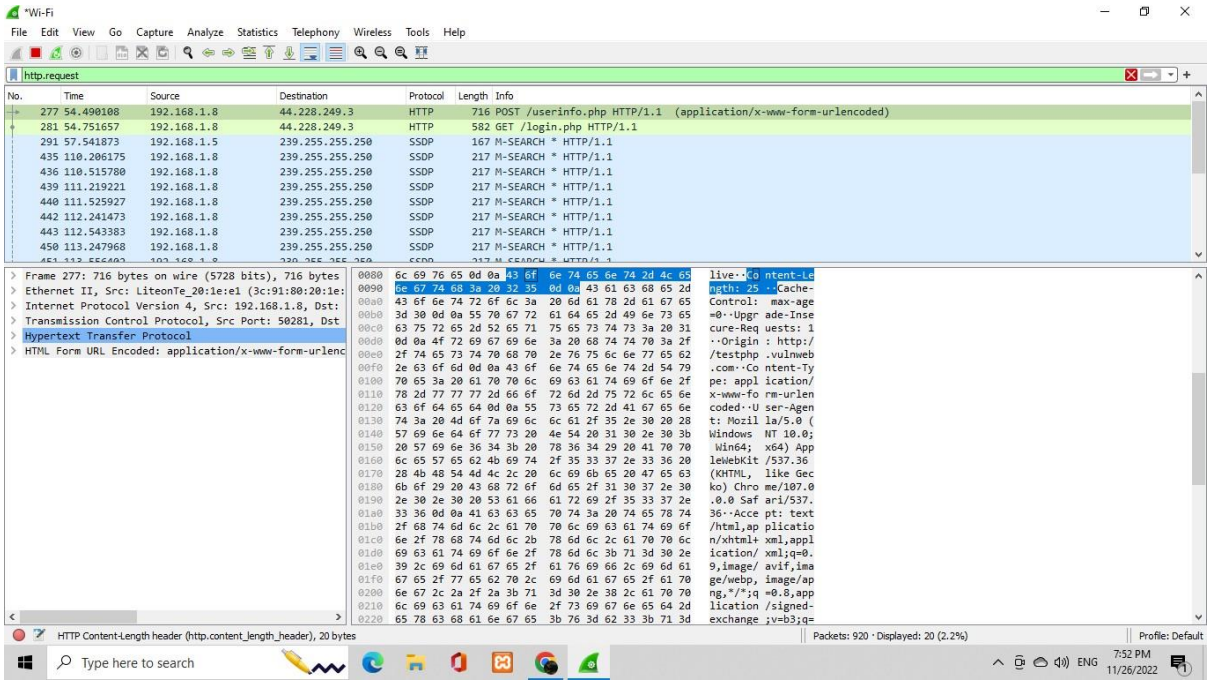
iii) ip.dst=’ip address’



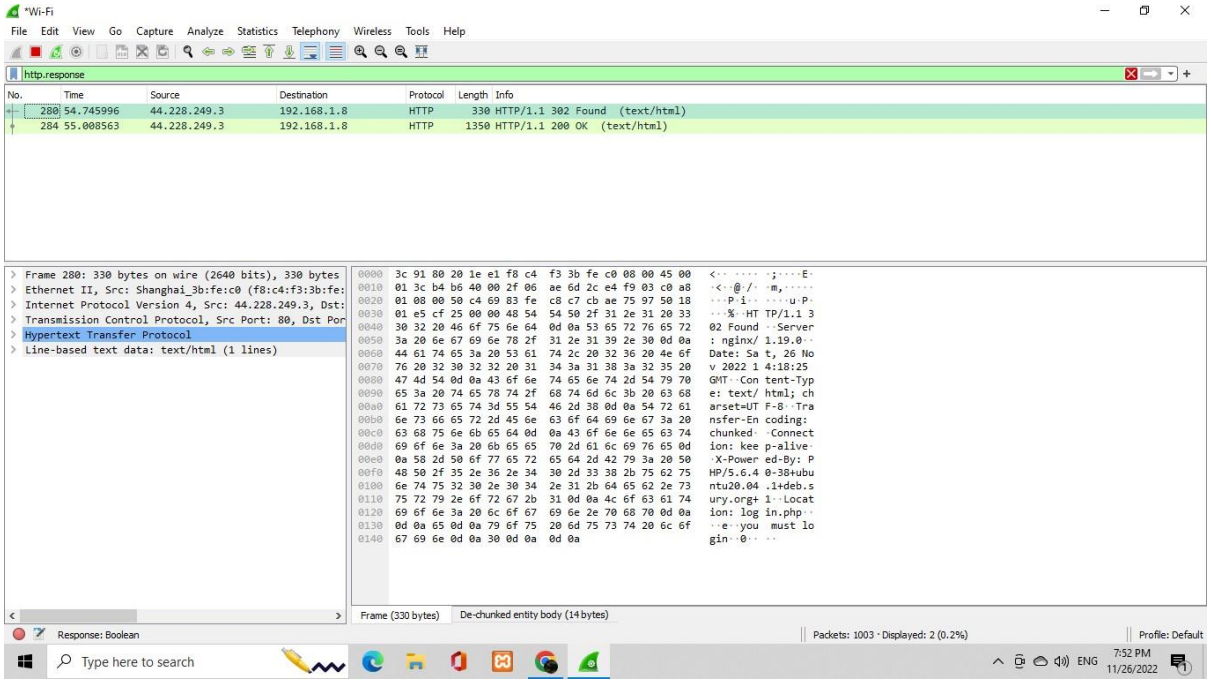
iv) http



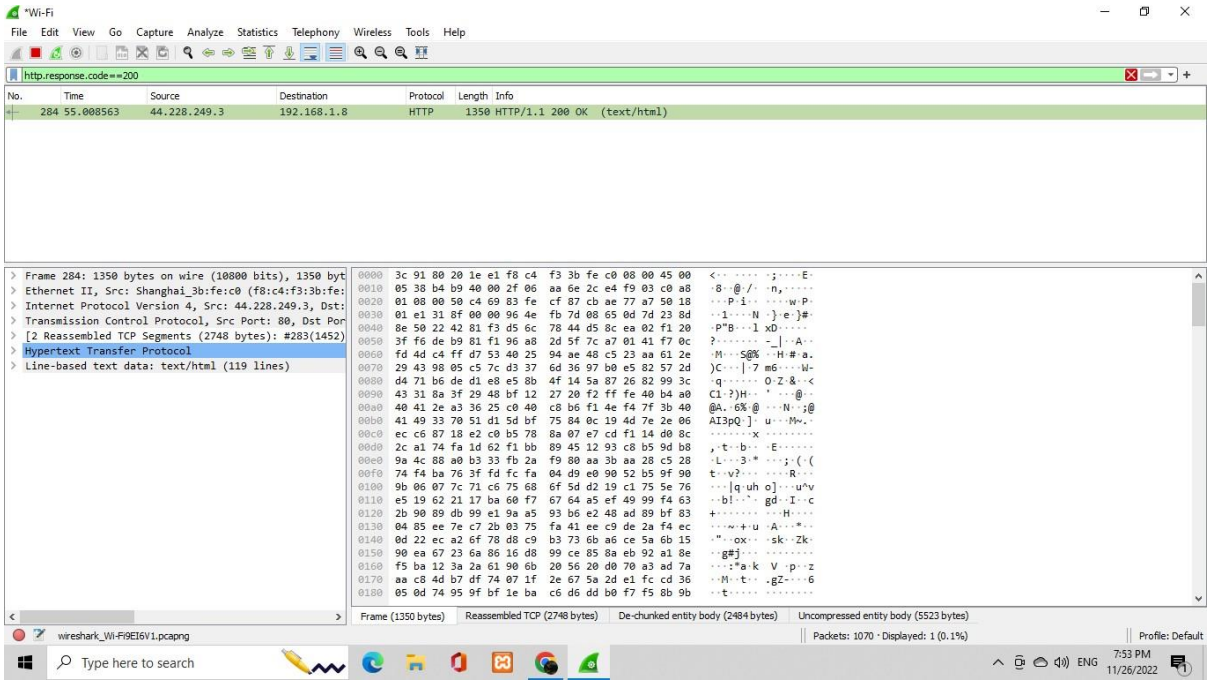
v) http.request



vi) http.response

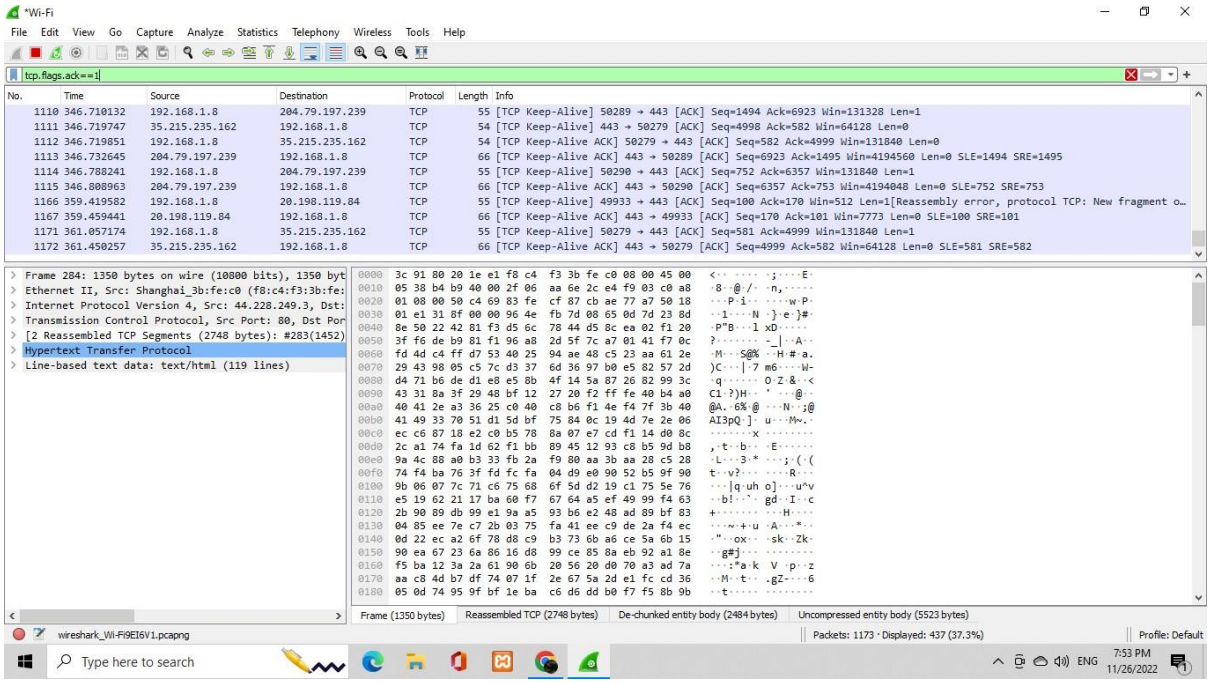


vii) http.response.code==200



TRAFFIC ANALYSIS:

i)tcp.flags.ack==1



RESULT:

Thus, network, packet and traffic analysis using wireshark tool was done successfully.

Ex.No: 11

DATE: **EMAIL FORENSICS**

AIM:

To perform email analysis and email forensics.

REQUIREMENT:

- Ip2location
- FTK imager

TOOL DESCRIPTION:

Ip2location:

- Ip2location provides web application interface to upload a list of IP address for both processing . Users can download the results in comma delimited text file.

FTK Imager:

- FTK or forensic tool kit is a computer forensics software made by Accessdata. It scan a hard drive looking for various information. It can , for example, potentially locate deleted emails and scan a disk for text strings to use them as a password dictionary to crack encryption.

TASKS:

- To perform email header analysis.
- To perform email file analysis**.**

PROCEDURE:

➢ To perform email header analysis

- Open browser and login into email.
- After login, open any one mail that you received.
- Inside the mail click the more options.
- Select the show original option and copy the header of the particular mail.
- Search Ip2location in the browser and open the email tracer.
- Paste the header in the search bar and click lookup.
- The information of the email header will be displayed in detailed format.

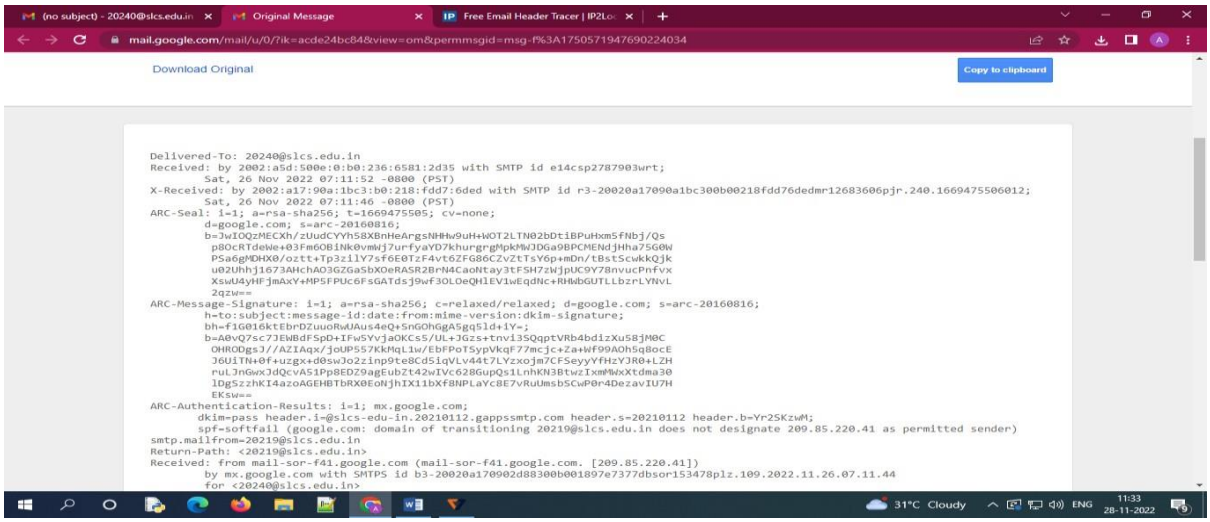
➢ To perform email file analysis

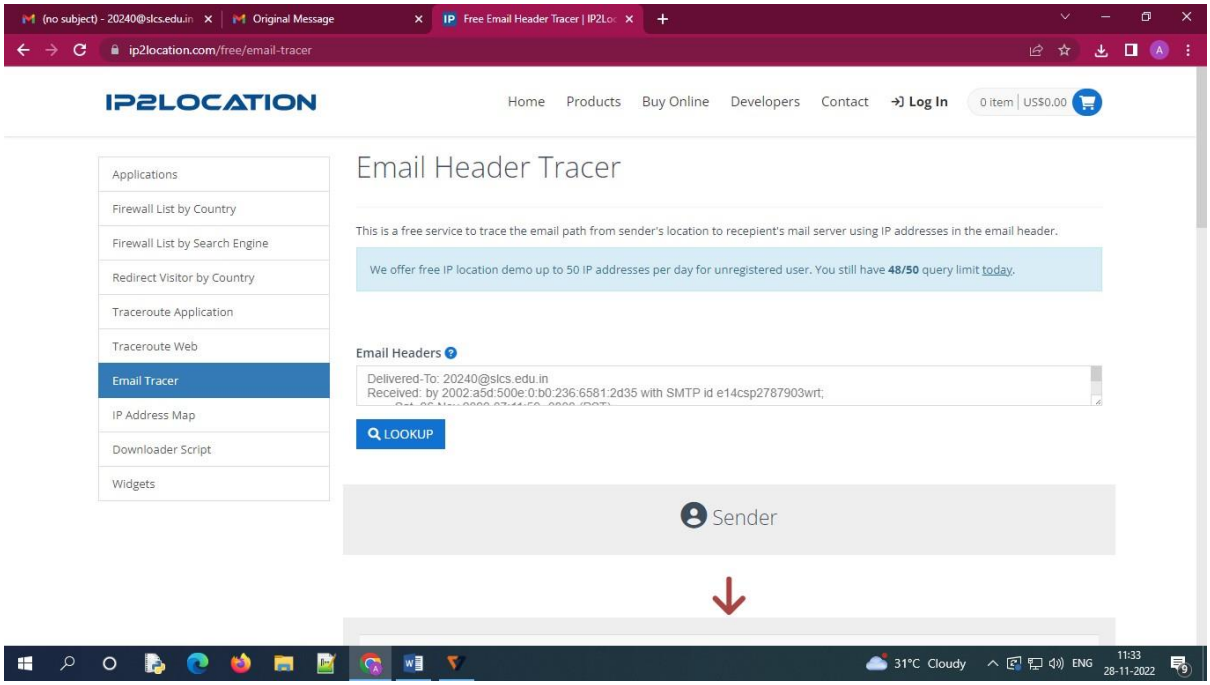
- Open browser and login into email.
- After login, open any one mail that you have received.
- Inside that mail, open the more option and click show original option.
- Download it and it will downloaded with .eml extension.

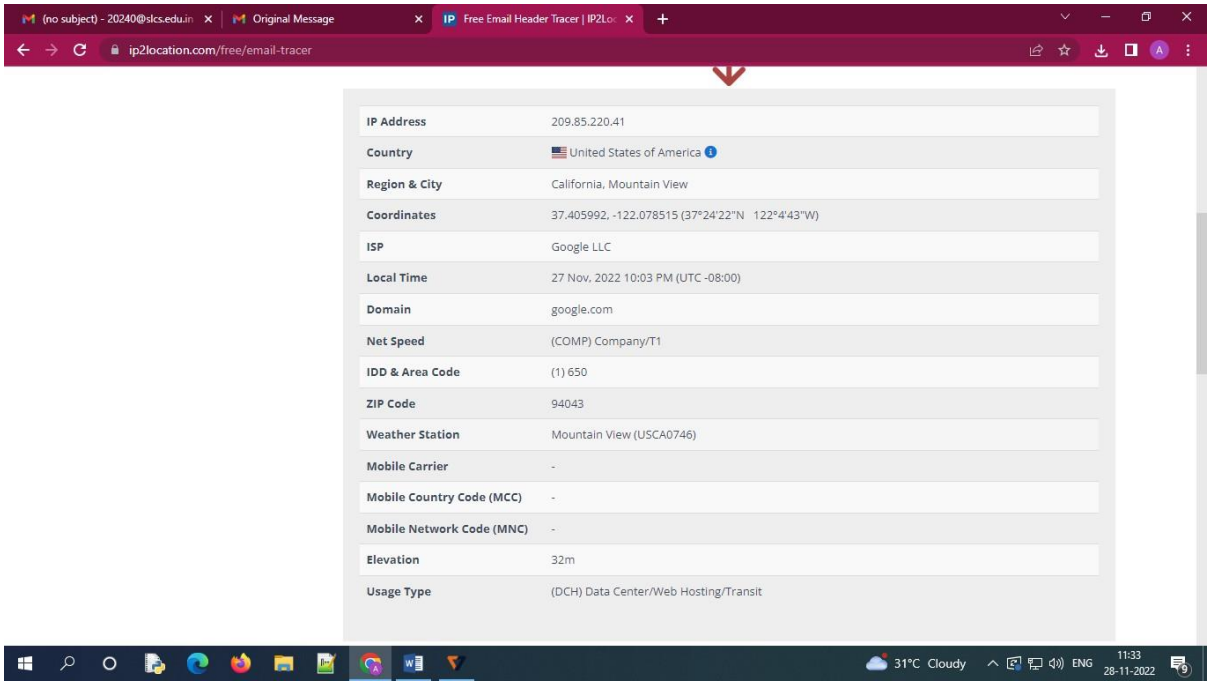
- Create a new folder and put the downloaded file inside it.
- Open FTK tool kit, click file and then click add evidence item option.
- And then click the content of a folder option and add the downloaded file there.
- Once it is opened, it will show the results of header, body, sender, receiver and more details about email.

OUTPUT:

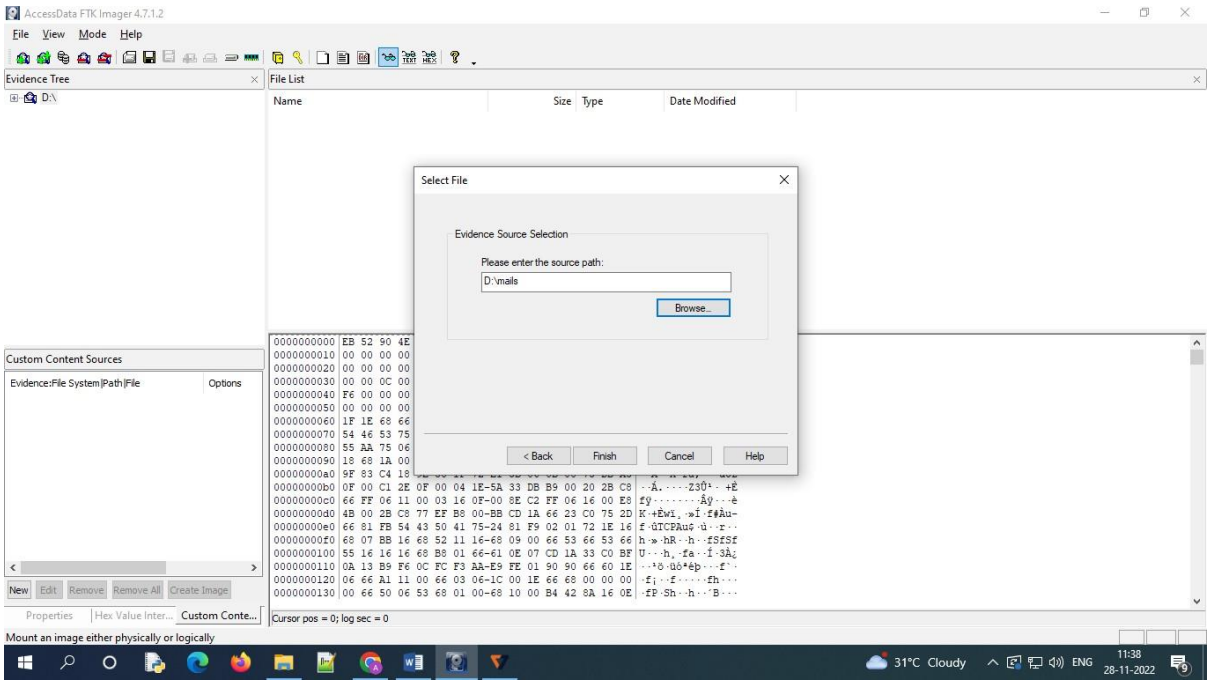
I) Email Analysis:

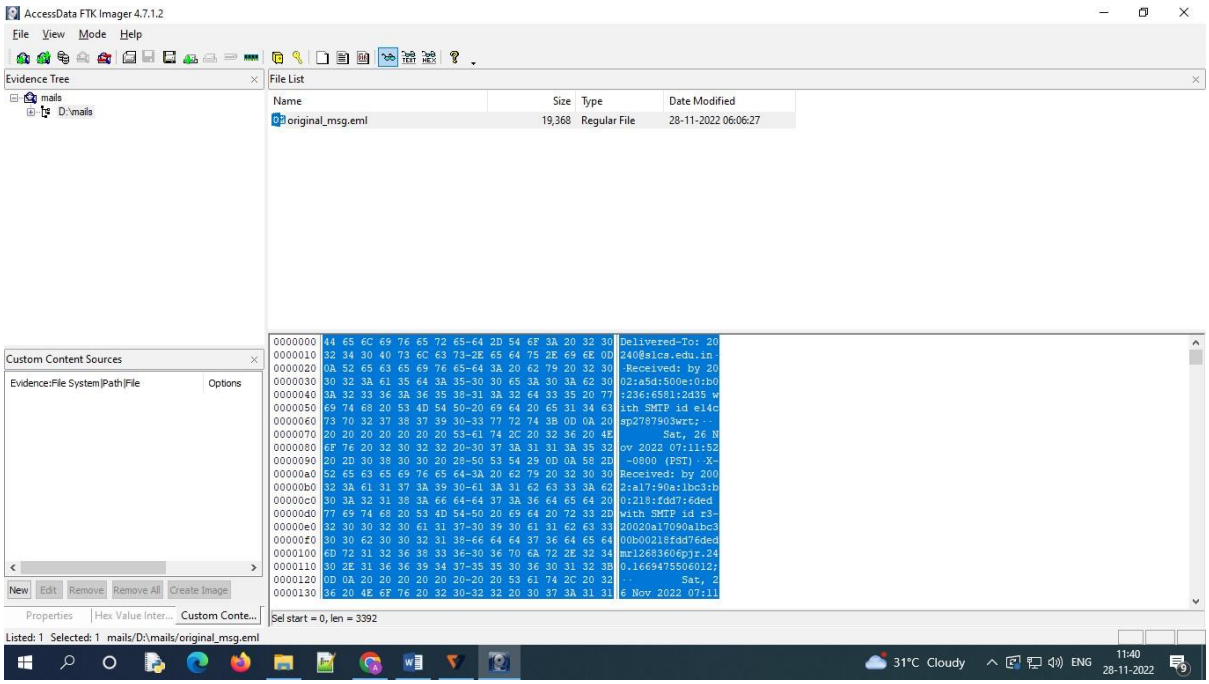






ii) Email file analysis:





RESULT:

Thus, the task to perform email analysis and email forensics was done successfully.

Ex.No:12

DATE: **NETWORK ANALYSIS , TRAFFIC MONITORING AND VULNERABILITY SCANNING**

AIM:

To analyse the network, monitor the network traffic and to scan the vulnerability using fast resolver and acunetix tools**.**

REQUIREMENTS:

- Fast resolver
- Acunetix

TOOL DESCRIPTION:

FAST RESOLVER:

- FastResolver is a small utility that resolves multiple host names into IP addresses and vice versa. It is a multithread application, so it can resolve dozens if addresses within a few seconds.

ACUNETIX:

- Acunetix is an automated web application security testing tool that audits your SQL injection, cross site scripting and other exploitable vulnerabilities.

NETSTAT:

- The network statistics (netstat) command is a networking tool used for troubleshooting and configuration, that can also serve as a monitoring tool for connections over the network. Both incoming and outgoing connections, routing tables, port listening, and usage statistics are common uses for this command.

TASKS:

- To perform resolving IP address into MAC.
- To monitor network traffic.
- To perform vulnerability scanning.

PROCEDURE:

➢ To resolve IP address into MAC

- Open browser and install fastresolver tool.
- Open the tool.
- Click the select host/IP addresses icon.
- Then a window will be opened.

- In that enter the range of IP address which you want to resolve.
- Then click retrieve MAC address option and then click ok.
- Then we can retrieve a list of MAC addresses of the computers in your network.

➢ **To monitor network traffic**

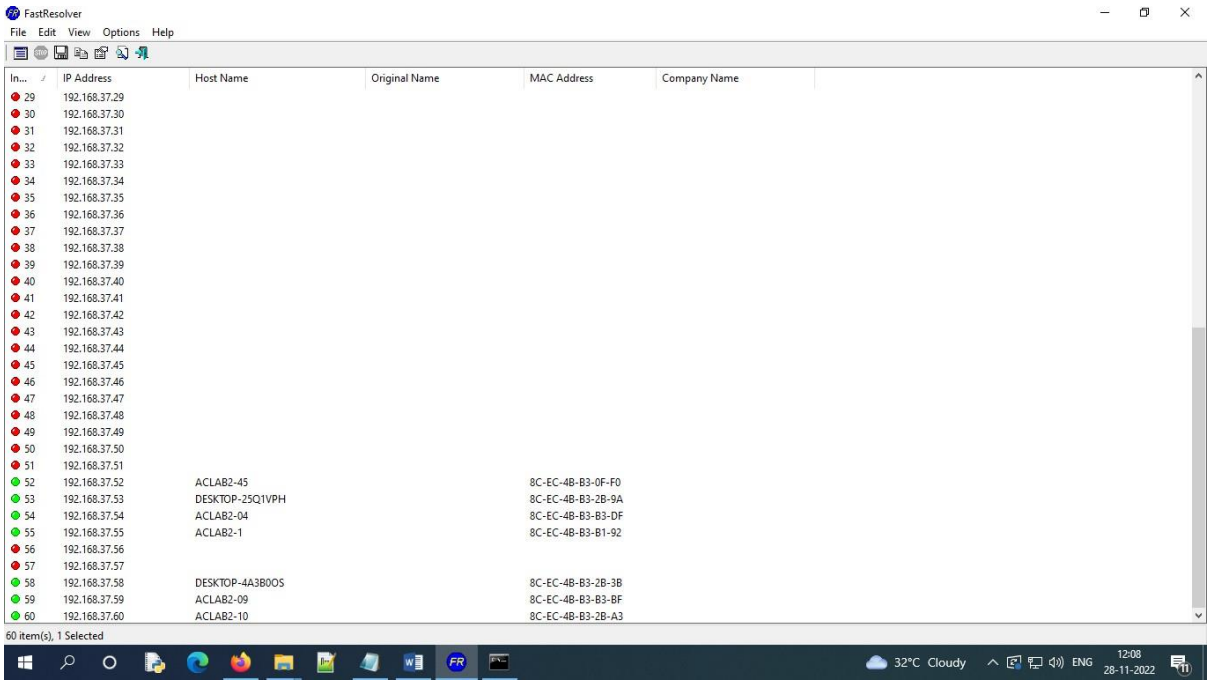
- Open command prompt in your PC.
- Type the command " netstat " and hit enter.
- It provides you about all active connections so you can find out which computer or networks is connected to your PC.

➢ **To perform vulnerability scanning**

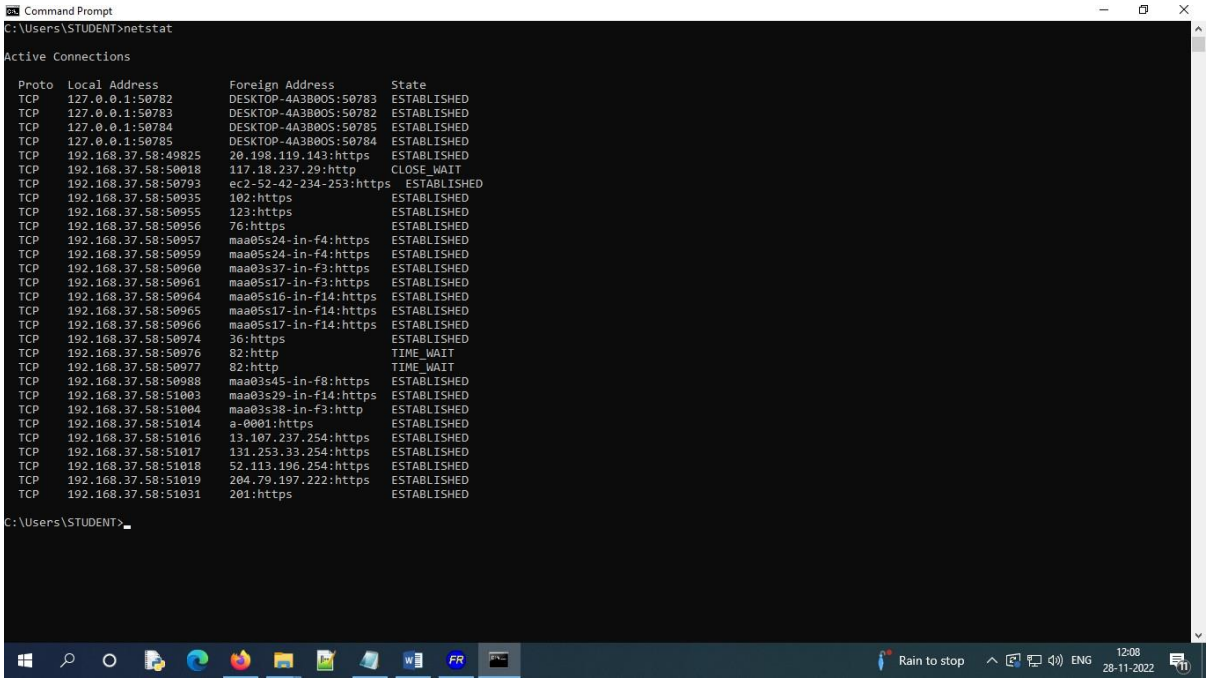
- Open browser and download acunetix tool.
- Open tool and click new scan option.
- Then a window will be opened.
- There click single website scan and give the link of the website which you want to scan for vulnerability and click next and next.
- And finally click finish.
- And the vulnerabilities of the given website will be shown there.

OUTPUT:

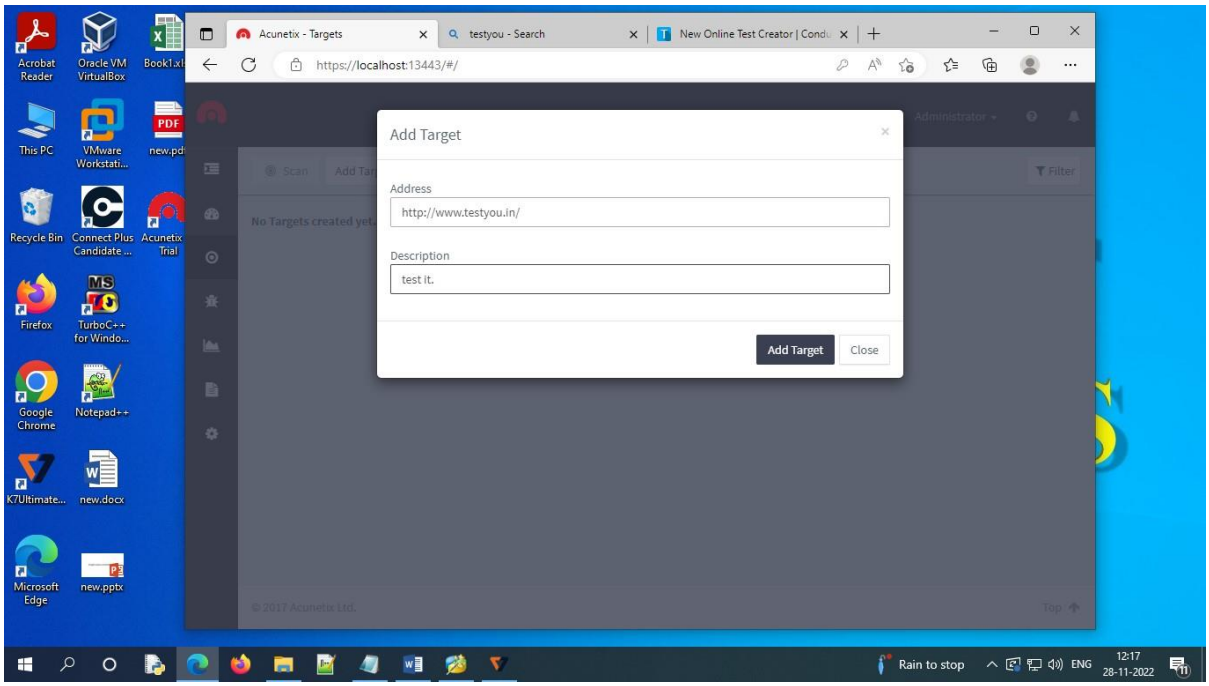
i) To resolve IP address into MAC

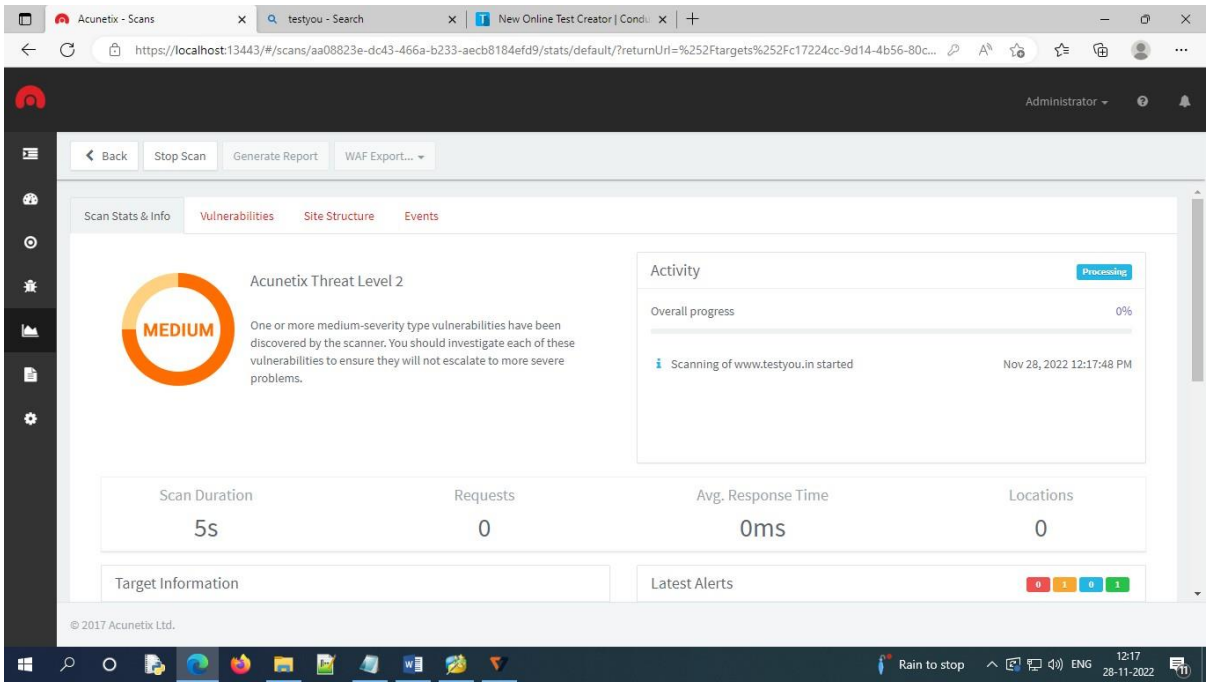


ii) To monitor network traffic



iii) To perform vulnerability scanning





RESULT:

Thus, performing network analysis, monitoring network traffic and vulnerability scanning was done successfully.

Ex.No:13

DATE: **SCANNING WITH NMAP TOOL**

AIM:

To perform scanning using nmap tool**.**

REQUIREMENTS:

- Kali linux

DESCRIPTION:

- Kali Linux is an open-source, Debian-based Linux distribution geared towards various information security tasks, such as Penetration Testing, Security Research, Computer Forensics and Reverse Engineering.

TASK:

- **To perform scanning using nmap tools.**

PROCEDURE:

1. Scanning using nmap tool

- To scan a single IP address

  COMMAND: **nmap 'target'**

- To scan multiple IP address

  COMMAND: **nmap 'target1' 'target2'**

- To scan a range of IP address

  COMMAND**: nmap 'target-target'**

- To scan a single port

  COMMAND: **nmap -p 'port_numnber' 'target'**

- To scan multiple port

  COMMAND: **nmap -p 'port-1','port-2','port-3' 'target'**

- To scan a range of ports

  COMMAND: **nmap -p 'port-1'-'port-n' 'target'**

- To scan all the ports

  COMMAND: **nmap -p 'target'**

- To read a target from the text file, create a text file and add some ip addresses to it. To open it and execute the nmap scan

  COMMAND: **nmap -iL filename**

- To save nmap result on the text file

  COMMAND: **nmap -0G /root/filename 'target'**

- To scan a ipV6 target

  COMMAND: **nmap -6 'target'**

- To perform ping scan

  COMMAND: **nmap -sP 'target'**

- To perform TCP/SYN ping scan

  COMMAND: **nmap -pS 'target'**

- To perform TCP/ACK ping scan

  COMMAND: **nmap -pA 'target'**

- To perform UDP ping scan

  COMMAND: **nmap -pU 'target'**

- To perform ICMP ping scan

  COMMAND: **nmap -pE 'target'**

- To perform traceroute scan

  COMMAND: **nmap -traceroute 'target'**

- To detect operating system running in the target

  COMMAND: **nmap -O 'target'**

- To perform service detection for the target system

  COMMAND: **nmap -sV 'target'**

- To perform aggressive scan

  COMMAND: **nmap -A 'target'**

- To scan live host

  COMMAND: **nmap -sn 'target'**

- To find active port

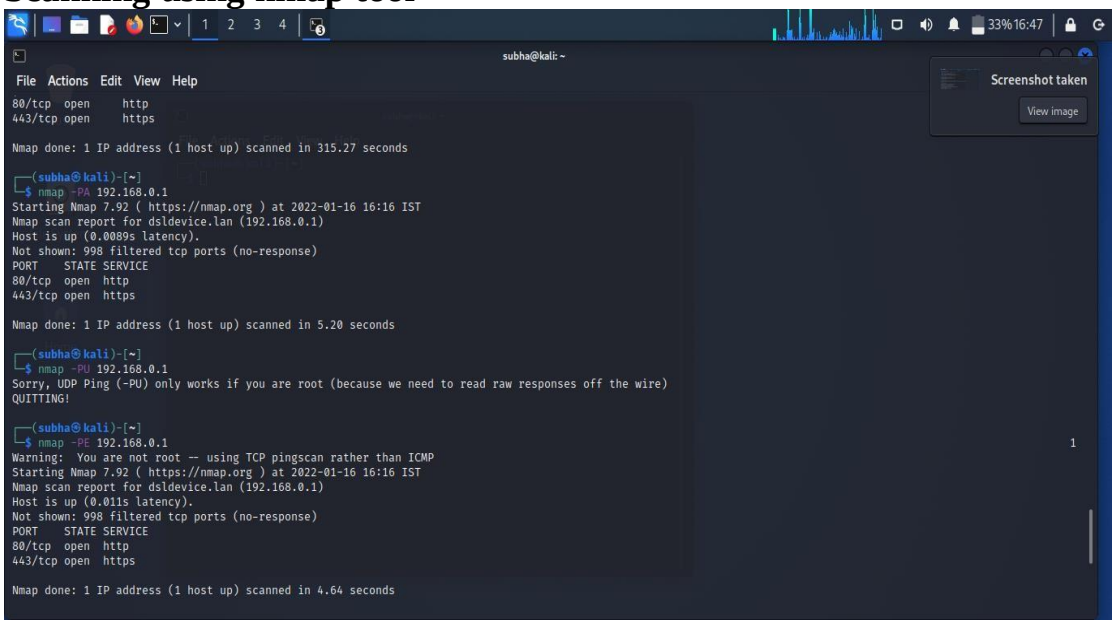
  COMMAND: **nmap -p- 'target'**

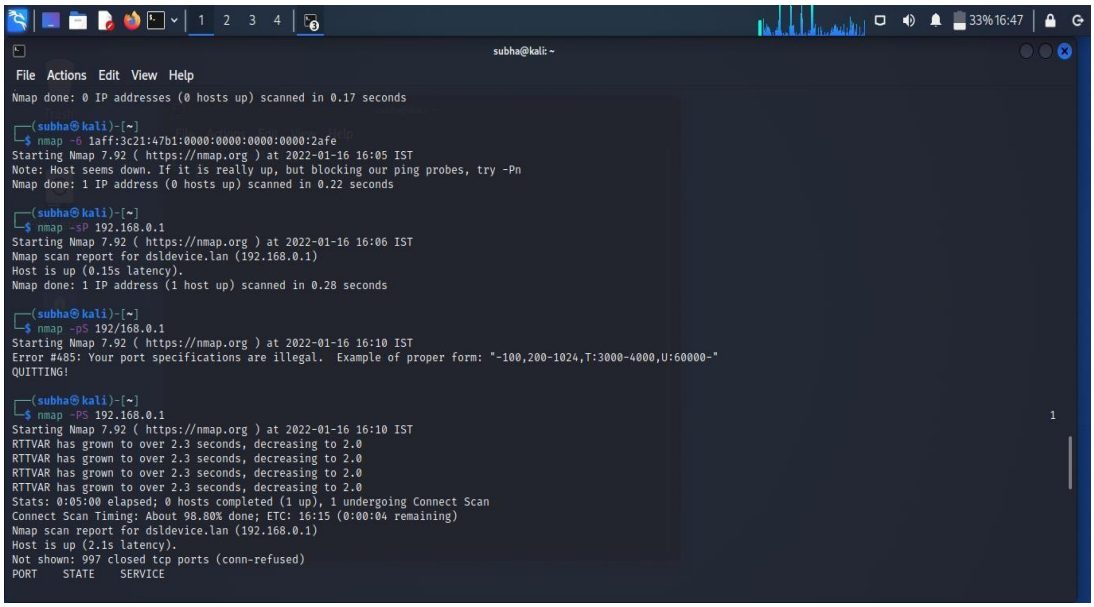
- To find vulnerability

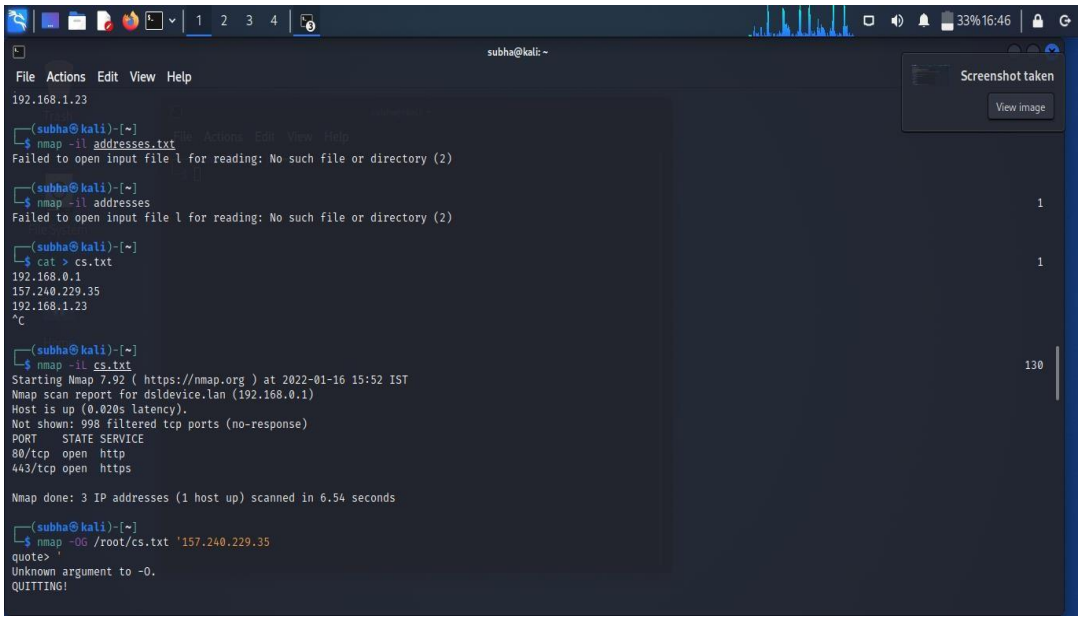
  COMMAND: **nmap -A ‘target’**

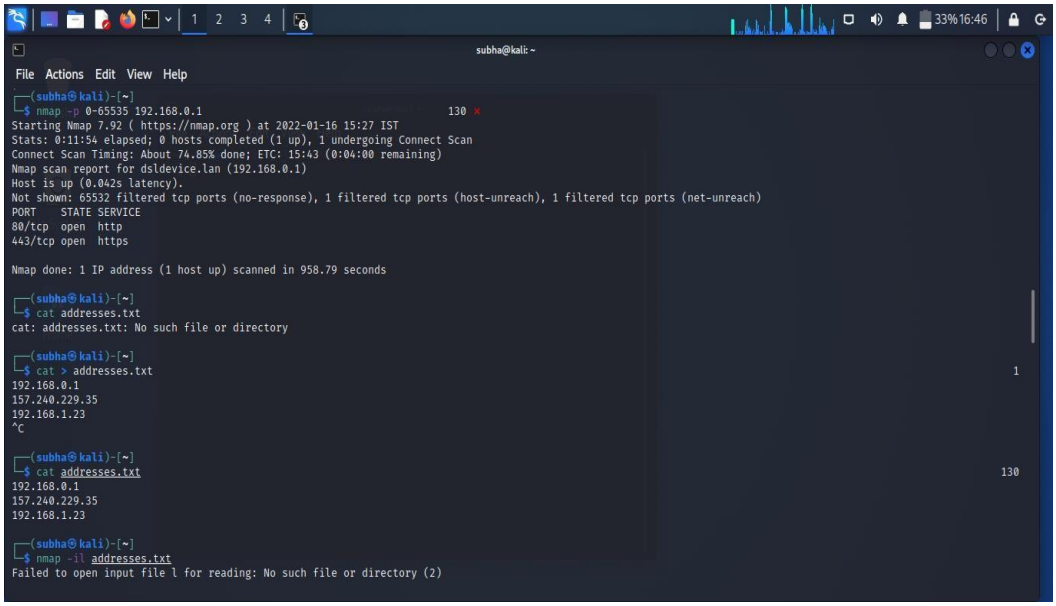
OUTPUT:

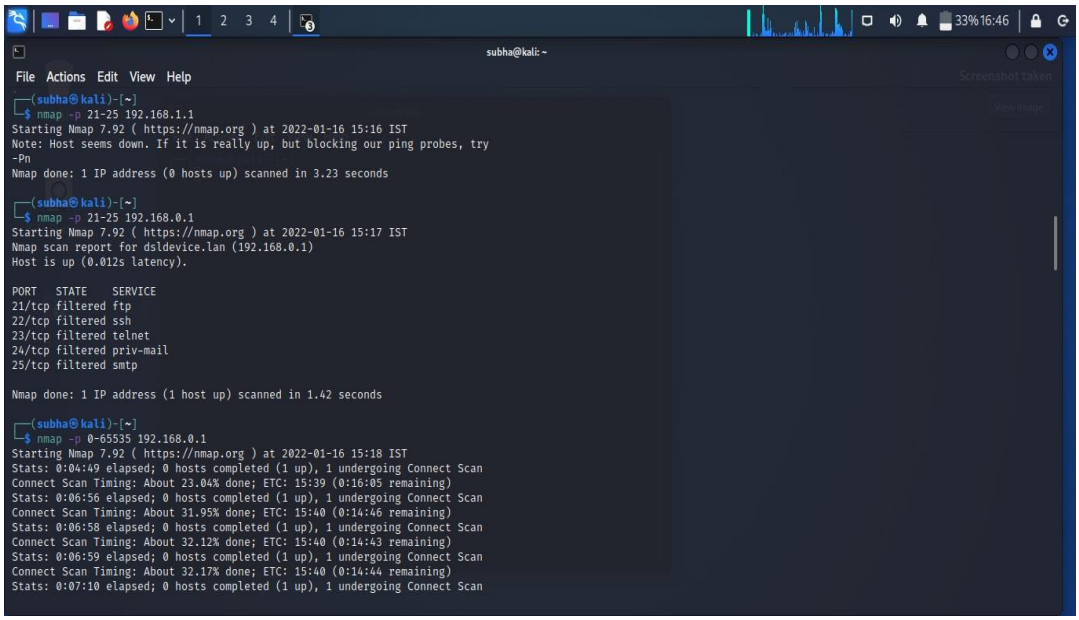
Scanning using nmap tool

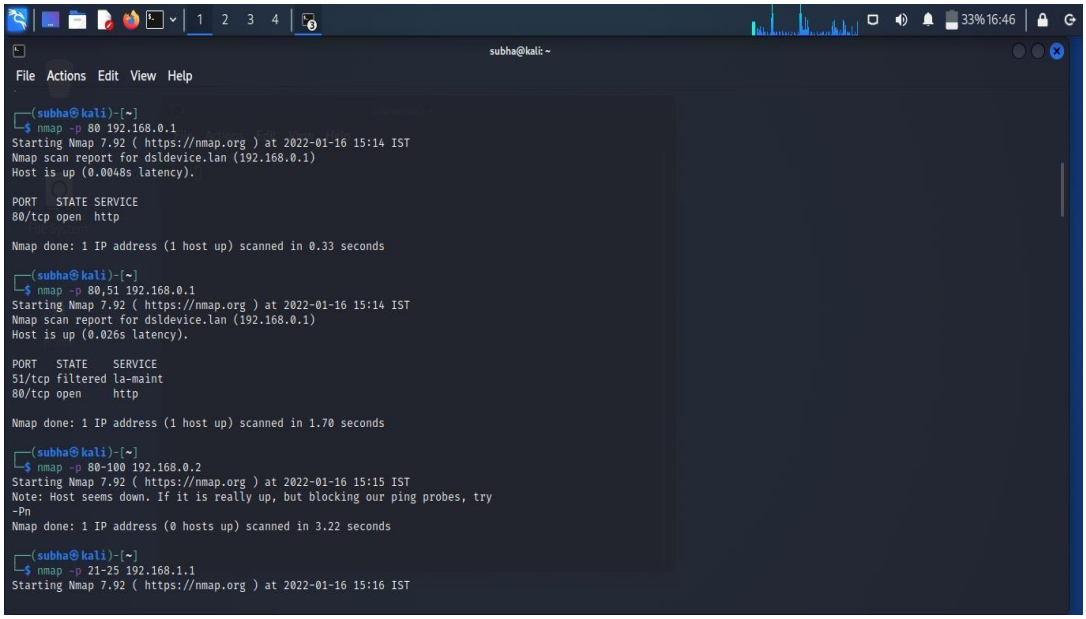


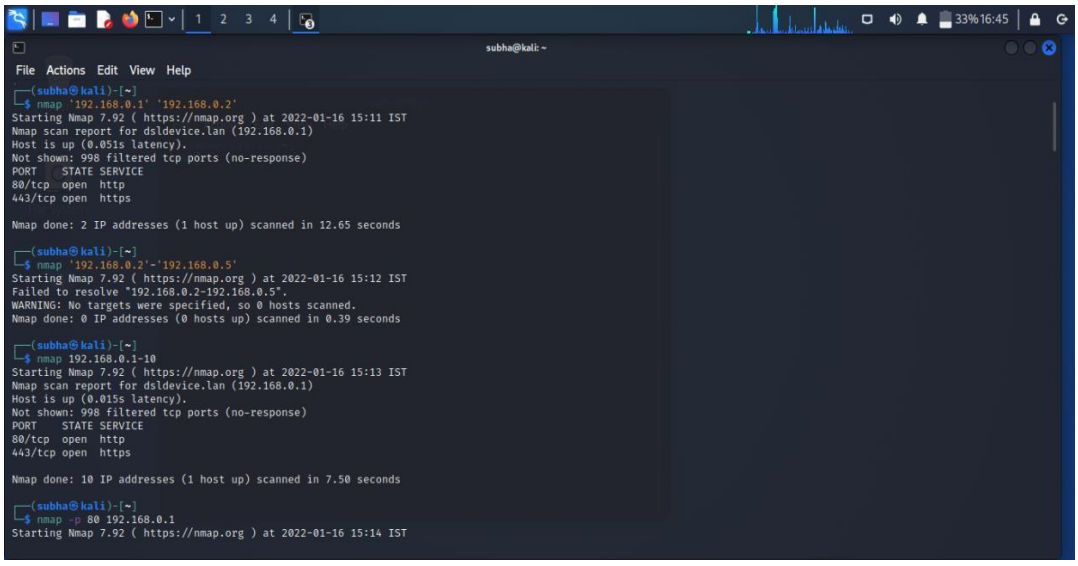


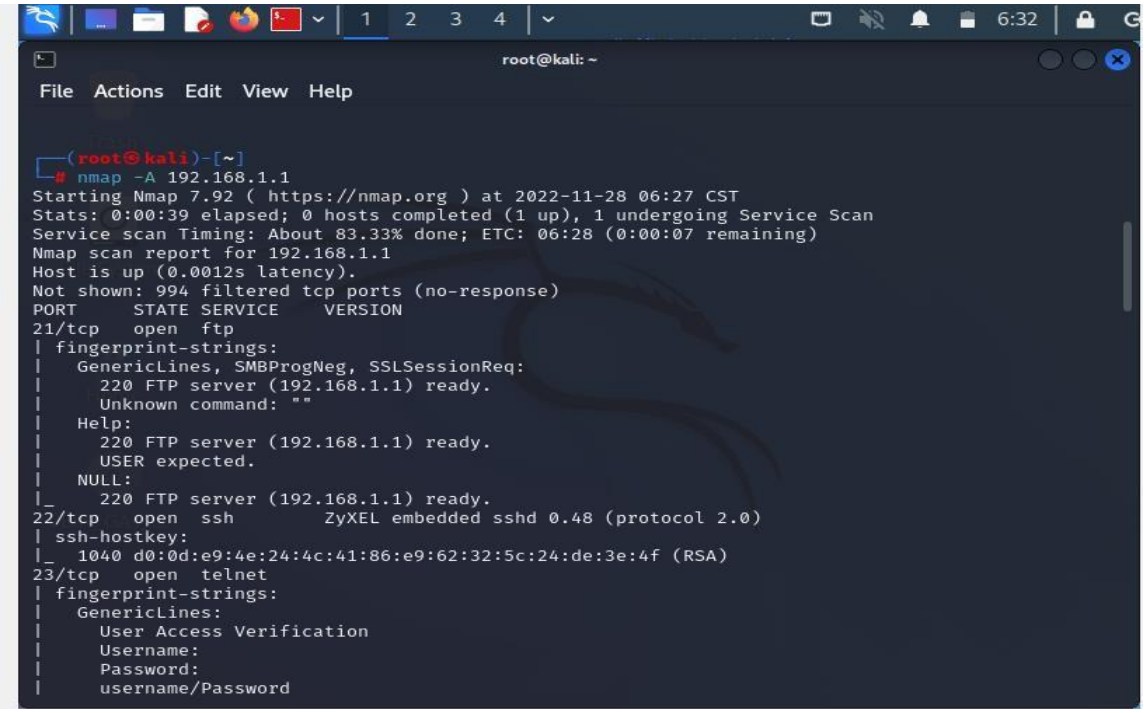

subha@kali: ~
File Actions Edit View Help
192.168.1.23
(subha㉿kali)-[~]
$ nmap -il addresses.txt
Failed to open input file l for reading: No such file or directory (2)
(subha㉿kali)-[~]
$ nmap -il addresses
Failed to open input file l for reading: No such file or directory (2)
(subha㉿kali)-[~]
$ cat > cs.txt
192.168.0.1
157.240.229.35
192.168.1.23
^C
(subha㉿kali)-[~]
$ nmap -iL cs.txt
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:52 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.020s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
443/tcp open https
Nmap done: 3 IP addresses (1 host up) scanned in 6.54 seconds
(subha㉿kali)-[~]
$ nmap -oG /root/cs.txt '157.240.229.35
quote> '
Unknown argument to -O.
QUITTING!
Screenshot taken
View image

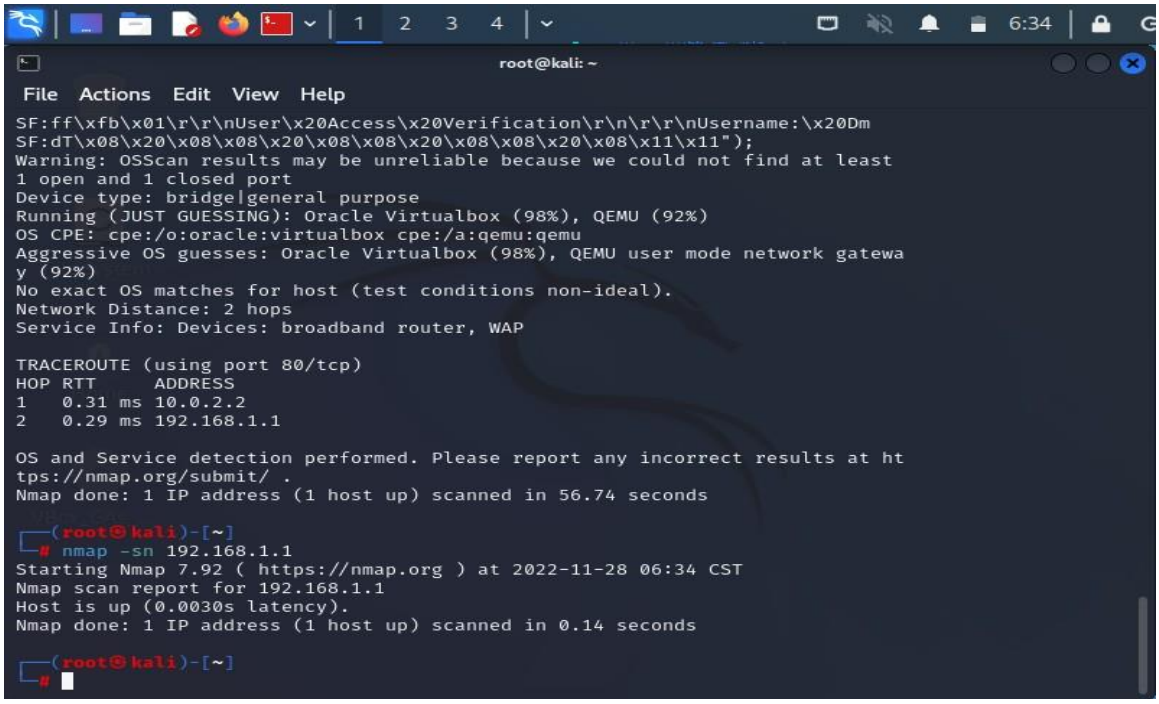

subha@kali: ~
File Actions Edit View Help
(subha㉿kali)-[~]
$ nmap -p 0-65535 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:27 IST
Stats: 0:11:54 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 74.85% done; ETC: 15:43 (0:04:00 remaining)
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.042s latency).
Not shown: 65532 filtered tcp ports (no-response), 1 filtered tcp ports (host-unreach), 1 filtered tcp ports (net-unreach)
PORT STATE SERVICE
80/tcp open http
443/tcp open https
Nmap done: 1 IP address (1 host up) scanned in 958.79 seconds
(subha㉿kali)-[~]
$ cat addresses.txt
cat: addresses.txt: No such file or directory
(subha㉿kali)-[~]
$ cat > addresses.txt
192.168.0.1
157.240.229.35
192.168.1.23
^C
(subha㉿kali)-[~]
$ cat addresses.txt
192.168.0.1
157.240.229.35
192.168.1.23
(subha㉿kali)-[~]
$ nmap -il addresses.txt
Failed to open input file l for reading: No such file or directory (2)


subha@kali: ~
File Actions Edit View Help
(subha㉿kali)-[~]
$ nmap -p 21-25 192.168.1.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:16 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.23 seconds
(subha㉿kali)-[~]
$ nmap -p 21-25 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:17 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.012s latency).
PORT STATE SERVICE
21/tcp filtered ftp
22/tcp filtered ssh
23/tcp filtered telnet
24/tcp filtered priv-mail
25/tcp filtered smtp
Nmap done: 1 IP address (1 host up) scanned in 1.42 seconds
(subha㉿kali)-[~]
$ nmap -p 0-65535 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:18 IST
Stats: 0:04:49 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 23.04% done; ETC: 15:39 (0:16:05 remaining)
Stats: 0:06:56 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 31.95% done; ETC: 15:40 (0:14:46 remaining)
Stats: 0:06:58 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 32.12% done; ETC: 15:40 (0:14:43 remaining)
Stats: 0:06:59 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 32.17% done; ETC: 15:40 (0:14:44 remaining)
Stats: 0:07:10 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan


subha@kali: ~
File Actions Edit View Help
(subha㉿kali)-[~]
$ nmap -p 80 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:14 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.0048s latency).
PORT STATE SERVICE
80/tcp open http
Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
(subha㉿kali)-[~]
$ nmap -p 80,51 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:14 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.026s latency).
PORT STATE SERVICE
51/tcp filtered la-maint
80/tcp open http
Nmap done: 1 IP address (1 host up) scanned in 1.70 seconds
(subha㉿kali)-[~]
$ nmap -p 80-100 192.168.0.2
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:15 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.22 seconds
(subha㉿kali)-[~]
$ nmap -p 21-25 192.168.1.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:16 IST


subha@kali: ~
File Actions Edit View Help
(subha㉿kali)-[~]
$ nmap '192.168.0.1' '192.168.0.2'
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:11 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.051s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
443/tcp open https
Nmap done: 2 IP addresses (1 host up) scanned in 12.65 seconds
(subha㉿kali)-[~]
$ nmap '192.168.0.2'-'192.168.0.5'
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:12 IST
Failed to resolve "192.168.0.2-192.168.0.5".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.39 seconds
(subha㉿kali)-[~]
$ nmap 192.168.0.1-10
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:13 IST
Nmap scan report for dsldevice.lan (192.168.0.1)
Host is up (0.015s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
443/tcp open https
Nmap done: 10 IP addresses (1 host up) scanned in 7.50 seconds
(subha㉿kali)-[~]
$ nmap -p 80 192.168.0.1
Starting Nmap 7.92 (https://nmap.org) at 2022-01-16 15:14 IST





RESULT:

Thus, scanning using nmap tool was done successfully**.**

Ex.No:15

DATE: **MOBILE FORENSICS**

AIM:

To perform mobile forensics using magnetic AXIOM Tools. And to complete the following tasks.

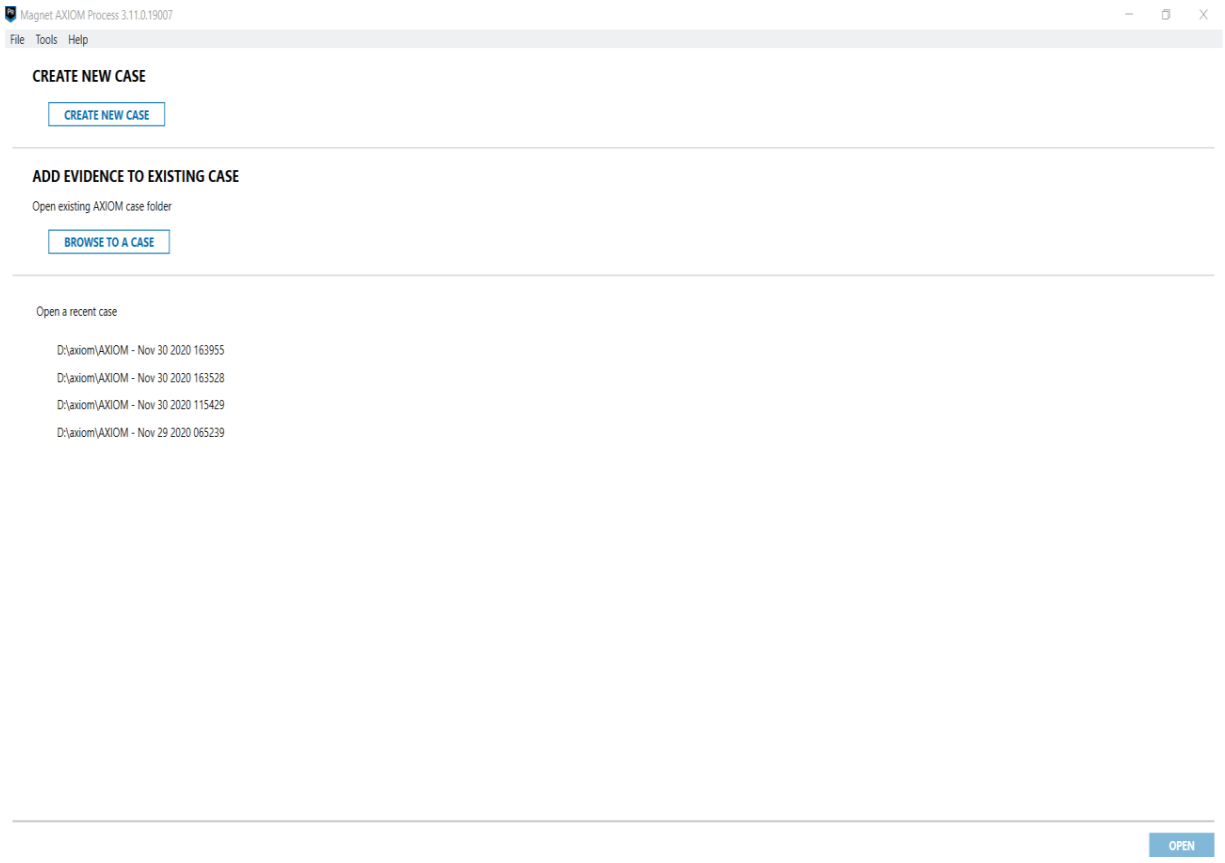
TASKS:

- Setup the Mobile to be Investigated and the AXIOM tool.
- Fetch call history of the Suspect mobile.
- Fetch the multimedia content (Pictures , Document, audio, Video) of the suspect Mobile.

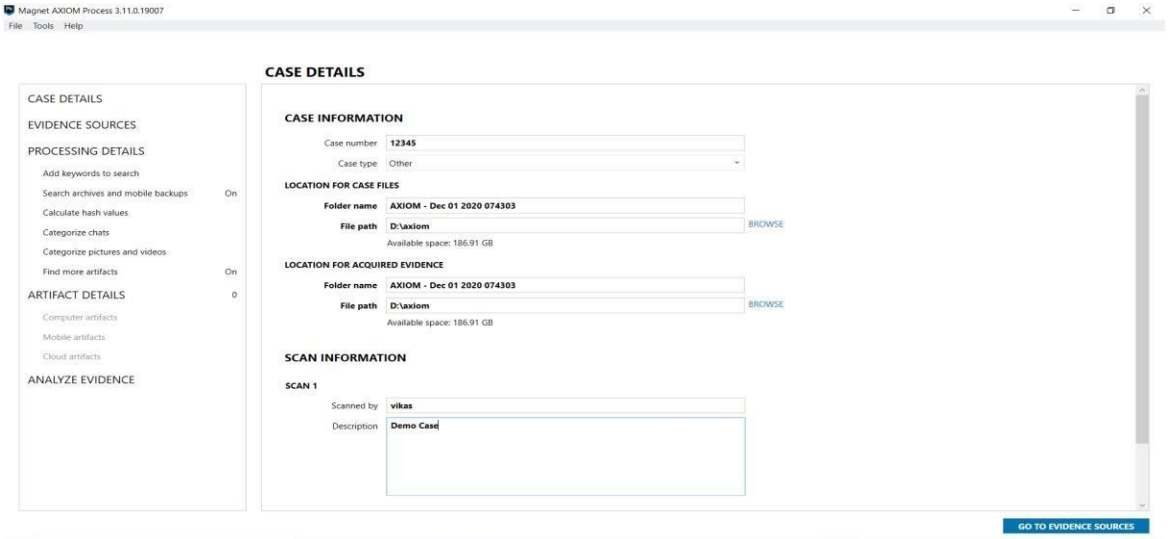
PROCEDURE:

Task 1. Setup the Mobile to be Investigated and the AXIOM tool.

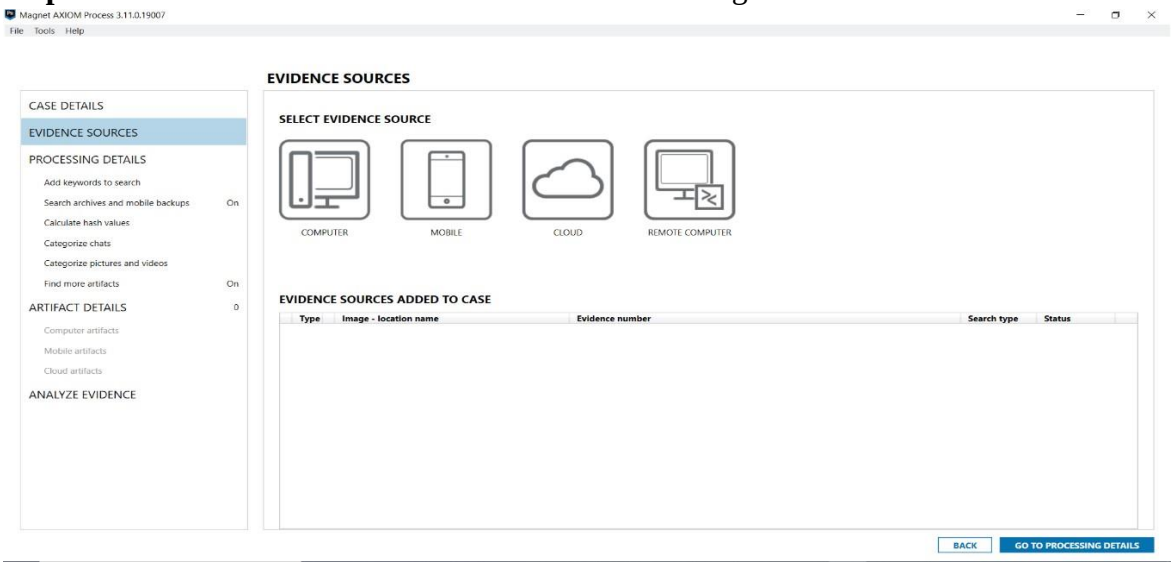
Step 1: - In Axiom process tool, click on create new case.



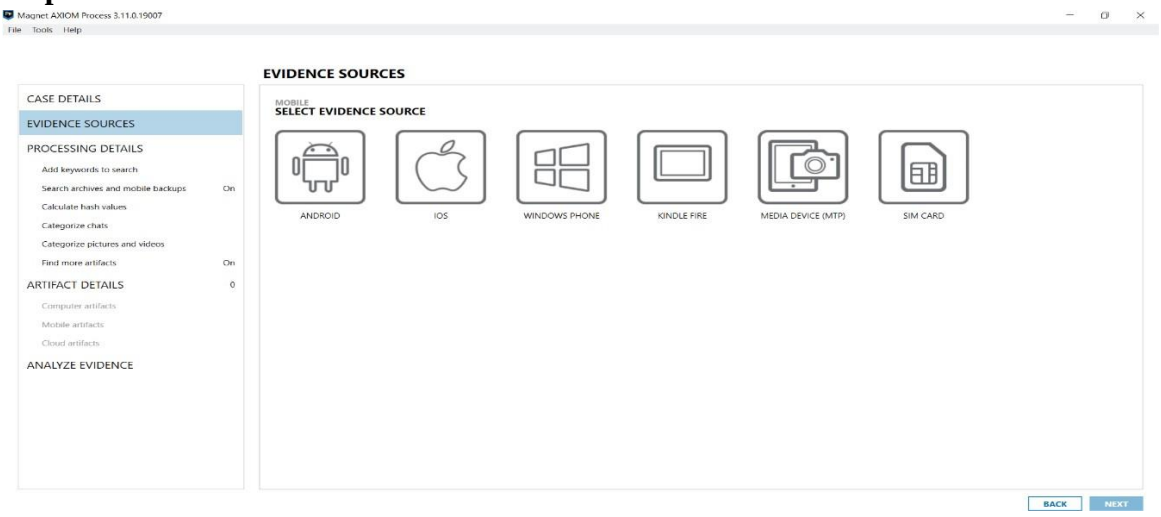
Step 2: Fill in the case details and click on go to evidence source.



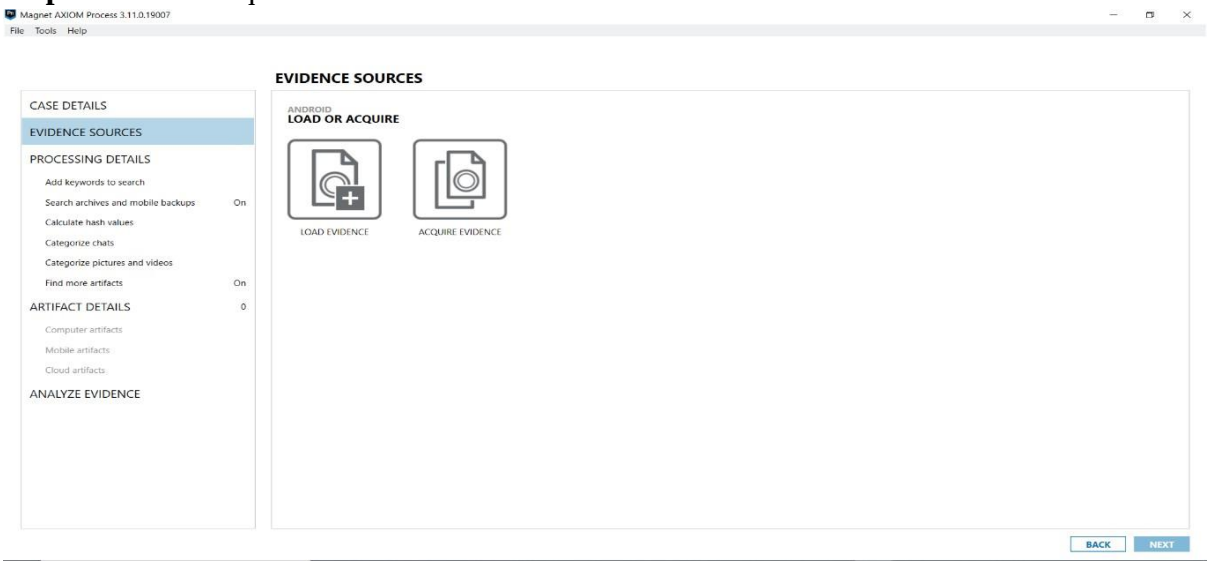
Step 3: -select evidence source as mobile .and click on go to evidence source.



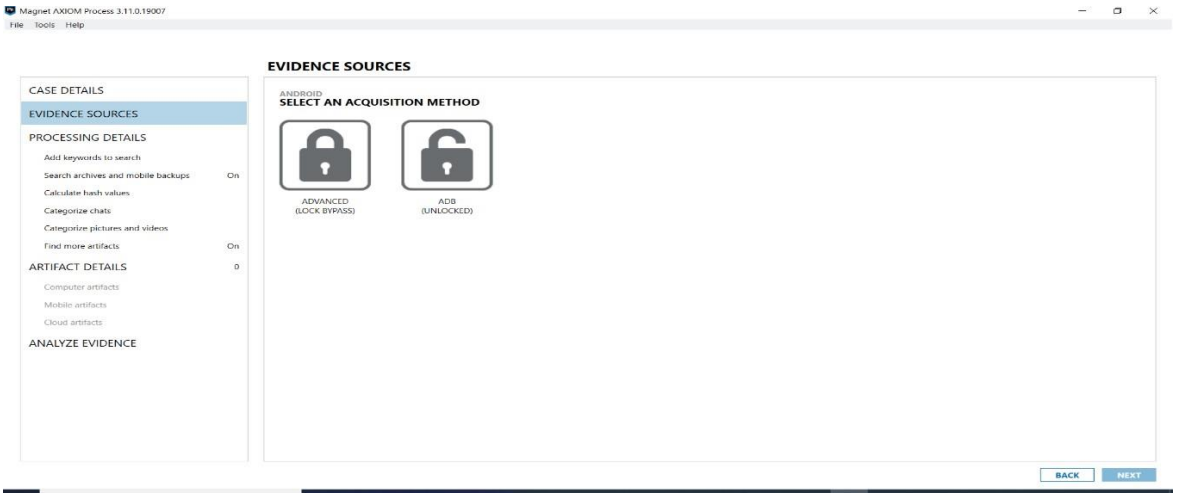
Step 4: -select evidence source as Android .and click on next.



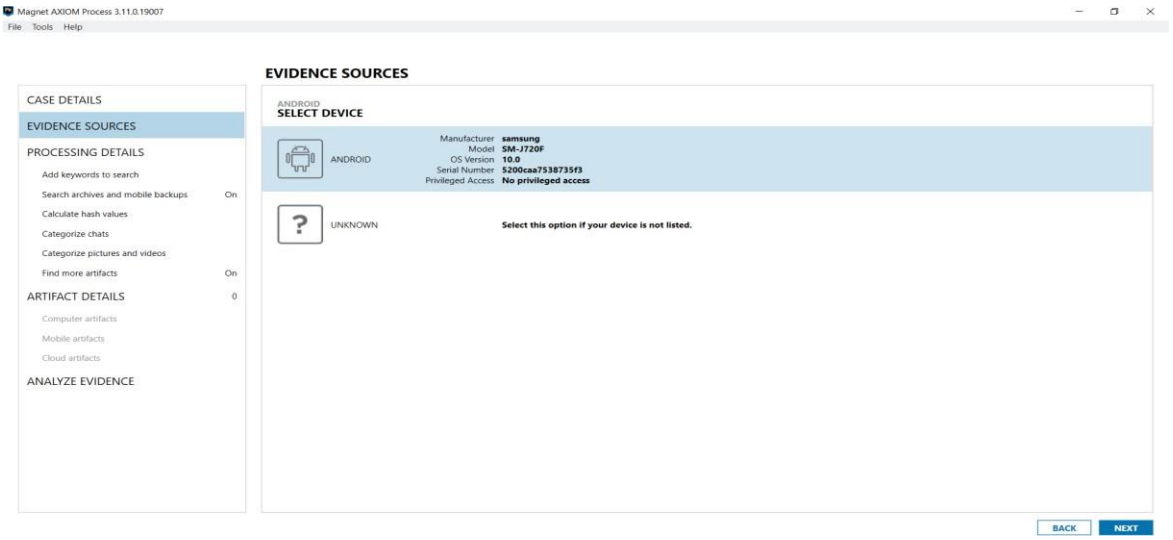
Step 5: -select Acquire Evidence .and click on next.



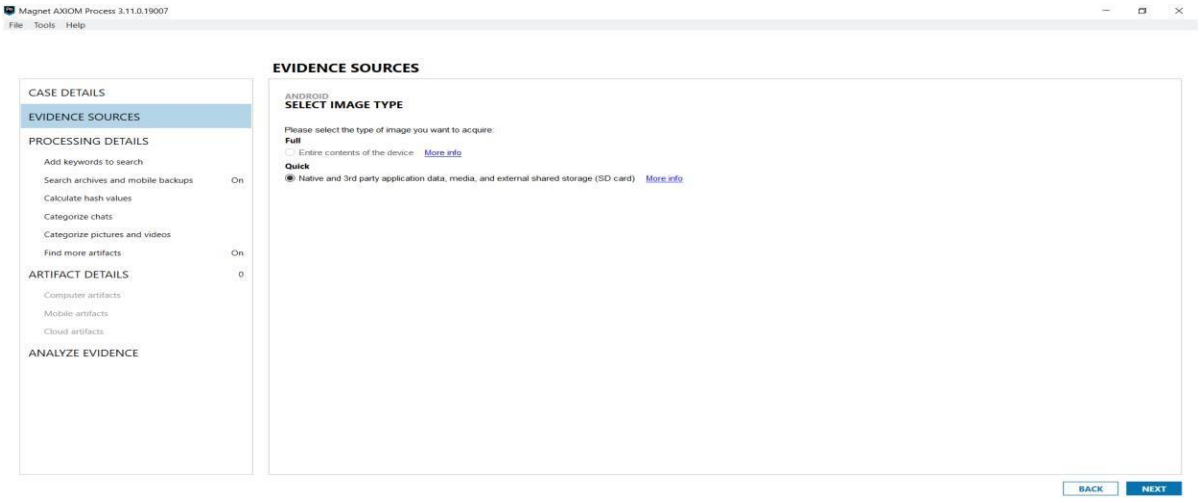
Step 6: -select ADB(Unlock) for unlocked device. and click on next.



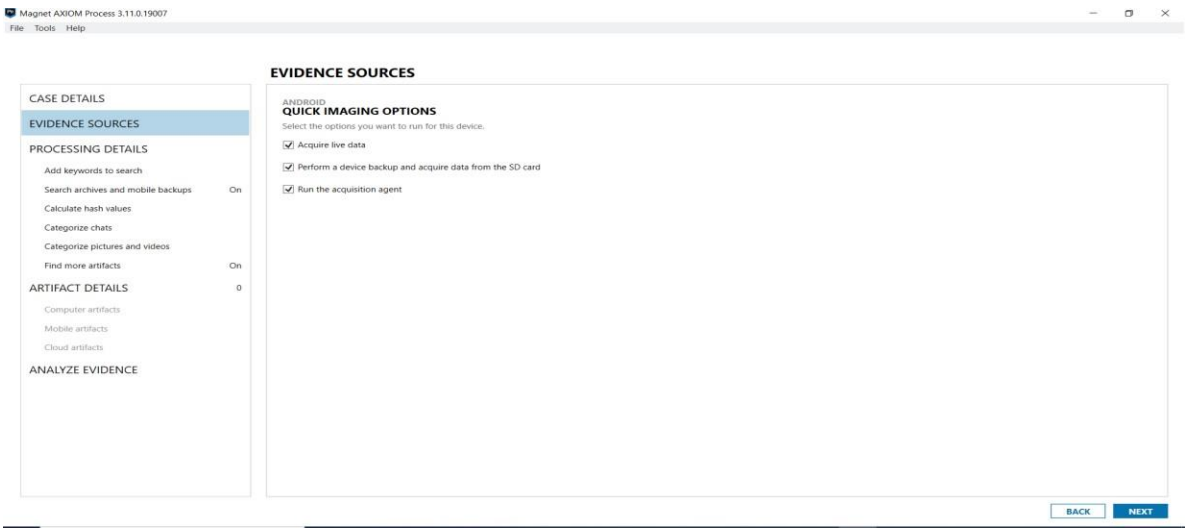
Step 7: -select detected device if not then unknown and follow the steps. and click on next button.



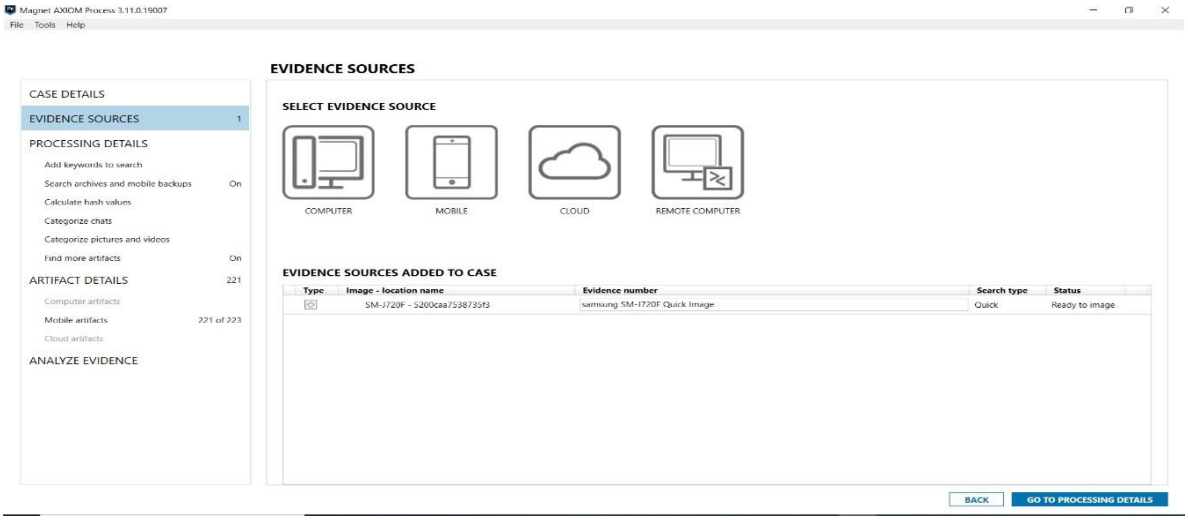
Step 8: -select Quick option button. and click on next button.



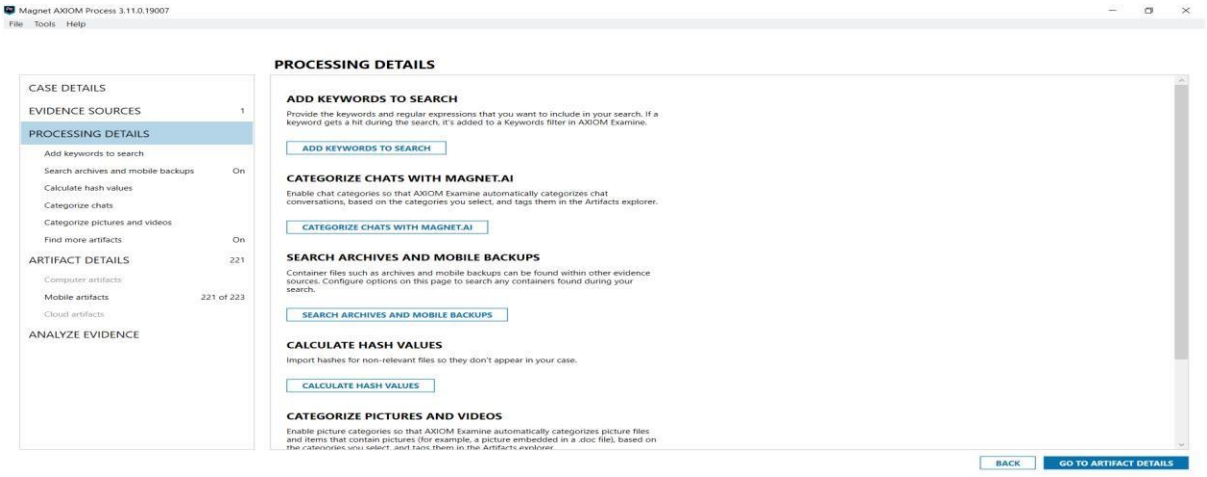
Step 9: -Click on next button.



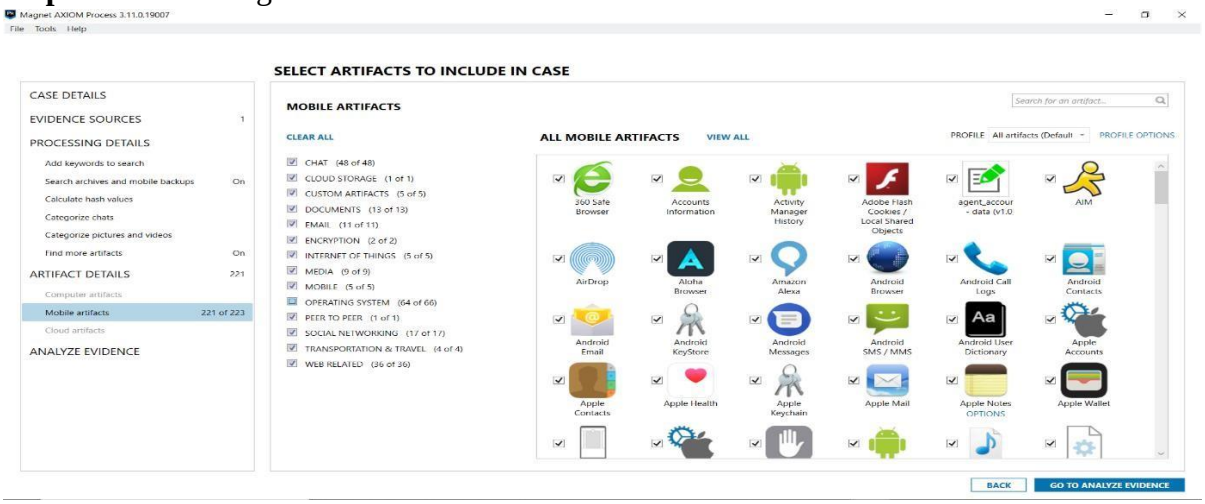
Step 10: -Click on go to evidence source.



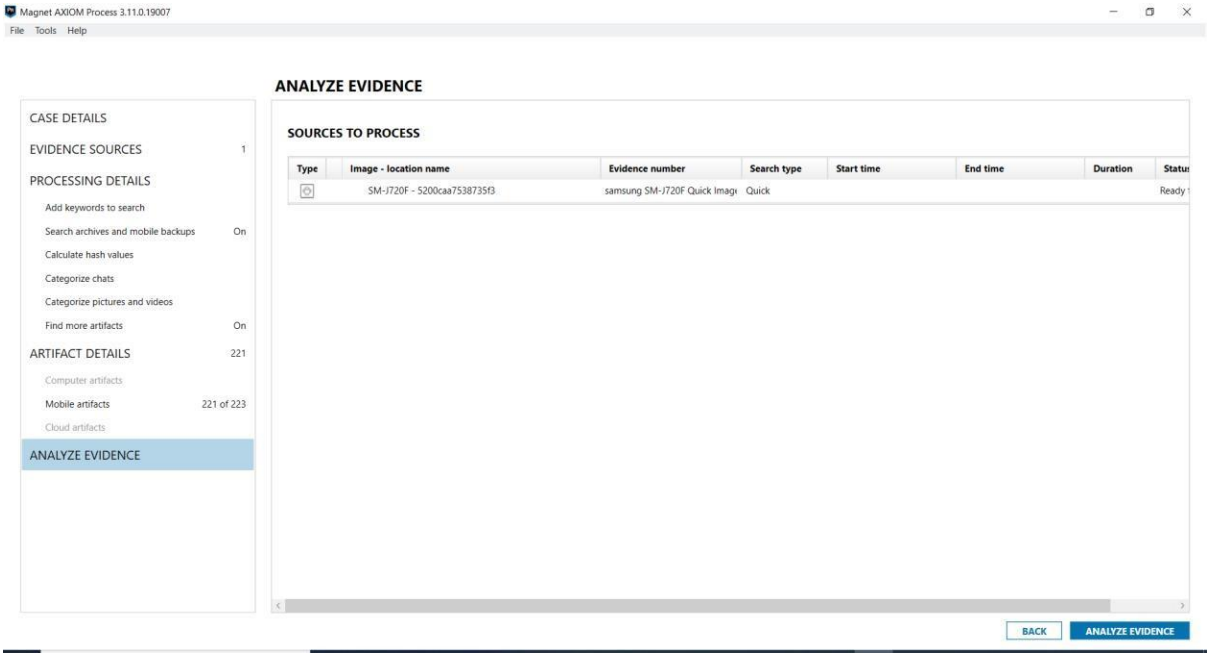
Step 11: -Click on go to evidence source. If you want to add filter you can do here.



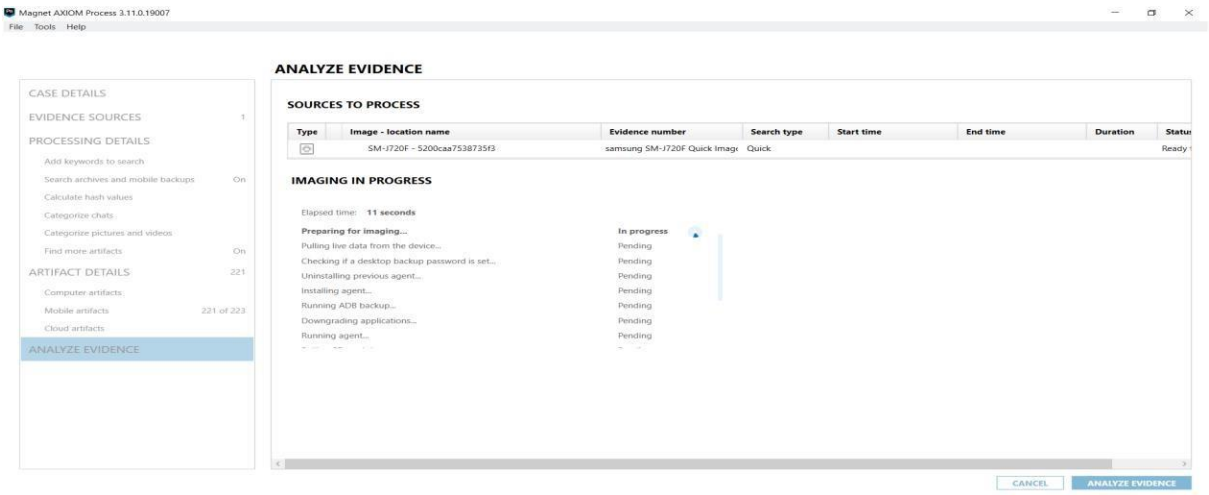
Step 12: -Click on go to evidence source.



Step 13: -Click on analyse evidence.

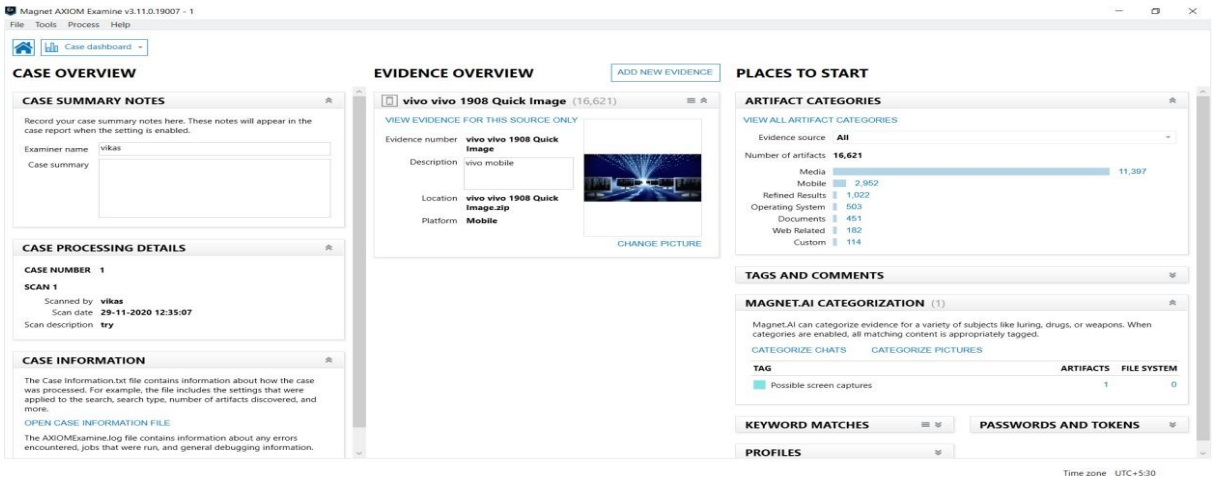


Step 14: -It will load and complete the task after a while.

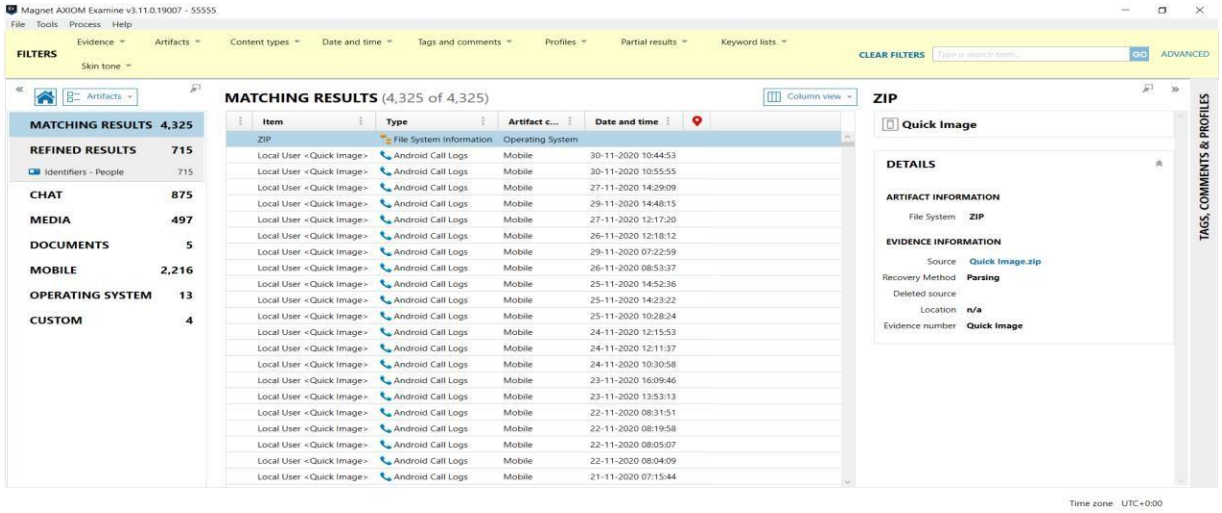


Task 2. Fetch call history of the Suspect mobile.

Step 1: - Now open Axiom Examine tool to retrieve call history. And click on view evidence for this source only.

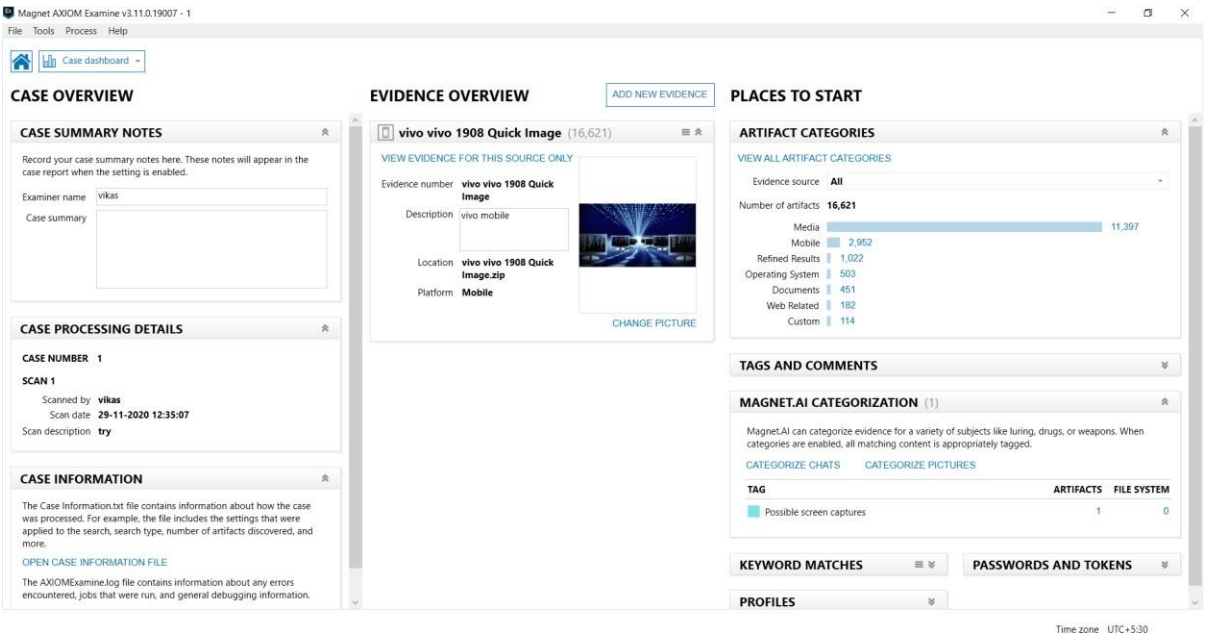


Step 2: -Here is the call history of the suspect.

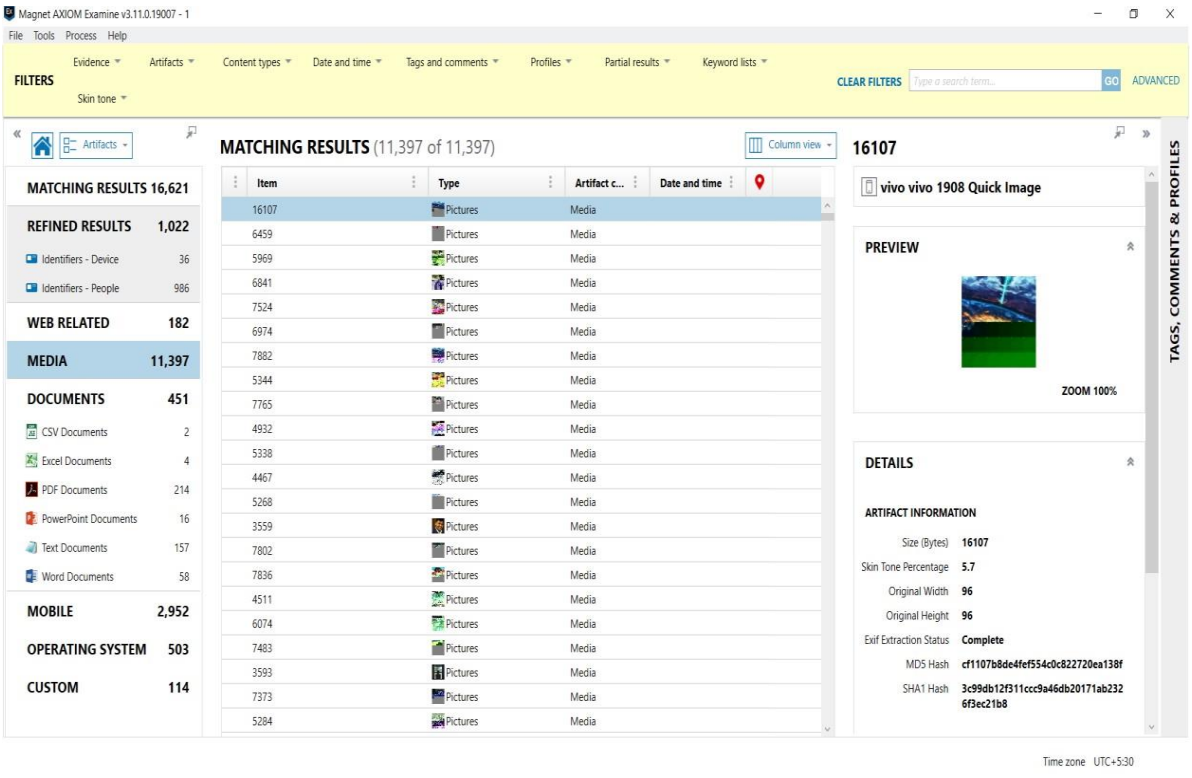


Task 3. Fetch the multimedia content (Pictures , Document, audio, Video) of the suspect Mobile.

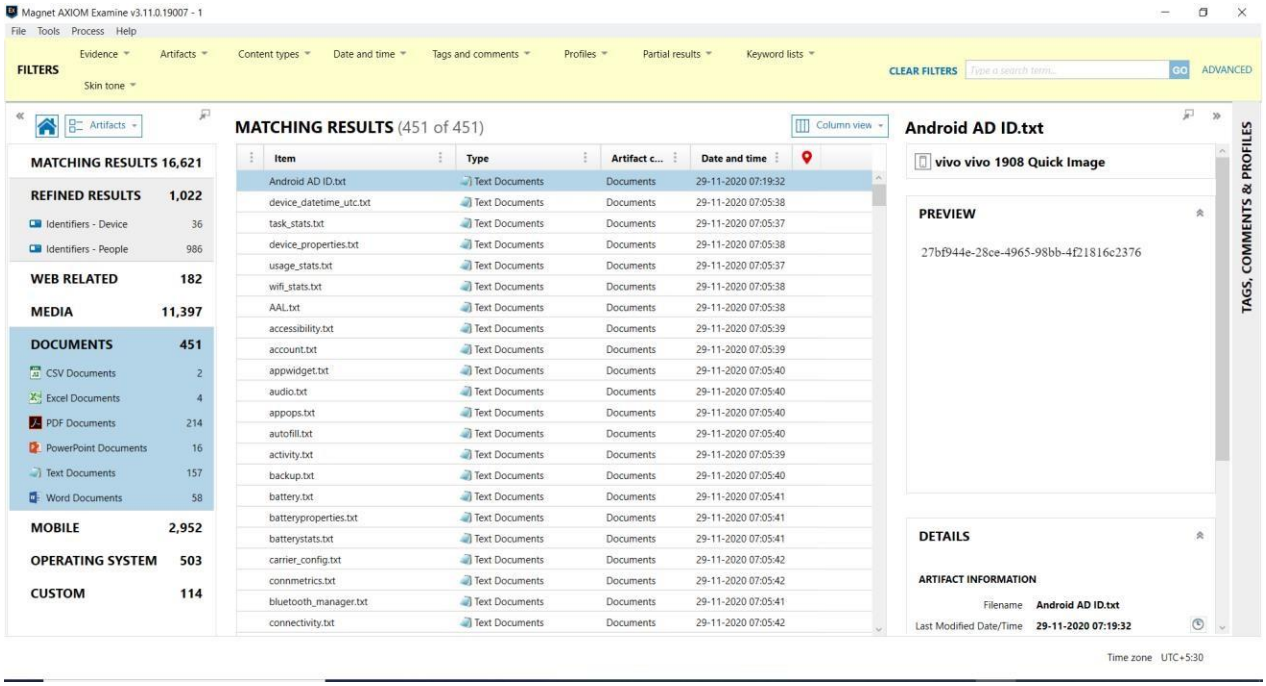
Step 1: - Now open Axiom Examine tool to retrieve Pictures , Document, audio, Video And click on view evidence for this source only.



Step 2: -Click on media to retrieve Pictures, audio, Video.



Step 3: - Click on document to retrieve Document.



Result: -

mobile forensics using magnetic AXIOM Tools.is done successfully and audio, video, image, document, call history is retrieved successfully and verified.